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OECD/NRC BENCHMARK BASED ON NUPEC PWR SUBCHANNEL AND BUNDLE TESTS (PSBT)

***Volume I: Experimental Database and Final Problem
Specifications***

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Foreword

The need to refine models for best-estimate calculations, based on good-quality experimental data, has been expressed in many recent meetings in the field of nuclear applications. The needs arising in this respect should not be limited to the currently available macroscopic methods but should be extended to next-generation analysis techniques that focus on more microscopic processes. One of the most valuable databases identified for the thermal-hydraulics modeling was developed by the Nuclear Power Engineering Corporation (NUPEC), Japan, which includes subchannel void fraction and departure from nucleate boiling (DNB) measurements in a representative Pressurized Water Reactor (PWR) fuel assembly. Part of this database is made available for this international benchmark activity entitled as the NUPEC PWR Subchannel and Bundle Tests (PSBT) benchmark. This international project is officially approved by the Japan Ministry of Economy, Trade, and Industry (METI), U.S. Nuclear Regulatory Commission (NRC), and endorsed by the OECD/NEA. The benchmark team is organized based on the collaboration between Japan and USA. A large number of international experts have agreed to participate in this program.

The fine-mesh high-quality subchannel void fraction and departure from nucleate boiling data encourages advancement in understanding and modeling complex flow behavior in real bundles. Considering that the present theoretical approach is relatively immature, the benchmark specification is designed so that it will systematically assess and compare the participants' analytical models on the prediction of detailed void distributions and DNB. The development of truly mechanistic models for DNB prediction is currently underway. The benchmark problem includes both macroscopic and microscopic measurement data. In this context, the subchannel grade void fraction data are regarded as the macroscopic data and the digitized computer graphic images are the microscopic data, which provides void distribution within a subchannel.

The NUPEC PSBT benchmark consists of two parts (phases). Each part is consisting of different exercises:

Phase I – Void Distribution Benchmark

- Exercise 1 – Steady-state single subchannel benchmark
- Exercise 2 – Steady-state bundle benchmark
- Exercise 3 – Transient bundle benchmark
- Exercise 4 – Pressure drop benchmark

Phase II – DNB Benchmark

- Exercise 1 – Steady-state fluid temperature benchmark
- Exercise 2 – Steady-state DNB benchmark
- Exercise 3 – Transient DNB benchmark

This report provides the specifications for the international OECD/NRC NUPEC PSBT benchmark problem. The specification report is prepared jointly by the Pennsylvania State University (PSU), USA and Japan Nuclear Energy Safety (JNES) Organization, in cooperation with US NRC and NEA/OECD. The work is sponsored by the US NRC, METI-Japan, NEA/OECD, and the Nuclear Engineering Program (NEP), Pennsylvania State University. The specifications cover the four exercises of Phase I, and the three exercises of Phase II. In addition, a CD-ROM has also been prepared with the complete NUPEC PWR database and is distributed along with the specifications to the participants, who have signed the NEA/OECD confidentiality agreement. The agreement as well as the other related information about the OECD/NRC PSBT benchmark can be found at:

<http://www.nea.fr/html/science/egrs/tb/PSBT/>.

Acknowledgments

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This report is the sum of many efforts including the funding agencies and their staff – the METI, Japan, US NRC and the Organization of Economic Co-operation and Development (OECD). Special appreciation goes to the report reviewer: Prof. Kostadin Ivanov from Pennsylvania State University (PSU). His comments, corrections, and suggestions were very valuable and significantly improved the quality of this report.

The authors wish to express their sincere appreciation for the outstanding support offered by the JNES personnel in providing the test data and discussing the test characteristics.

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Chapter 1

INTRODUCTION

1.1 Background

Over the last four years the Pennsylvania State University (PSU) under the sponsorship of U.S. Nuclear Regulatory Commission (NRC) has prepared, organized, conducted and summarized the OECD/NRC Benchmark based on NUPEC BWR Full-size Fine-mesh Bundle Tests (BFBT). The international benchmark activities have been conducted in cooperation with the Nuclear Energy Agency (NEA), OECD and Japan Nuclear Energy Safety (JNES) organization, Japan. From 1987 to 1995, the Nuclear Power Engineering Corporation (NUPEC) in Japan performed a series of void measurement tests using full-size mock-up tests for both BWRs and PWRs. For BWRs, based on state-of-the-art computer tomography (CT) technology, the void distribution was visualized at the mesh size smaller than the subchannel under actual plant conditions. NUPEC also performed steady-state and transient critical power test series based on the equivalent full-size mock-ups. Considering the reliability not only of the measured data, but also other relevant parameters such as the system pressure, inlet sub-cooling and rod surface temperature, these test series supplied the first substantial database for the development of truly mechanistic and consistent models for void distribution and boiling transition.

Consequently, the JNES has made available the BWR NUPEC database for the purposes of the OECD/NRC BFBT international benchmark. This international benchmark has encouraged advancement in the uninvestigated fields of two-phase flow theory with very important relevance to the nuclear reactors safety margins evaluation. The BFBT benchmark is made up of two parts (phases), each part consisting of four different exercises and has been used for validation of CFD, subchannel and system thermal-hydraulic codes. The BFBT benchmark activity has been very successful with about thirty (30) organizations from fifteen (15) countries participating in different benchmark exercises. The BFBT activity is being completed with final comparative reports to be published as US NRC NUREG and NEA/OECD reports. A special Nuclear Engineering and Design journal issue is being prepared devoted to the BFBT benchmark, which will document the models and results of the participants.

Based on the success of the OECD/NRC BFBT benchmark the JNES, Japan has decided to release also the data based on the NUPEC PWR Subchannel and Bundle Tests (PSBT) for an international benchmark and has asked PSU to organize and conduct this benchmark activity. Void fraction measurements and departure from nucleate boiling (DNB) tests were performed at NUPEC under the conditions simulating PWR thermal-hydraulic conditions, including the steady-states and the transients such as the power increase, the flow reduction, the depressurization and the temperature increase.

1.2 Objective

The established international OECD/NRC PWR Subchannel and Bundle Tests (PSBT) benchmark, based on the NUPEC database, encourages advancement in subchannel analysis of fluid flow in rod bundles, which has very important relevance to the nuclear reactor safety margin evaluation. This benchmark specification is being designed so that it can systematically assess and compare the participants' numerical models for the prediction of detailed subchannel void distributions and departure from nucleate boiling (DNB) to full scale experimental data on a prototypical PWR rod bundle. Currently the numerical modeling of subchannel void distribution has limited theoretical approach that can be applied to a wide range of geometrical and operating

conditions. In the past decade, experimental and computational technologies have tremendously improved the study of the flow structure. Over the next decade, it can be expected that mechanistic approaches will be more widely applied to the complicated fluid phenomena inside fuel bundles. The development of truly mechanistic models for DNB prediction is currently underway. These models must include processes such as void distribution, droplet deposition, liquid film entrainment, and spacer grid behavior. The benchmark specification requires participants to explain their modeling correlations between the measured DNB and the flow dominant processes.

It should be recognized that the purpose of this benchmark is not only the comparison of currently available computational approaches but above-all the encouragement to develop novel next-generation approaches that focus on more microscopic processes. In this context, the subchannel grade void fraction data are regarded as the macroscopic data and the digitized computer graphic images are the microscopic data, which provides the detailed void distribution within the subchannel. This benchmark specification is supplemented by a CD-ROM with complete NUPEC PWR Benchmark Database, which is distributed to all participants who have signed the confidentiality agreement.

1.3 Outline of the PSBT Specification

Chapter 1 of this specification discusses the main objectives of the international OECD/NRC PSBT benchmark. A definition of the benchmark phases and exercises is provided.

Chapter 2 discusses the NUPEC PWR PSBT facility and the specific methods used in the void distribution and DNB measurements.

Chapter 3 provides specifications of fuel assembly types and heater rods as well as spacer data and thermo-mechanical properties of structural materials.

Chapter 4 defines the NUPEC PSBT data to be used for the benchmark exercises. In total seven exercises of two phases will be performed in the framework of this benchmark.

Chapter 5 describes the requested output and the output format that the benchmark team will collect from the participants.

Chapter 6 summarizes the conclusions of the specification.

1.4 Definition of Benchmark Phases

The test facility design and data from NUPEC includes both macroscopic and microscopic measurements. There are two separate phases, each consisting of different exercises. These phases and exercises are discussed below.

1.4.1 Phase I - Void Distribution Benchmark

The purpose of this benchmark phase is three-fold:

- 1) To provide data for validation of numerical models of void distribution;
- 2) To provide data over a wide range of geometrical and operating conditions for validation of void distribution models;
- 3) To provide data for development of mechanistic approaches widely applicable to the fluid phenomena inside fuel bundles.

Phase I includes four exercises:

Exercise 1 – Steady-state single subchannel benchmark, which can be used to improve the current models of void generation and void distribution within a subchannel.

Exercise 2 – Steady-state bundle benchmark, which can be applied to subchannel and system thermal-hydraulic codes.

Exercise 3 – Transient bundle benchmark, which can be used to determine the capabilities of subchannel and system thermal-hydraulic codes of predicting void generation during transients.

Exercise 4 – Pressure drop benchmark. Code-to-code comparisons will also be performed.

1.4.2 Phase II - Departure from Nucleate Boiling (DNB) Benchmark

The purpose of this benchmark is to develop truly mechanistic models for DNB prediction.

Phase II includes three exercises:

Exercise 1 – Steady-state fluid temperature benchmark, which can be applied to subchannel and system thermal-hydraulic codes as well as CFD codes.

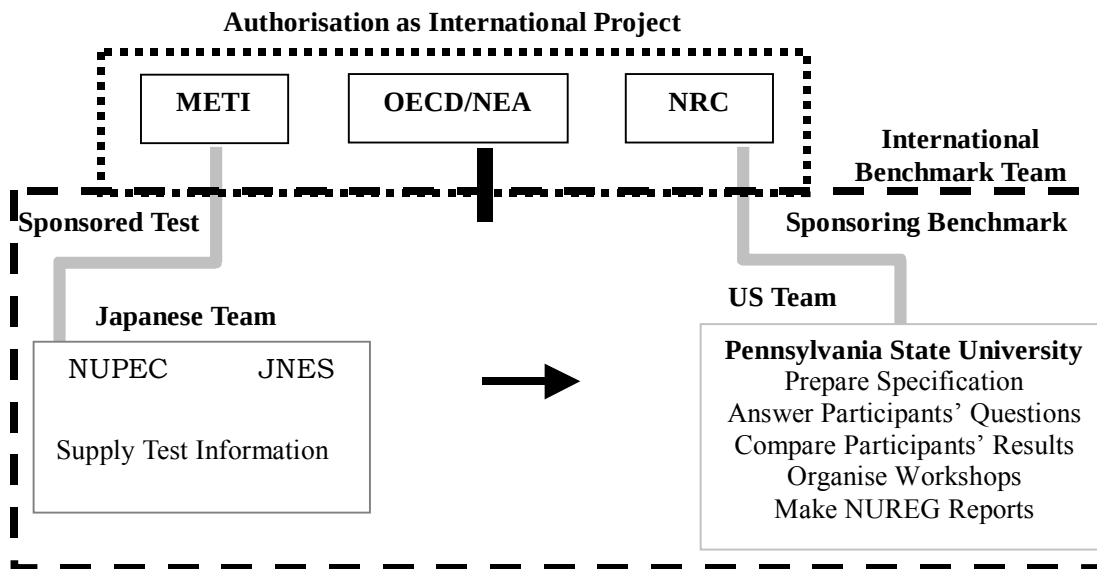
Exercise 2 – Steady-state DNB benchmark, which can be used to improve the current models of DNB prediction in PWR bundles at steady-state.

Exercise 3 – Transient DNB benchmark, which can be used to improve the current models of DNB prediction in PWR bundles during transients.

1.5 Benchmark Team and Sponsorship

The benchmark activities are being performed as an international project supported by US NRC and METI (Japan), and endorsed by the OECD/NEA. The benchmark team is organized based on the collaboration between USA and Japan as shown in Figure 1. This approach has been very successful in the OECD/NRC BFBT benchmark, which has been completed. It shall be recognized that METI had sponsored the NUPEC PWR subchannel and bundle test project. JNES on behalf of METI provides the test information and measured data. OECD/NEA provides supporting activities for this international PSBT benchmark project. The US team, headed by Prof. Avramova and Prof. Ivanov from PSU, co-ordinates the benchmark project in cooperation with US NRC. The PSU team is preparing the benchmark specification, organizing the technical content of the workshops (the logistics of the workshops is being managed by NEA/OECD), answering the questions issued by participants, comparing and analyzing participants' results and preparing comparison reports as NEA/OECD and NUREG reports. The basic activity of the US benchmark team (PSU team) is sponsored by the US NRC. In addition to the above described activities the PSU team is performing PSBT calculations using the PSU version of COBRA-TF and working in cooperation with US NRC on the development and validation of a subchannel model in TRACE using the PSBT data.

Figure 1.5.1 PSBT Benchmark Team



Chapter 2 TEST FACILITIES

2.1 General

The void distribution and DNB measurements took place at the facility described in Section 2.2. The facility is able to simulate the high-pressure, high-temperature fluid conditions found in PWR's. Although the same test loop was used for both phases, different test sections were constructed to represent a single subchannel and a complete rod bundle.

2.2 Test Loop

The NUPEC test facility shown in Figure 2.2.1 consists of a high pressure and high temperature recirculation loop, a cooling loop, and instrumentation and data recording systems. The recirculation loop consists of a test section, circulation pump, preheater, steam drum (acting as a pressurizer), and a water mixer. The design pressure is 19.2 MPa and the design temperature is 362 °C. The operating conditions of the test facility are shown in Table 2.2.1 and Table 2.2.2. Reference rated operating conditions of a PWR are listed in Table 2.2.3. Note that the units given for pressure in Table 2.2.3 may require conversion to more conventional units. The following conversions are to be used:

$$1 \frac{\text{kg}}{\text{cm}^2\text{a}} = \begin{cases} 0.980665 \text{ bar} \\ 14.223 \text{ psia} \end{cases}$$

Table 2.2.1 Range of NUPEC PWR Test Facility Operating Conditions

Quantity	Range
Pressure	4.9 – 16.6 MPa
Mass Velocity	550 – 4150 kg/m ² s
Inlet Coolant Temperature	140 – 345 °C

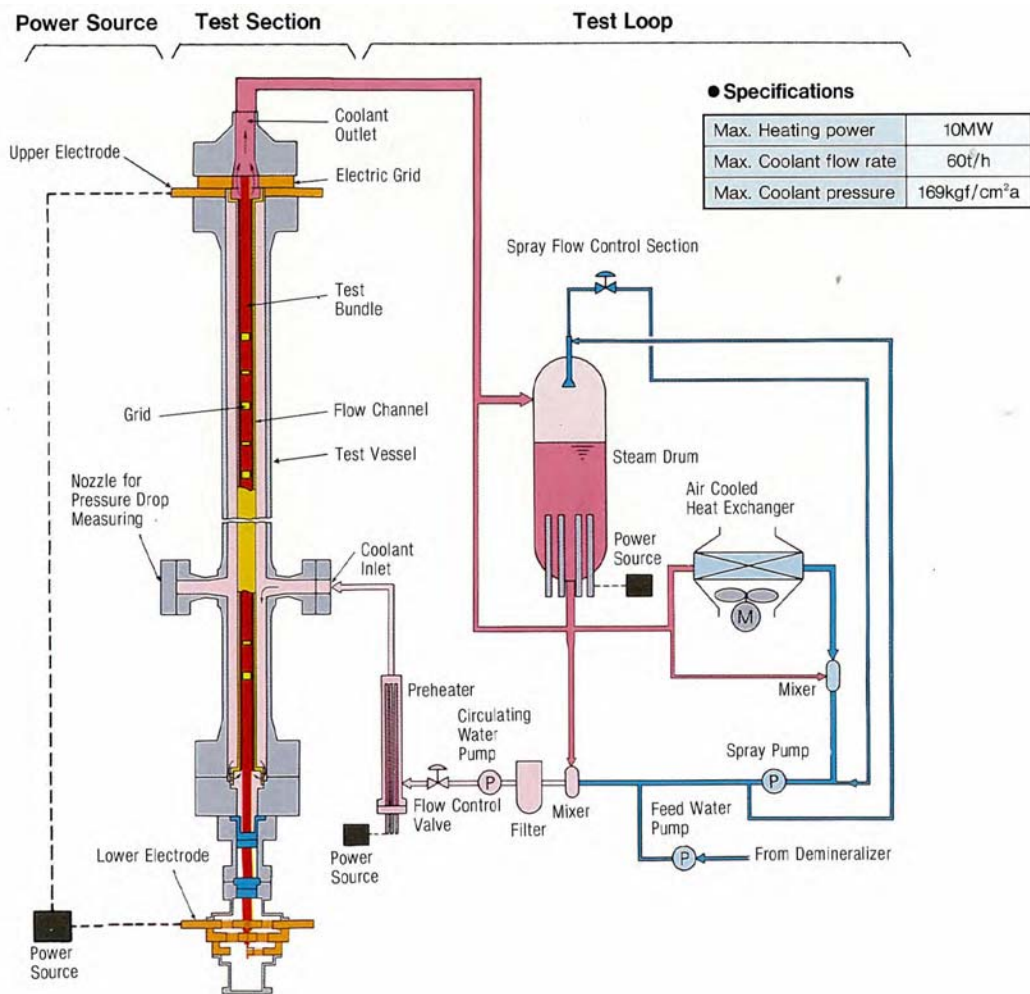
Table 2.2.2 Transient Parameters of NUPEC PWR Test Facility

Transient Scenario	Transient Change
Depressurization	-0.03 MPa/s
Temperature Increase	1 °C/s
Flow Reduction	-25 %/s
Power Increase	15 %/s

Table 2.2.3 Reference Rated Operating Conditions of PWR

Test section	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² h)	Heat flux (10 ⁶ kcal/m ² h)	Power (MW)	Inlet temperature (°C)
5×5 Rod bundle	158	12	0.5	2.7	290
6×6 Rod bundle	158	12	0.5	3.9	290

Figure 2.2.1 System Diagram of NUPEC PWR Test Facility



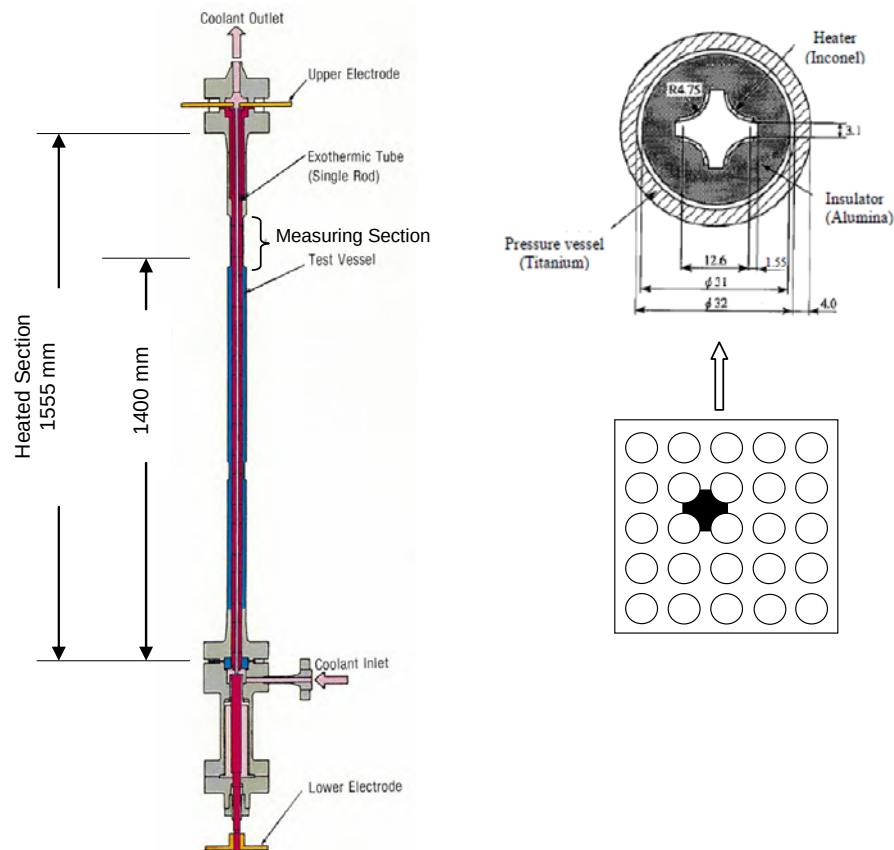
2.3 Test Section

Three different test sections were used to perform the void distribution measurement tests; one for the subchannel void distribution test, one for the rod bundle void distribution test, and one for the bundle DNB measurements.

2.3.1 Single Subchannel Void Distribution Measurements

Figure 2.3.1.1 shows the test section used for the central subchannel void measurement. It simulates one of the subchannel types found in a PWR assembly. Coolant flows in the pressure vessel horizontally through the coolant inlet nozzle located just below the heated section. Similar test sections were used for the central with thimble, side, and corner subchannel types. The effective heated length is 1555 mm, and the void measurement section begins at 1400 mm from the bottom of the heated section [1] .

Figure 2.3.1.1 Test Section for Central Subchannel Void Distribution Measurement



2.3.2 Bundle Void Distribution Measurements

An electrically heated rod bundle was used to simulate a partial section and full length of a PWR fuel assembly. Figure 2.3.2.1 shows the test section used for the rod bundle void measurements. The effective heated length is 3658 mm, which is broken into three sections (upper, middle, lower), measuring at 3177 mm, 2669 mm, and 2216 mm, respectively. Coolant flows into the pressure vessel horizontally through the coolant inlet nozzle and down through the section between the flow channel and the pressure vessel. The coolant continues into the flow channel, flowing from the bottom of the pressure vessel up through the test assembly, where the bottom of the heated section is located 630 mm above the bottom of the pressure vessel. Table 2.3.2.1 provides data regarding the structure of the test vessel and flow channel. Table 2.3.2.2 shows the tolerances for the manufacture of the test assemblies.

Figure 2.3.2.1 Test Section for Rod Bundle Void Distribution Measurement

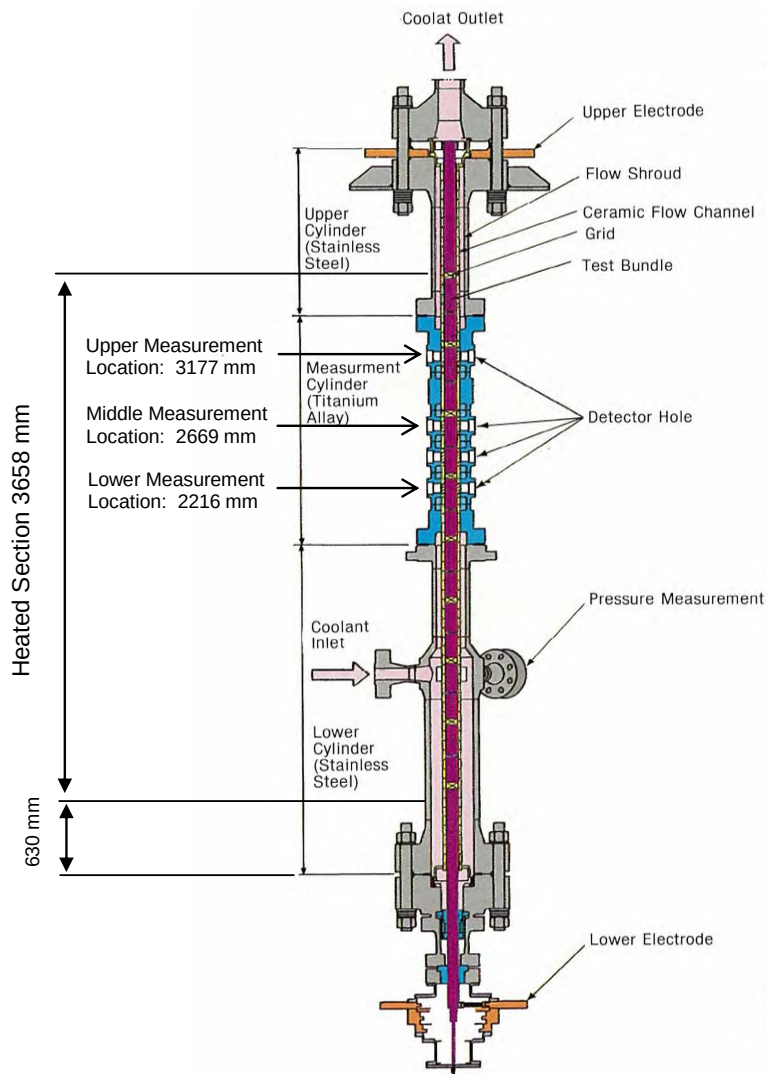


Figure 2.3.2.2 Cross Section of Test Vessel

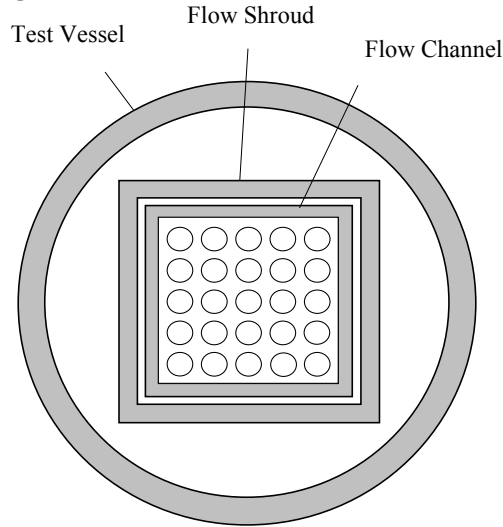


Table 2.3.2.1 Test Vessel and Flow Channel Structure Data

	Item	Data
Flow Channel	Inner length (mm)	64.9
	Thickness (mm)	5.0
	Material	Alumina
Flow Shroud	Inner length (mm)	75.1
	Thickness (mm)	10
	Material	SUS304
Test Vessel (Upper Cylinder)	Outer diameter(mm)	216.3
	Thickness (mm)	23.0
	Material	SUS304
Test Vessel (Lower Cylinder)	Outer diameter(mm)	318.5
	Thickness (mm)	33.3
	Material	SUS304

Table 2.3.2.2 Manufacturing Tolerances for Test Assembly

Item	Tolerance
Rod bundle	
Heater rod diameter	0.02 mm
Heater rod displacement	0.45 mm
Flow channel inner width	0.05 mm
Flow channel displacement	0.20 mm
Power distribution	3 %

2.3.3 Bundle DNB Measurements

The DNB measurements were performed for full-length partial 5×5 and 6×6 array rod bundles, which simulate 17×17 PWR fuel assemblies. Measurements were performed both for steady-state and transients. The measurements were performed at the facility shown in Figure 2.2.1 using the assemblies described in Table 3.3.1 .

2.4 Void Distribution Measurement Methods

The detailed explanation for the void fraction measurement procedure is provided in References [2] and [3]. A brief description follows.

A gamma-ray transmission method was used to measure the density of the flow, which was converted to the void fraction of the gas-liquid two-phase flow. Figure 2.4.1 shows the procedure used to perform the void fraction measurements for the entire rod bundle. The top half of the figure shows the procedure used to perform the void fraction measurements for a single subchannel. In the subchannel experiments [3] a narrow gamma-ray beam CT scanner was used to measure the subchannel averaged void fraction and a wide gamma-ray beam was used to measure the chordal averaged void fraction. For each subchannel type - corner, side, or center – a relationship between the subchannel averaged and the chordal averaged void fractions was individually derived. These relationships were then used to correct the subchannel averaged void fraction measured with the wide beam in the bundle tests.

The void measurement systems shown in Figure 2.4.1 consist of gamma-ray sources (^{137}Cs); detectors; collimators, and signal processing units. The attenuation of the gamma-rays, which depends on the void fraction, was detected. The intensity of the gamma-ray source was determined to obtain the count-rate of the signal processing (30×10^4 cps).

The CT scanner system was used to determine the distribution of density/void fraction over the subchannel at steady-state flow and to define the subchannel averaged void fraction. The system was operated by translate/rotate method. At each translation/rotation location, the intensity of gamma-ray attenuated by the object, the so-called “projection data”, was detected. An image reconstruction was then performed by a filtered back-projection algorithm to obtain the distribution of the linear attenuation coefficient. A sufficient measuring time was given in order to avoid the effect of the flow motion.

Two densitometer systems – in x- and y-directions - were used in the chordal averaged void fraction measurements in the single subchannel tests. Each of them consisted of a gamma source and a detector. They were located at the same tables (the same elevation) as CT, which was fixed during the measurements.

A multi-beam system was used to measure each subchannel void fraction of the rod bundle. Six transmission data of x-direction and six transmission data of y-direction between the rod and rod/channel wall were used to reconstruct the void fraction of the 36 subchannels by an iterative method. These subchannel void fractions corresponded to the chordal measurements of the single subchannel tests. The relationships between the subchannel averaged void fraction and the chordal averaged void fraction obtained in the single subchannel tests were used to determine the subchannel averaged void fractions in the rod bundle tests. Such measurements were performed simultaneously at three axial elevations.

Table 2.4.1 shows the estimated accuracy of the void fraction measurements. Table 2.4.2 describes the sources of error in the void measurement process (these values correspond to one standard deviation, 1σ).

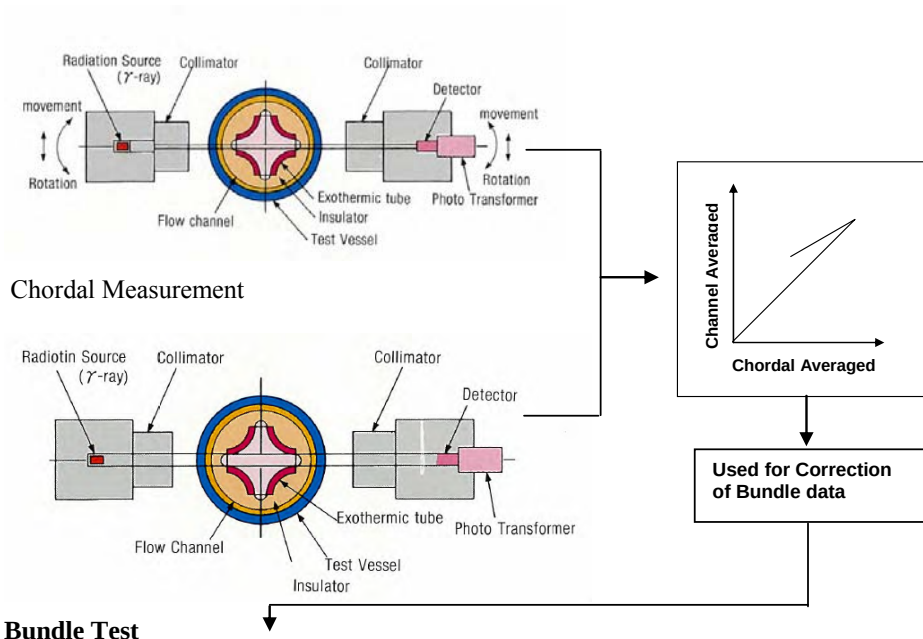
Figure 2.4.2 was used to determine the uncertainties inherent in the void measurements and to correct the measured values based on the pressure of the test case. The correlation between the chordal and CT averaged values is given by the best-fit curves and was introduced for the high and low-pressure conditions, respectively. The reference averaged density was 500 kg/m^3 . The uncertainty of the correlation was determined to be less than 18 kg/m^3 , which was regarded as three standard deviations (3σ). Therefore, one standard deviation (1σ) is 6 kg/m^3 . Table 2.4.3 provides information on the gamma beams used for these measurements.

Table 2.4.4 shows the time required to perform the void fraction measurements.

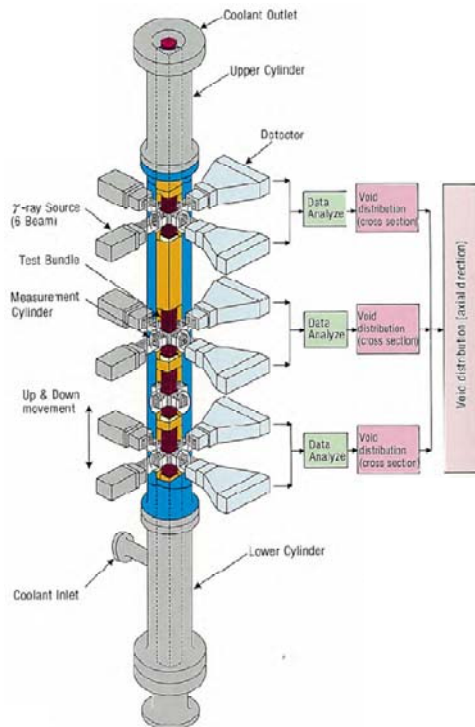
Figure 2.4.1 Void Fraction Measurement Procedure

Subchannel Test

CT Measurement

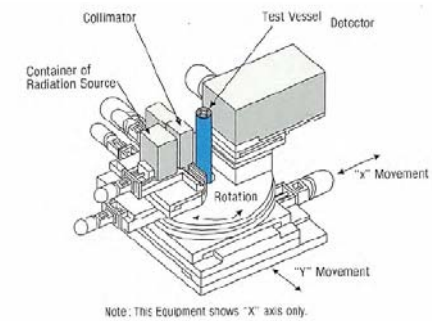


Bundle Test



Void Measurement Equipment

Subchannel Test



Bundle Test

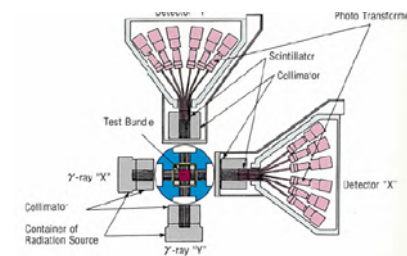


Table 2.4.1 Estimated Accuracy for Void Fraction Measurements

Quantity	Accuracy
Process parameters	
Pressure	1 %
Flow	1.5%
Power	1 %
Fluid temperature	1 Celsius
Void fraction measurement	
CT measurement	
Gamma-ray beam width	1 mm
Subchannel averaged (steady-state)	3% void
Spatial resolution of one pixel	0.5 mm
Chordal measurement	
Gamma-ray beam width (center)	3 mm
Gamma-ray beam width (side)	2 mm
Subchannel averaged (steady-state)	4% void
Subchannel averaged (transient)	5% void

Table 2.4.2 Error Sources for Void Measurements

Error source		Chordal Averaged		CT
		Steady-state	Transient	Averaged
γ -ray Measurement	Effect of surrounding condition (magnetic-field and temperature) on measurement system	0.1%	0.1%	0.1%
	Randomness of γ -ray source decay	0.02%	0.2%	0.1%
	Correction error due to back ground	0.0%	0.0%	0.0%
	Correction error due to counting loss	<0.5%	<0.5%	<0.1%
	Calibration error	0.1%	0.1%	0.1%
	Correction error due to attenuation by surrounding water	0.0%	0.0%	-
	Correction error due to scattering from multi γ -rays	<0.2%	<0.2%	-
	Total	<0.55%	<0.6%	<0.2%
Subchannel Density	Transfer to density	<9 kg/m ³	<10 kg/m ³	<15 kg/m ³
	Distribution error to Subchannel	<5 kg/m ³	<5 kg/m ³	-
Correlation error from Chordal averaged to CT averaged		<6 kg/m ³	<6 kg/m ³	-
Subchannel Density		<20 kg/m ³	<21 kg/m ³	<15 kg/m ³
Subchannel Void*		0.040	0.042	0.030
Uncertainty (1 σ)		4 %	5 %	3 %

* Reference averaged density is 500 kg/m³ (see Figure 2.4.1).

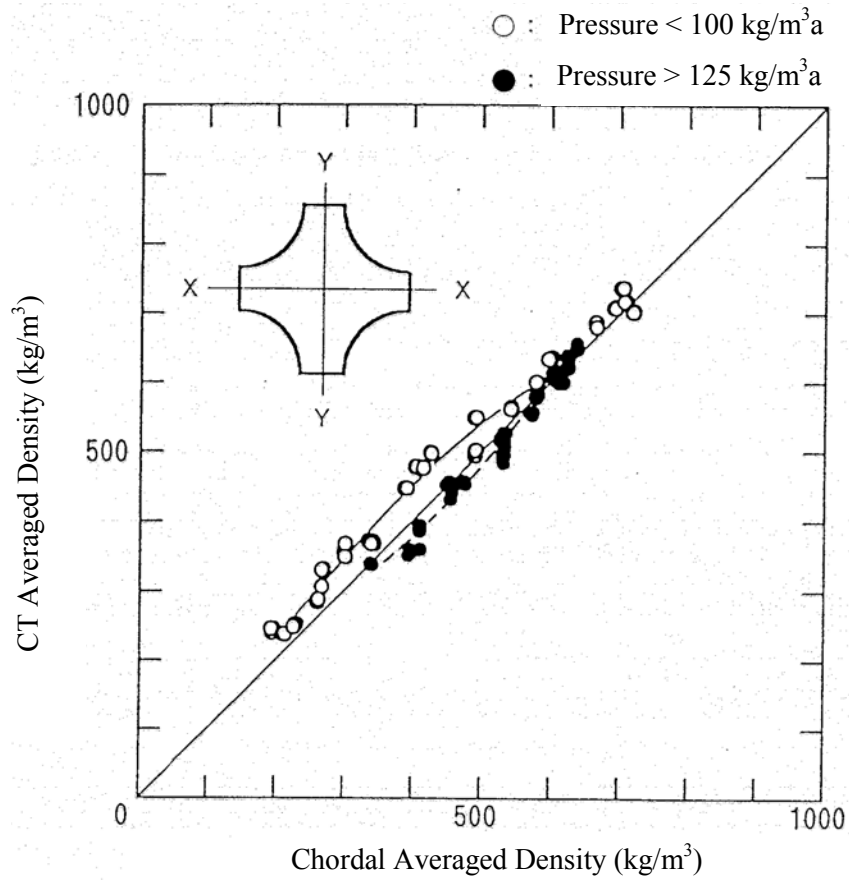
Table 2.4.3 Number of Gamma Ray Beams

Test assembly	CT Measurement	Chordal Measurement
Subchannel	2 (X and Y direction)	2 (X and Y direction)
Rod bundle	-	6 beam $\times$ 2 $\times$ 3 section (total 36 beams)

Table 2.4.4 Time Required to Perform Void Fraction Measurements

Item		CT Measurement	Chordal Measurement
Steady-state	Time needed	5 s/step $\times$ $^{T}33 \times ^R 17$ step (it takes 2 h)	100 s sampling cycle 0.1 s
	Measurement	2 times	3 times
Transient	Time needed	-	200 s
	Measurement	-	1 time

Figure 2.4.2 Relation between Chordal and CT Averaged Densities in bundle S1



2.5 DNB Measurement Methods

The thermocouples described in Table 3.2.2.2 are used to determine the heat flux axially along the rod bundle. These thermocouples are attached to the inner surface of the heater rods and determine the boiling transition. The bundle power is increased gradually by fine steps to the vicinity of DNB power, which is based on preliminary analysis. The occurrence of DNB is confirmed by a rod temperature rise of more than 11°C (20°F) as measured by the thermocouples. The DNB power is defined as the power corresponding to the step just before the step where the temperature increased. Table 2.5.1 shows the estimated accuracies of different process parameters for the DNB measurements.

Table 2.5.1 Estimated Accuracy for DNB Measurements

Quantity	Accuracy
Process parameters	
Pressure	1 %
Flow	1.5%
Power	1 %
Fluid temperature	1 Celsius

2.6 Studies on the Accuracy of the Void Fraction

A deviation exists between the measured void fraction and the void fraction calculated using the measured fluid densities. Solving the standard equation for mixture density, $\bar{\rho} = \alpha\rho_g + (1 - \alpha)\rho_f$, for the void fraction gives $\alpha = (\bar{\rho} - \rho_f)/(\rho_g - \rho_f)$, where the liquid and vapor densities (ρ_f and ρ_g , respectively) are assumed to be at saturation. As it can be seen in Figure 2.6.1 the “measured” void fraction is consistently greater than the “calculated” void fraction.

Following the results of the study performed on the calculation of void fraction, the benchmark team evaluated the experimentally-determined quality for the selected cases in Test Series S1 and S2, the only subchannel series for which quality was provided in the experimental database. Since temperature data at the measurement location was not available, the mixture enthalpy was determined using the measured density and pressure. These two parameters, along with steam table data, were used to determine the fluid temperature at the measurement location. The temperature and the given pressure were then used to determine the mixture enthalpy.

Once this mixture enthalpy was determined, and the fluid and vapor enthalpies (h_f and h_g , respectively) were found assuming saturation properties at the previously determined flow temperature, the equilibrium quality was evaluated using the equation $x = (h_{\text{mixture}} - h_f)/(h_g - h_f)$.

The differences between the experimental and “calculated” qualities (determined using the equation above, referred to as “Method 1”) for these two test series are shown in Figure 2.7.1. This procedure differs from that used for the original (NUPEC, referred to as “Method 2”) calculations of quality, which assumed conservation of energy in the test section. In this method, the mixture quality was expressed by

$$h_{\text{mix}} [\text{kJ} / \text{kg}] = h_{\text{in}} [\text{kJ} / \text{kg}] + \frac{z_{\text{local}} [\text{mm}]}{z_{\text{total}} [\text{mm}]} Q [\text{kW}] \times \frac{3600 \left[\frac{\text{s}}{\text{h}} \right]}{A [\text{m}^2] G \left[\frac{\text{kg}}{\text{m}^2 \text{h}} \right]}$$

where h_{in} is taken at the inlet temperature and pressure. The above expression for quality can then be used, with the fluid and vapor enthalpies taken at saturation for the given pressure. The differences between the recalculated values and the experimental values are reflected in Figure 2.7.2 through Figure 2.7.6.

Figure 2.6.1 Deviation of Measured Void Fraction from Calculated Void Fraction

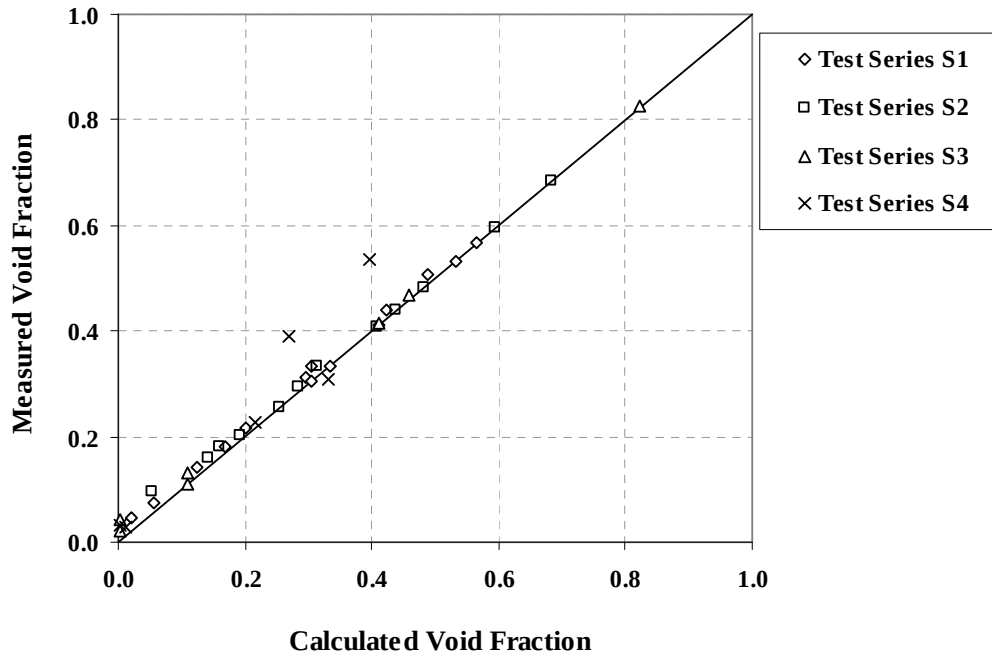


Figure 2.7.1 Deviation of Experimental Quality from Calculated Quality (Method 1; S1 and S2)

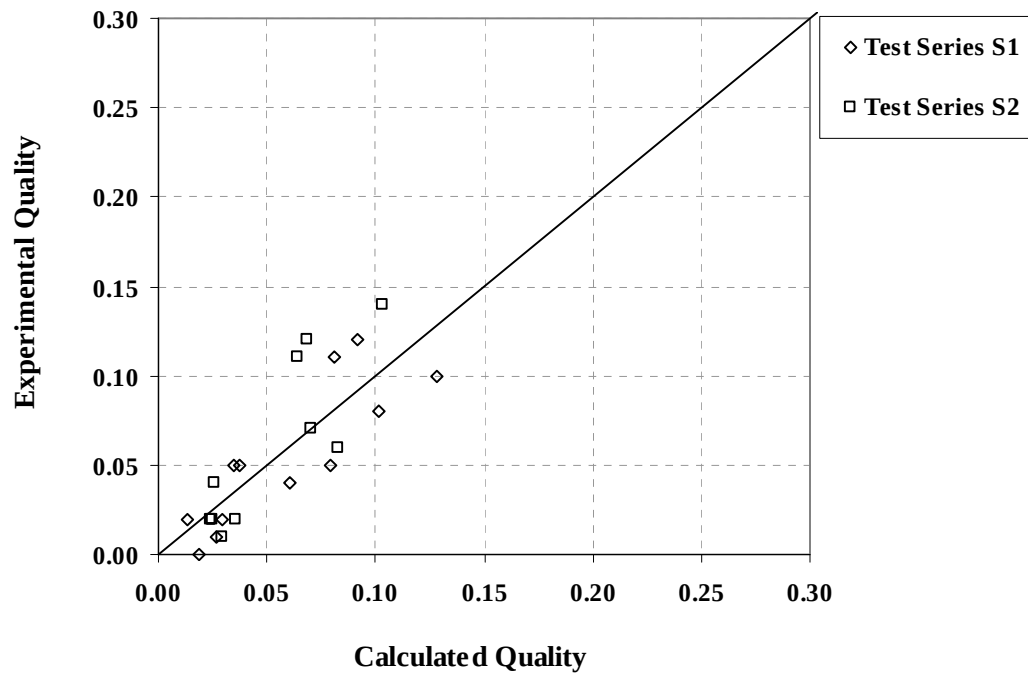


Figure 2.7.2 Deviation of Experimental Quality from Calculated Quality (Method 2; S1 and S2)

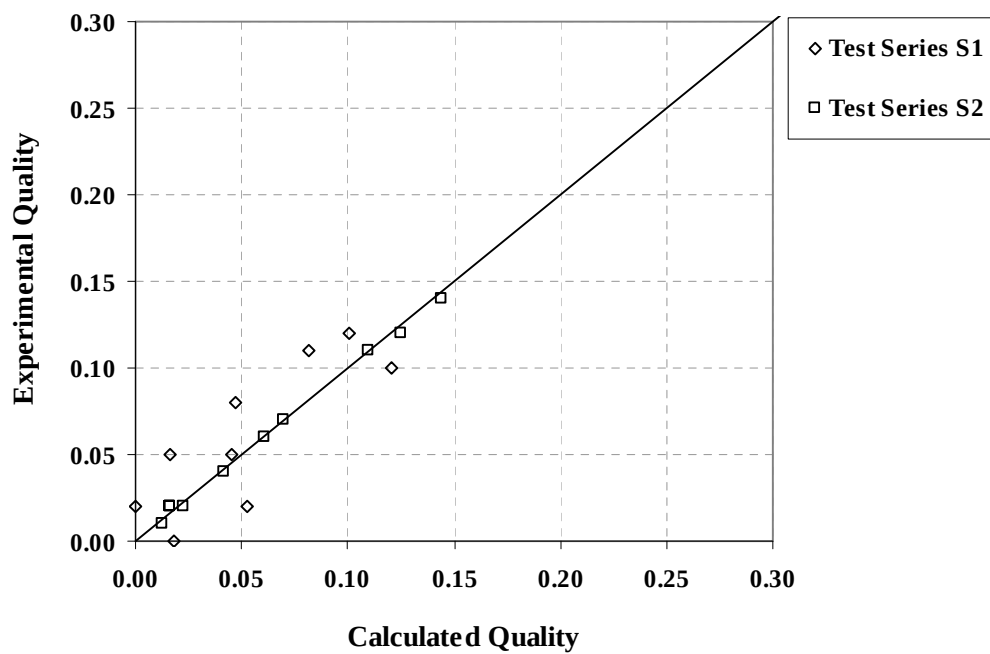


Figure 2.7.3 Deviation of Experimental Quality from Calculated Quality (Method 2; B5)

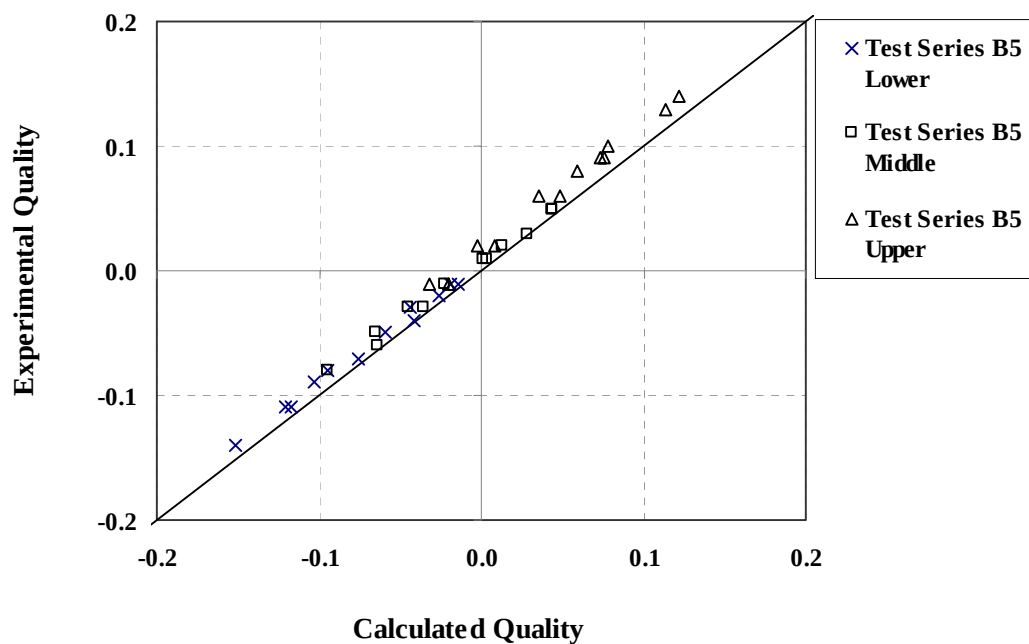


Figure 2.7.4 Deviation of Experimental Quality from Calculated Quality (Method 2; B6)

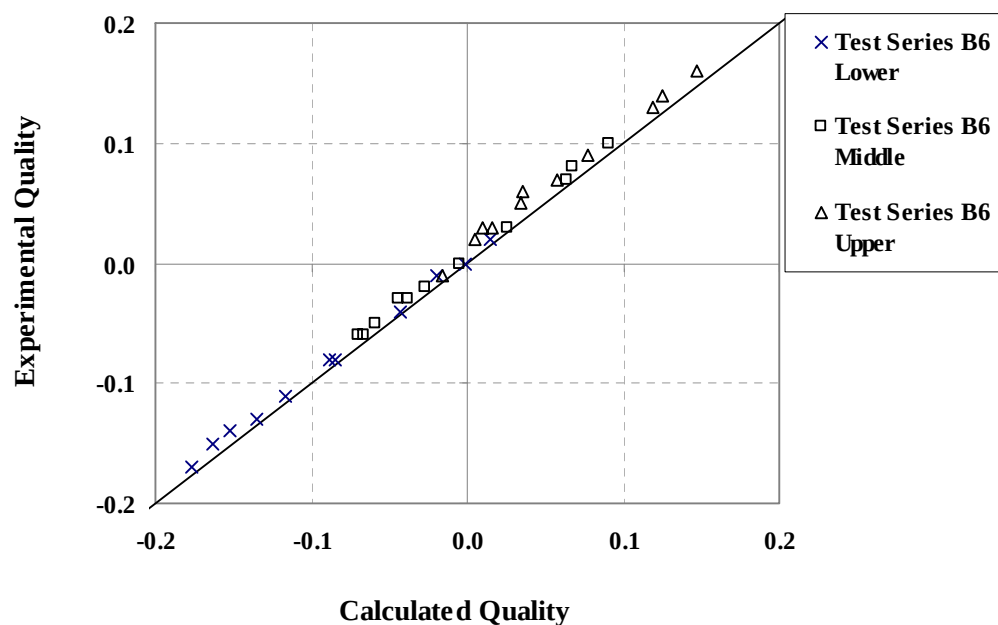


Figure 2.7.5 Deviation of Experimental Quality from Calculated Quality (Method 2; B7)

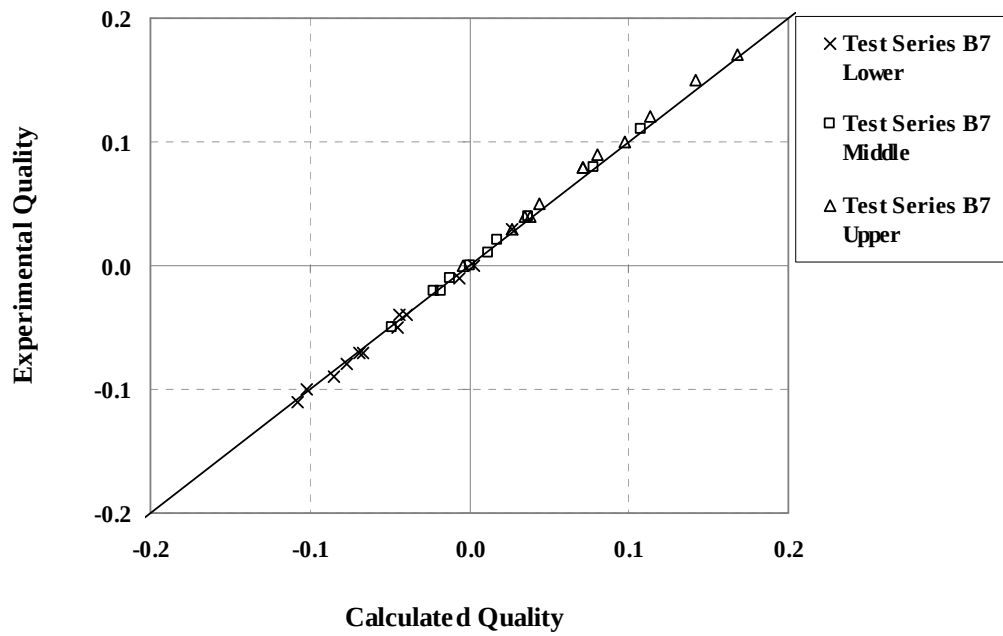
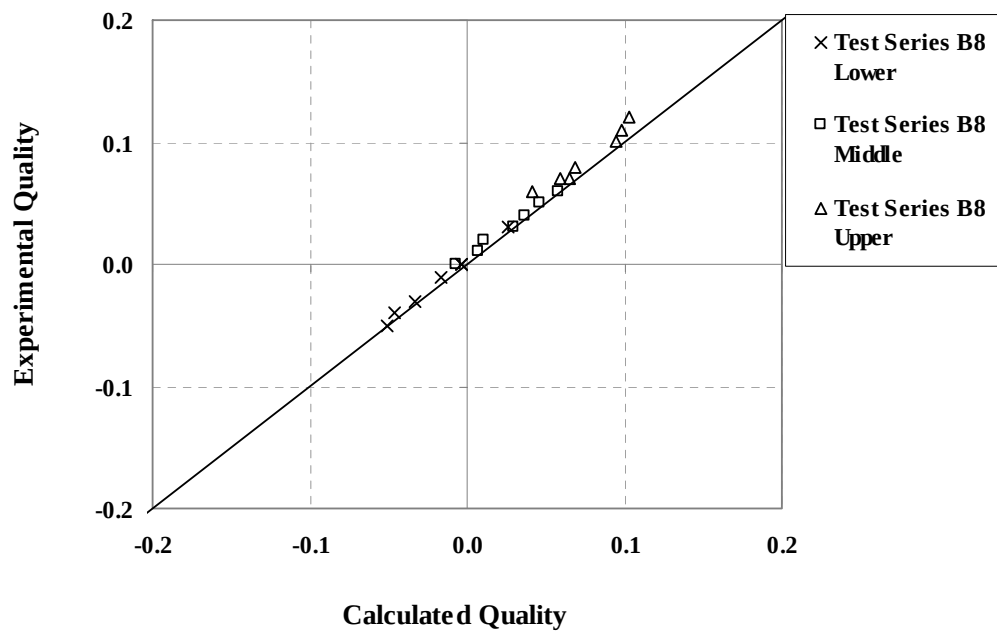


Figure 2.7.6 Deviation of Experimental Quality from Calculated Quality (Method 2; B8)



Chapter 3 TEST ASSEMBLY DATA

3.1 General

This chapter provides the participants with geometry and material data for the PWR rod bundles and the single subchannel apparatus utilized in the void distribution and DNB measurements. The power distribution patterns, radial and axial, as well as the tests conditions are described. Data for the fuel assemblies and spacers' dimensions, heater rods specifications, and material properties are also given.

3.2 Void Distribution Data

Two different test assemblies were used for the void distribution measurements: a single subchannel, and a full-size PWR fuel assembly. These assemblies and their characteristics are outlined in Table 3.2.1 . The thimble rods in the bundle assemblies are filled with stratified water. In such rods, the bottom of the tube structure is open to the flow while a pin hole is located at the top of the structure.

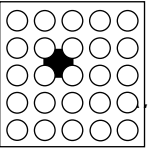
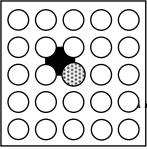
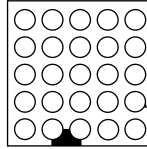
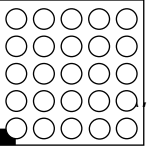
Table 3.2.1 Test Assemblies for Void Fraction Measurements

Assembly	Reference fuel type	Type of cell		Power distribution	
				Radial	Axial
S1	17×17 M	Subchannel	Center (Typical)	-	Uniform
S2			Center (Thimble)	-	Uniform
S3			Side	-	Uniform
S4			Corner	-	Uniform
B5	17×17 M	5×5 Rod bundle	Typical cell	A	Uniform
B6			Typical cell	A	Cosine
B7			Thimble cell	B	Cosine

3.2.1 Single Subchannel Specification

Table 3.2.1.1 outlines the four different subchannels used to perform the void distribution measurements. Figure 3.2.1.1 shows a cross-sectional view of the subchannel test assembly.

Table 3.2.1.1 Geometry and Power Shape for Test Assembly S1, S2, S3, and S4

Item	Data			
Assembly (Subjected subchannel)				
	S1	S2	S3	S4
Subchannel type	Center (Typical)	Center (Thimble)	Side	Corner
Number of heaters	4×1/4	3×1/4	2×1/4	1×1/4
Axial heated length (mm)	1555	1555	1555	1555
Axial power shape	Uniform	Uniform	Uniform	Uniform

■: Subjected subchannel ○ : Heated rod ⊗: Thimble rod

Figure 3.2.1.1 Cross Sectional View of Subchannel Test Assembly

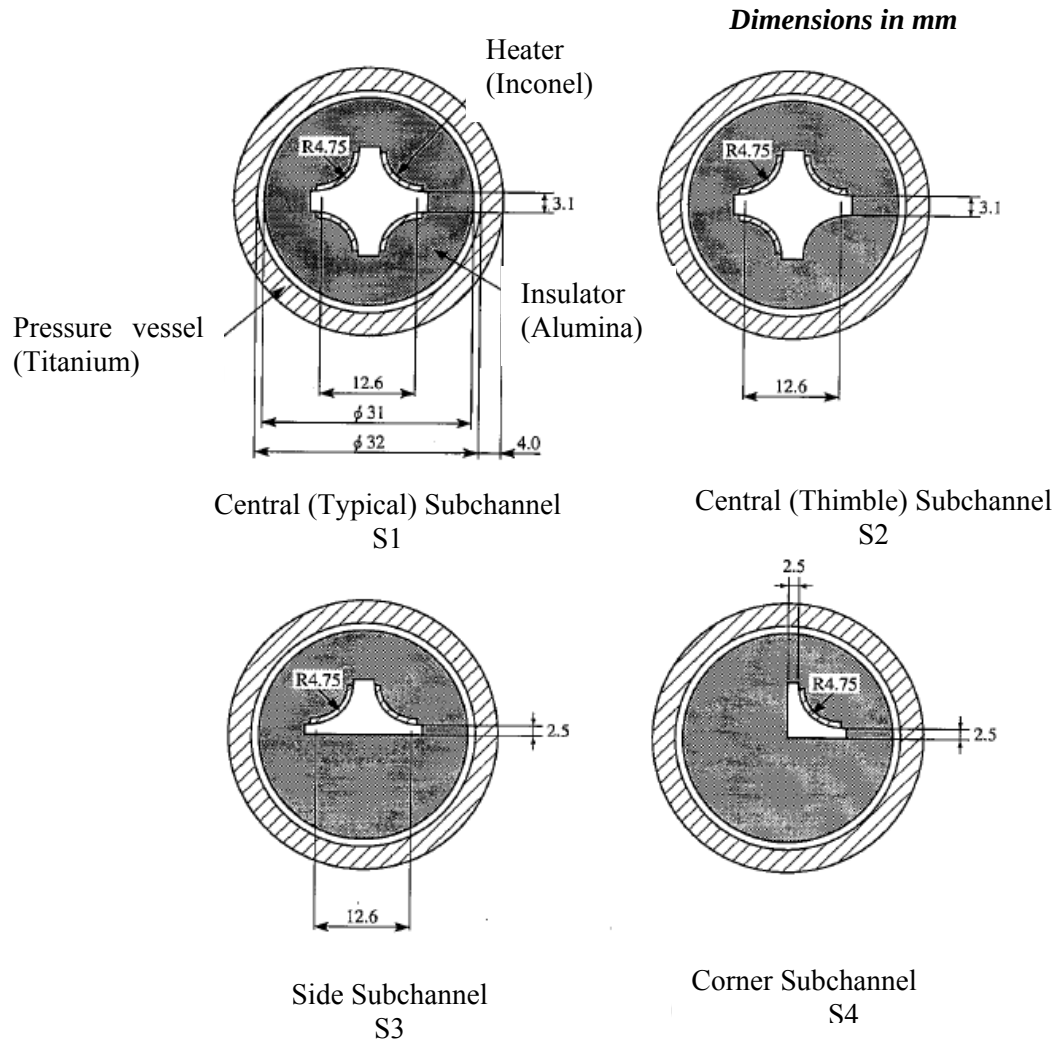


Table 3.2.1.2 Geometric Characteristics of Subchannel Assemblies

	Subchannel Type			
	Typical (S1)	Thimble (S2)	Side (S3)	Corner (S4)
Flow area, mm ²	107.098	107.098	68.464	42.592
Heated perimeter, mm	29.845	22.384	14.923	7.461
Wetted perimeter, mm	54.645	54.645	44.923	33.161

3.2.1.1 Heater Rod Structure for Subchannel Test Assembly

Table 3.2.1.1.1 describes the properties of the heater rod used in the subchannel test assembly. Figure 3.2.1.1 provides a diagram of the cross-sectional structure of the heater rods in the subchannel test assembly.

Table 3.2.1.1.1 Heater Rod Structure for Subchannel Test Assembly

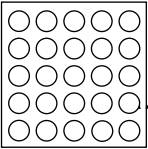
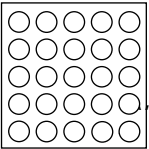
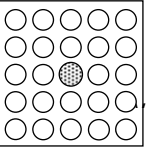
Item		Data
Heater	Outer radius (mm)	4.75
	Thickness (mm)	0.85
	Material	Inconel 600
	Heating Method	Direct Heating
Insulator	Outer diameter (mm)	31
	Material	Alumina
Pressure vessel	Inner Diameter (mm)	32
	Thickness (mm)	4
	Material	Titanium

3.2.2 Bundle Specification

Table 3.2.2.1 outlines the three different bundles used to perform the void distribution measurements. Figure 3.2.2.1 shows the locations of the ten pressure taps inserted into the rod bundle to measure pressure loss. Figure 3.2.2.2 and Figure 3.2.2.3 show the two different radial power distributions, named A and B, respectively. All powers shown are relative powers.

Table 3.2.2.2 shows the axial power distribution (based on a cosine shape) that was used in the bundle tests.

Table 3.2.2.1 Geometry and Power Shape for Test Assembly B5, B6, and B7

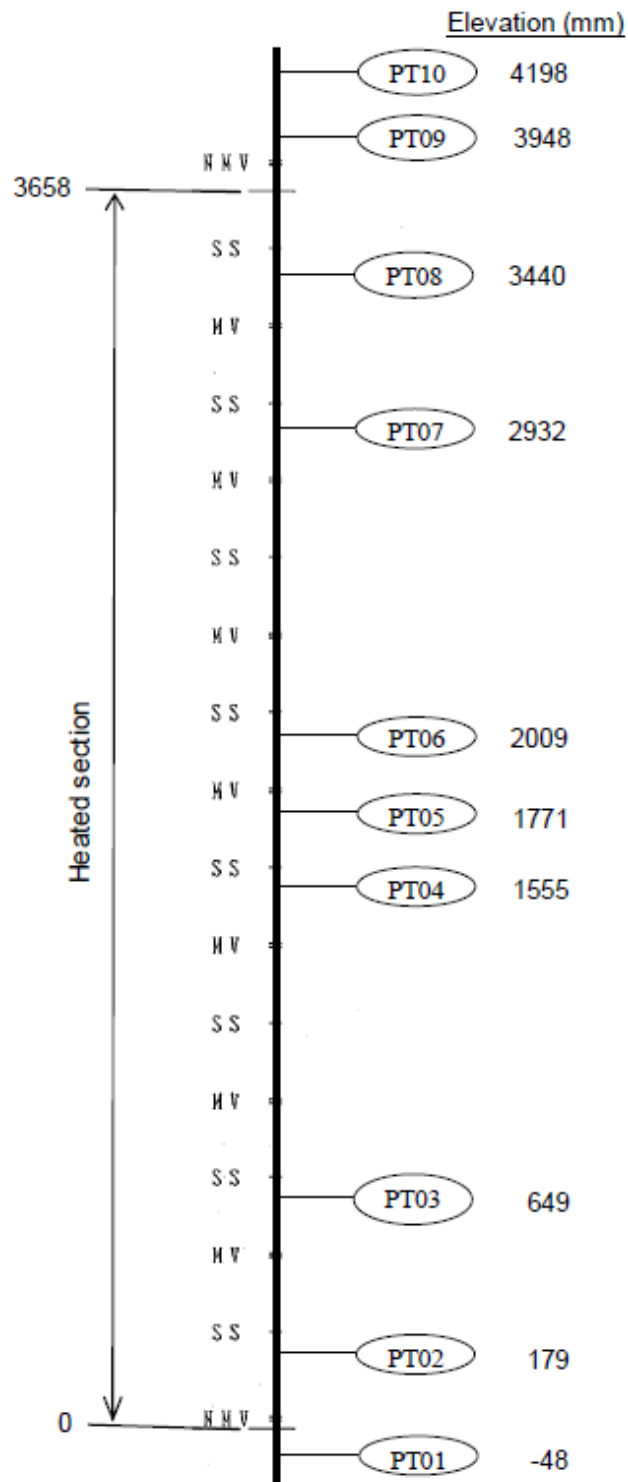
Item	Data		
Assembly			
	B5	B6	B7
Rods array	5×5	5×5	5×5
Number of heated rods	25	25	24
Number of thimble rods	0	0	1
Heated rod outer diameter (mm)	9.50	9.50	9.50
Thimble rod outer diameter (mm)	-	-	12.24
Heated rods pitch (mm)	12.60	12.60	12.60
Axial heated length (mm)	3658	3658	3658
Flow channel inner width (mm)	64.9	64.9	64.9
Radial power shape	A	A	B
Axial power shape	Uniform	Cosine	Cosine
Number of MV spacers	7	7	7
Number of NMV spacers	2	2	2
Number of simple spacers	8	8	8
MV spacer location (mm)	471, 925, 1378, 1832, 2285, 2739, 3247		
NMV spacer location (mm)	2.5, 3755		
Simple spacer location (mm)	237, 698, 1151, 1605, 2059, 2512, 2993, 3501		

○: Heated rod ●: Thimble rod

MV: Mixing vane, NMV: No mixing vane

Spacer location is distance from bottom of heated length to spacer bottom face.

Figure 3.2.2.1 Location of Pressure Taps



**Figure 3.2.2.2 Radial Power Distribution
Type A**

0.85	0.85	0.85	0.85	0.85
0.85	1.00	1.00	1.00	0.85
0.85	1.00	1.00	1.00	0.85
0.85	1.00	1.00	1.00	0.85
0.85	0.85	0.85	0.85	0.85

**Figure 3.2.2.3 Radial Power Distribution
Type B**

0.85	0.85	0.85	0.85	0.85
0.85	1.00	1.00	1.00	0.85
0.85	1.00	0.00	1.00	0.85
0.85	1.00	1.00	1.00	0.85
0.85	0.85	0.85	0.85	0.85

Table 3.2.2.2 Axial Power Distribution (Cosine)

Node	Relative Power
	<i>Cosine</i>
(Bottom)	
1	0.42
2	0.47
3	0.56
4	0.67
5	0.80
6	0.94
7	1.08
8	1.22
9	1.34
10	1.44
11	1.51
12	1.55
13	1.55
14	1.51
15	1.44
16	1.34
17	1.22
18	1.08
19	0.94
20	0.80
21	0.67
22	0.56
23	0.47
24	0.42
(Top)	

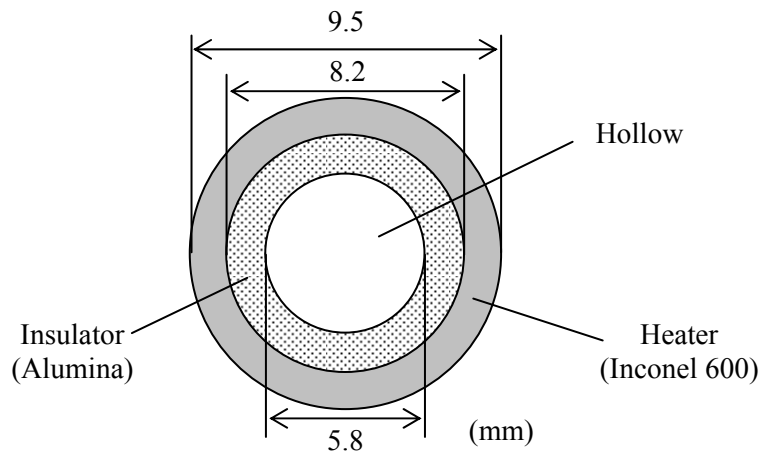
3.2.2.1 Heater Rod Structure for Bundle Test Assembly

Table 3.2.2.1.1 describes the properties of the heater rods used for the rod bundle test assembly. The insulator and the heater are made of Inconel and Alumina respectively, and the gap between the heater and insulator could be assumed zero contact. The rod is a single-ended, grounded electrical heater rod, which simulates the heat generation from a fuel rod. Figure 3.2.2.1.1 shows a cross-sectional view of the bundle heater rod.

Table 3.2.2.1.1 Heater Rod Structure for Bundle Test Assembly

Item		Data
Heater	Outer diameter (mm)	9.5
	Thickness (mm)	0.65
	Material	Inconel 600
	Heating Method	Direct Heating
Insulator	Outer diameter (mm)	8.2
	Inner diameter (mm)	5.8
	Material	Alumina

Figure 3.2.2.1.1 Cross Section of Bundle Heater Rod



3.2.2.2 Spacer Grid Data

There were three types of spacers instrumented along the axial length: simple spacer (SS), spacer with no mixing vanes (NMV), and spacer with mixing vanes (MV). The simple spacer has only dimples while NMV and MV have dimples and springs. The grids straps are made out of Inconel 600 allow.

Three-dimensional views of the different spacer types are given in Figures 3.2.2.2.1 through 3.2.2.2.3.

Figures 3.2.2.2.4 through 3.2.2.2.6 show the dimensions of the simple spacer, the non-mixing vanes spacer, and the mixing vanes spacer, respectively. Note that the dimensions on Figure 3.2.2.2.4 are in millimetres, while the dimensions in Figures 3.2.2.2.5 and 3.2.2.2.6 are in inches. The drawings of the spacers were developed using the SolidWorks® mechanical design software package.

Table 3.2.2.2.1 provides the bundle average spacer pressure loss coefficients for the three types of grids. Depending on the participants' computer code, and using the provided spacer data, each participant may choose the subchannel grids loss coefficients or other required input values.

Table 3.2.2.2.1 Bundle Average Spacer Pressure Loss Coefficients

Spacer Type	Loss Coefficient
Simple Spacer (SS)	0.4
Non-Mixing Vanes Spacer (NMV)	0.7
Mixing Vanes Spacer (MV)	1.0

Figure 3.2.2.2.1 Three-Dimensional View of the Simple Spacer

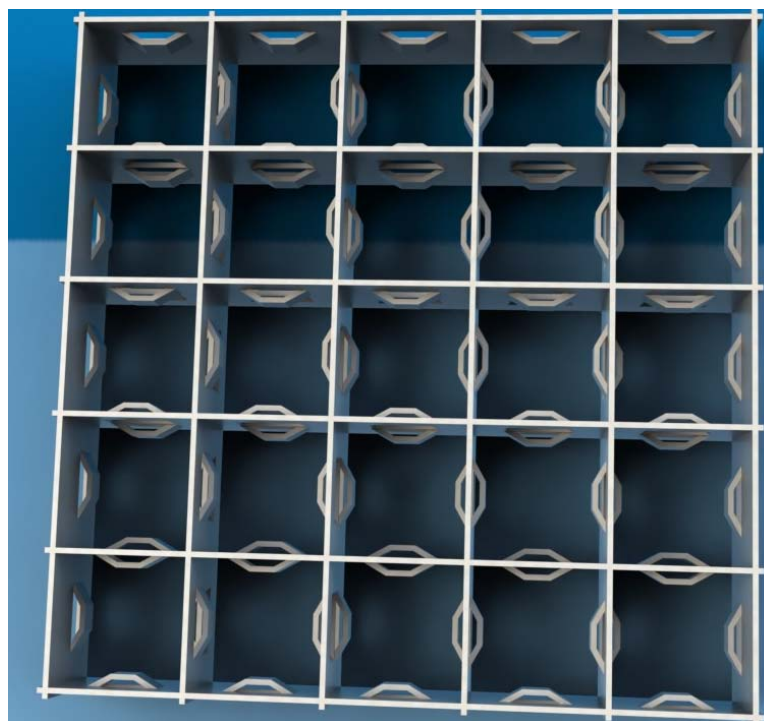
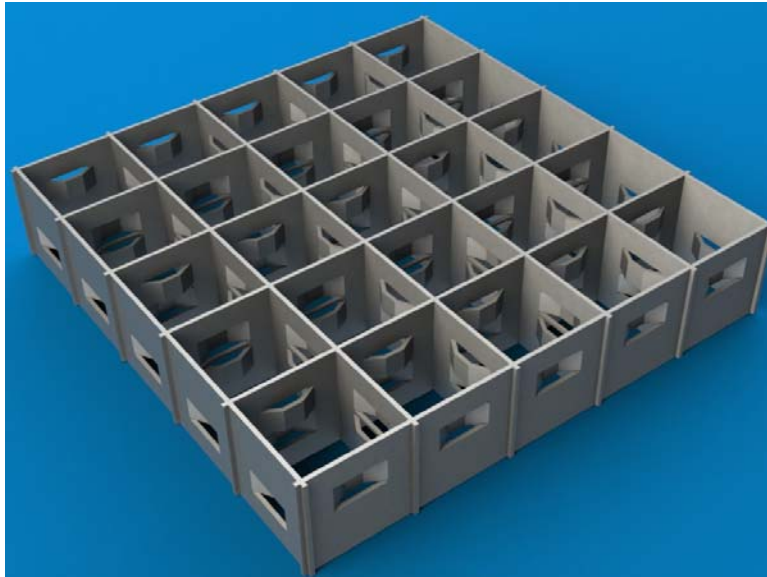


Figure 3.2.2.2.2 Three-Dimensional View of the Non-Mixing Vane Spacer

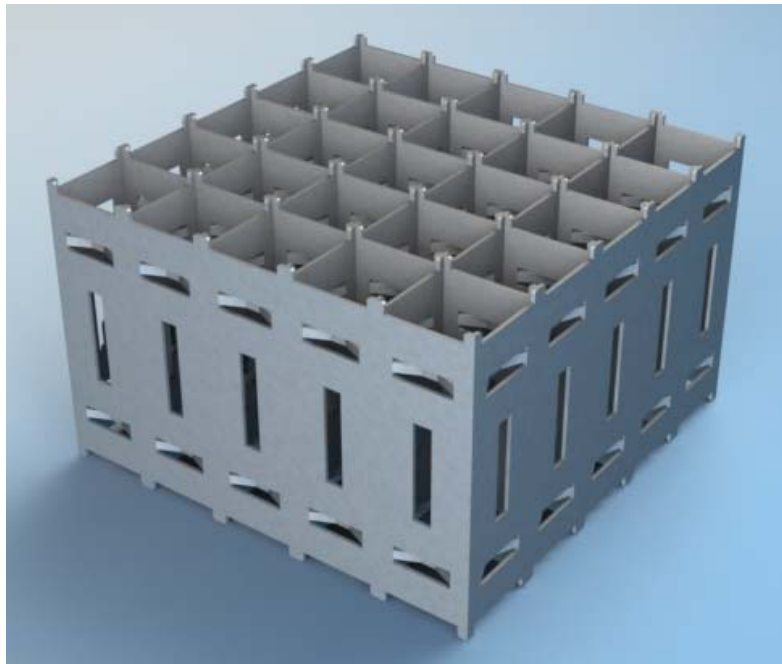


Figure 3.2.2.2.3 Three-Dimensional View of the Mixing Vane Spacer

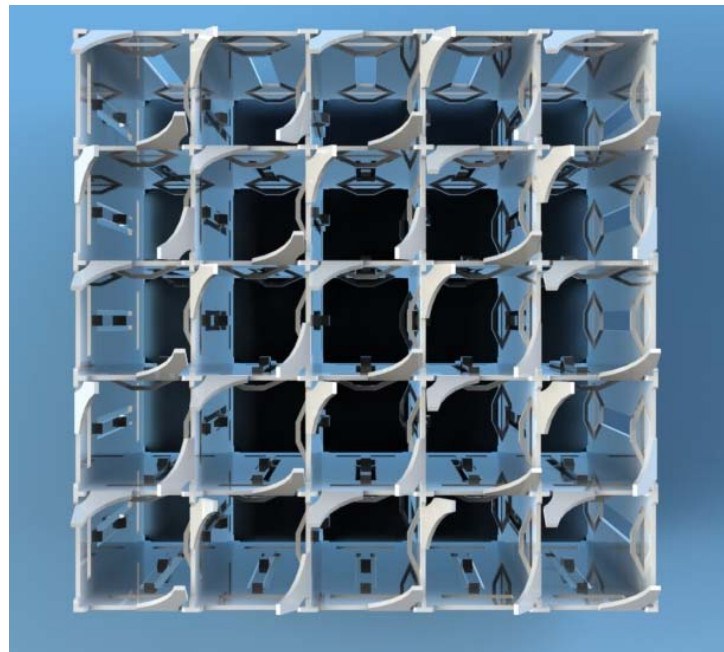
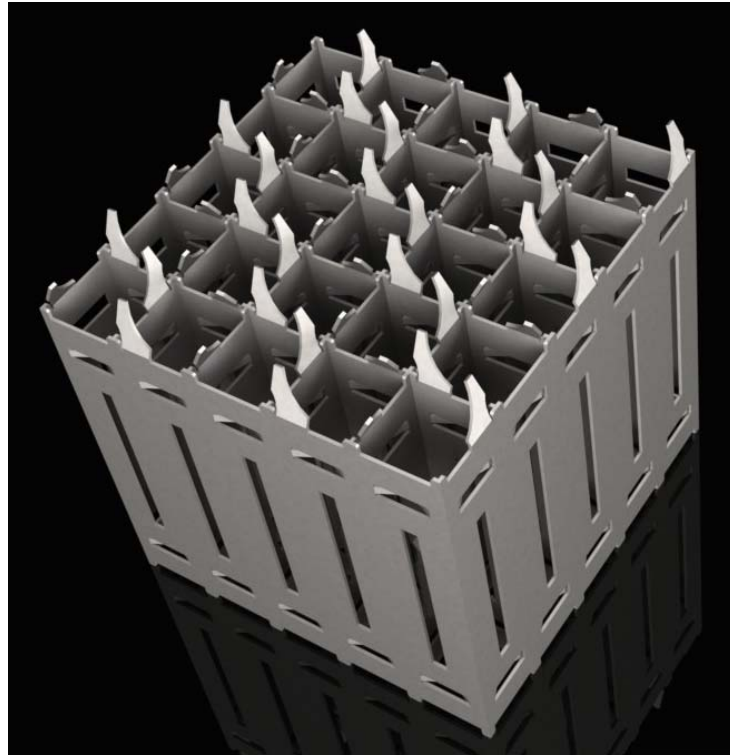
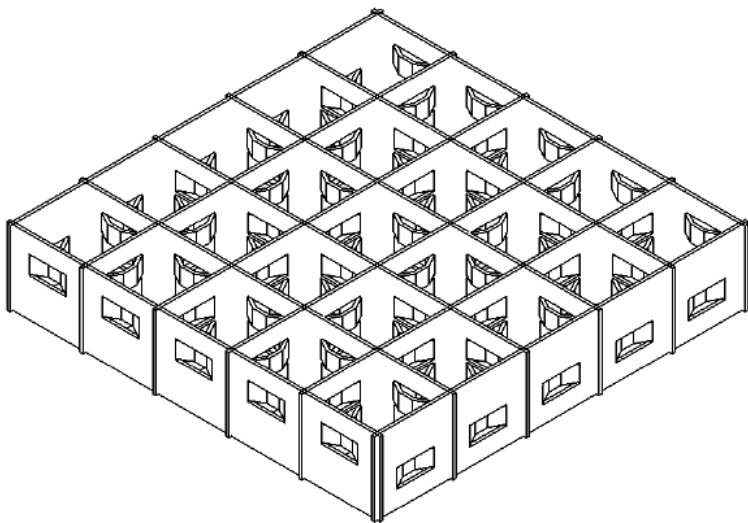
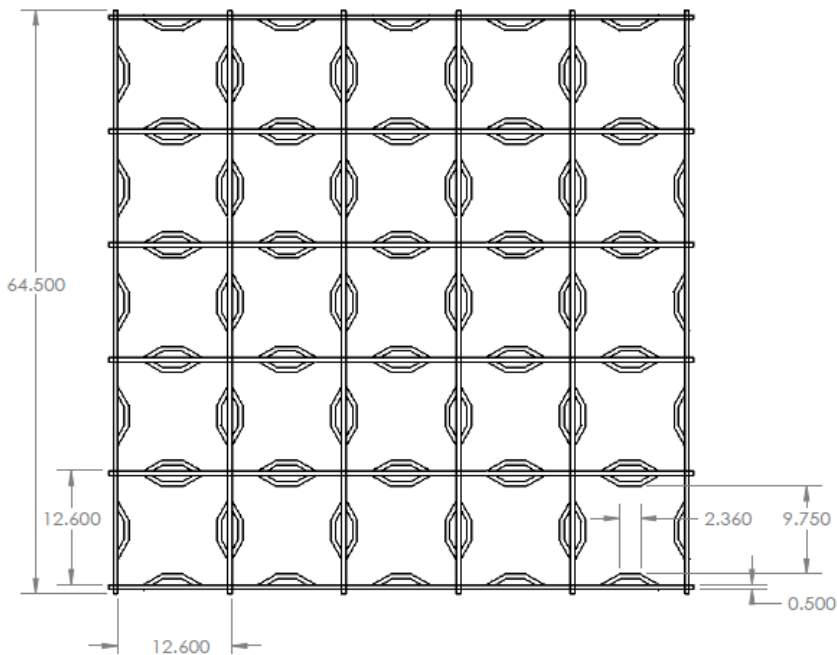


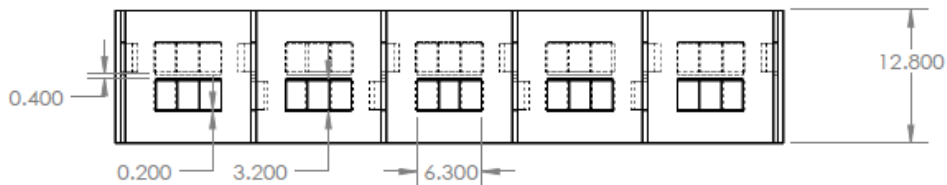
Figure 3.2.2.2.4 Dimensions of the Simple Spacer



a) Overall Schematic

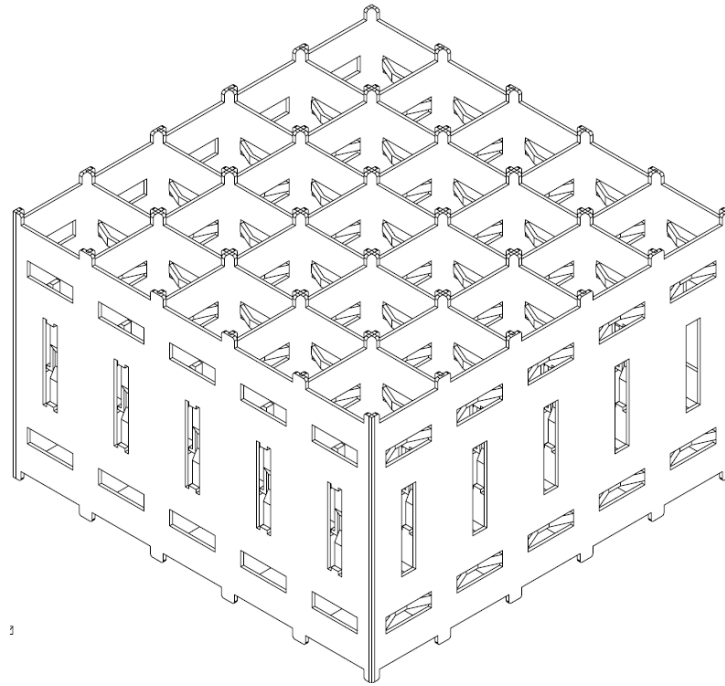


b) Top View

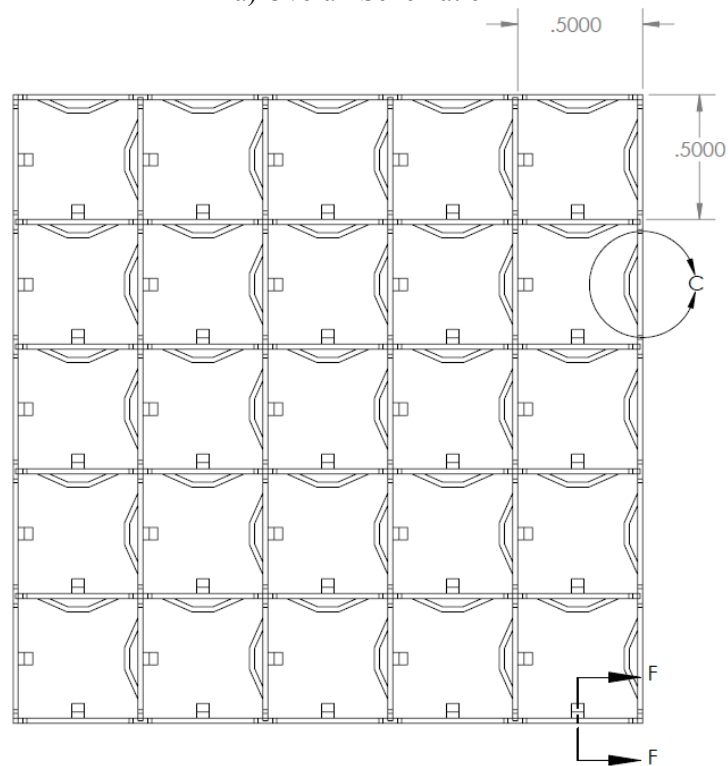


c) Side View

Figure 3.2.2.5 Dimensions of the Non-Mixing Vane Spacer

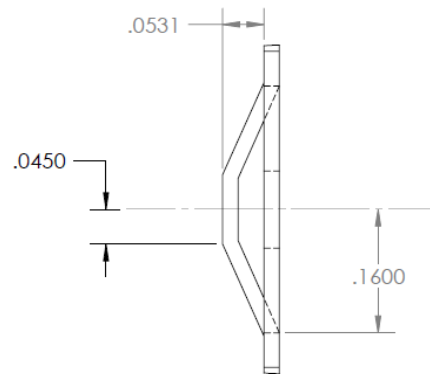


a) Overall Schematic

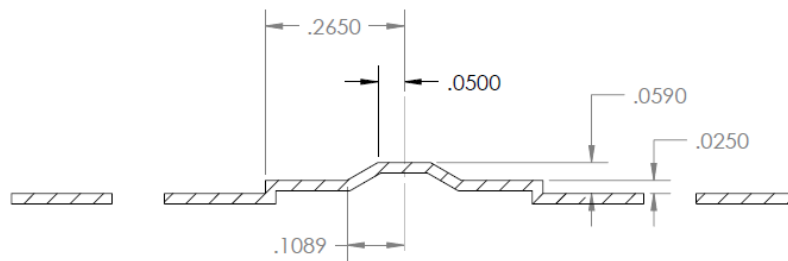


b) Top View

Figure 3.2.2.2.5 Dimensions of the Non-Mixing Vane Spacer – cont.



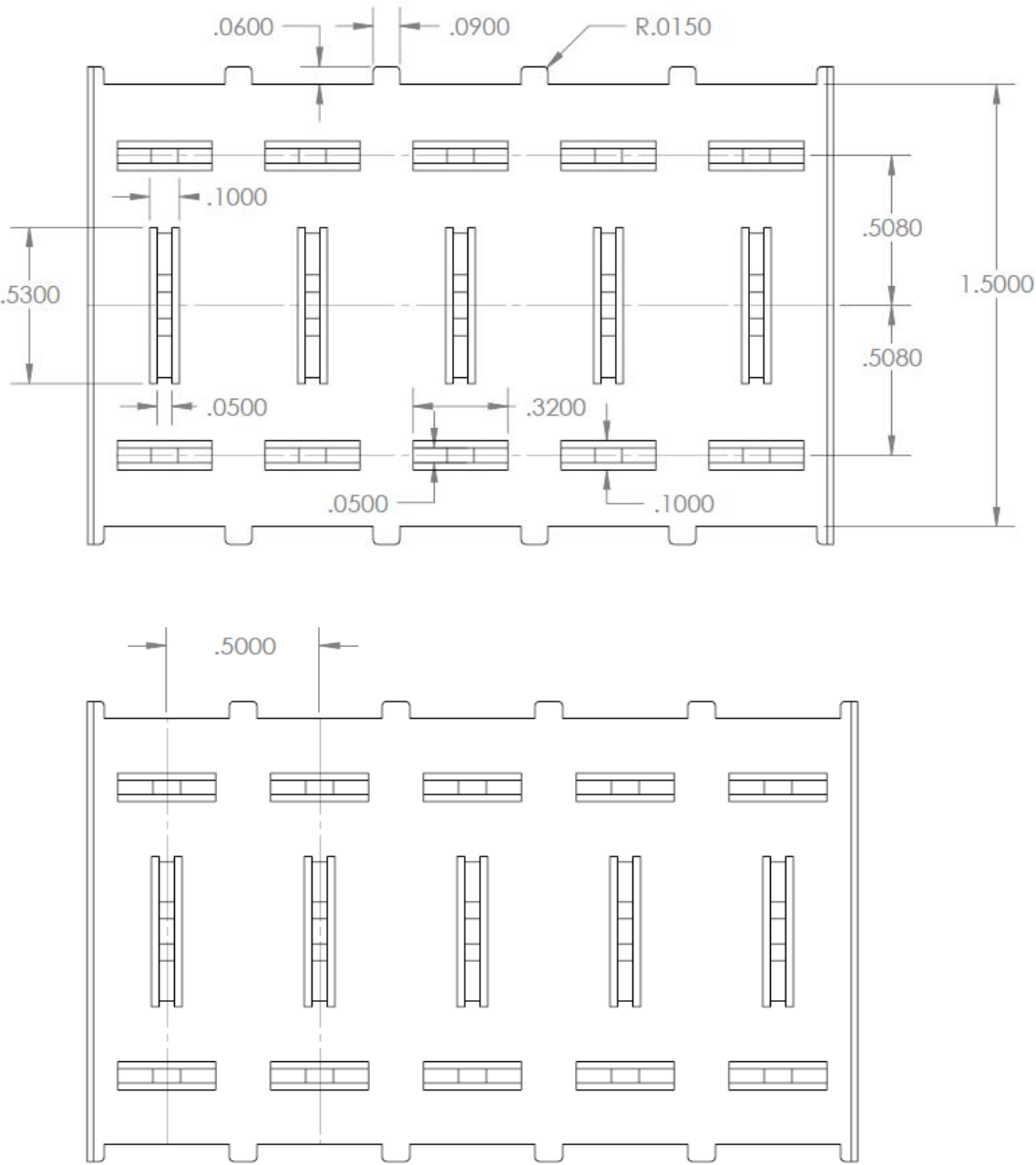
DETAIL C
SCALE 6 : 1



SECTION F-F
SCALE 4 : 1

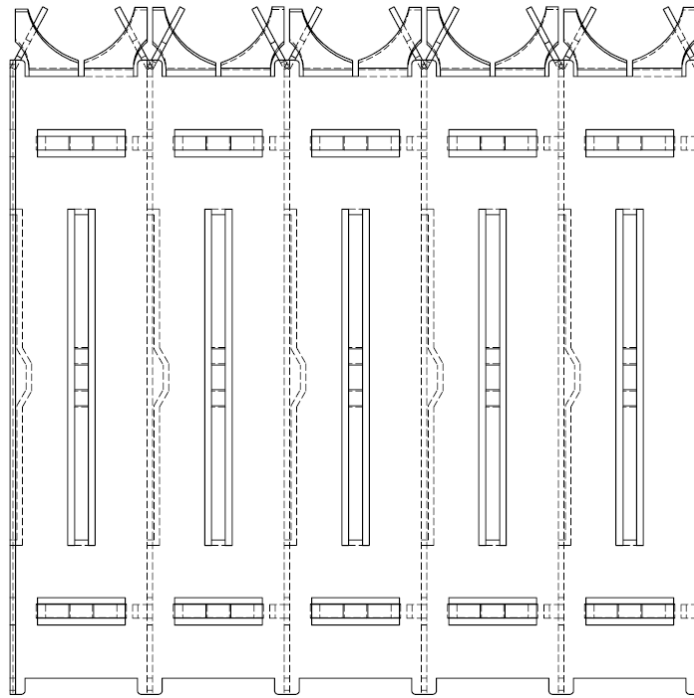
c) Springs and Dimples Dimensions

Figure 3.2.2.5 Dimensions of the Non-Mixing Vane Spacer – cont.

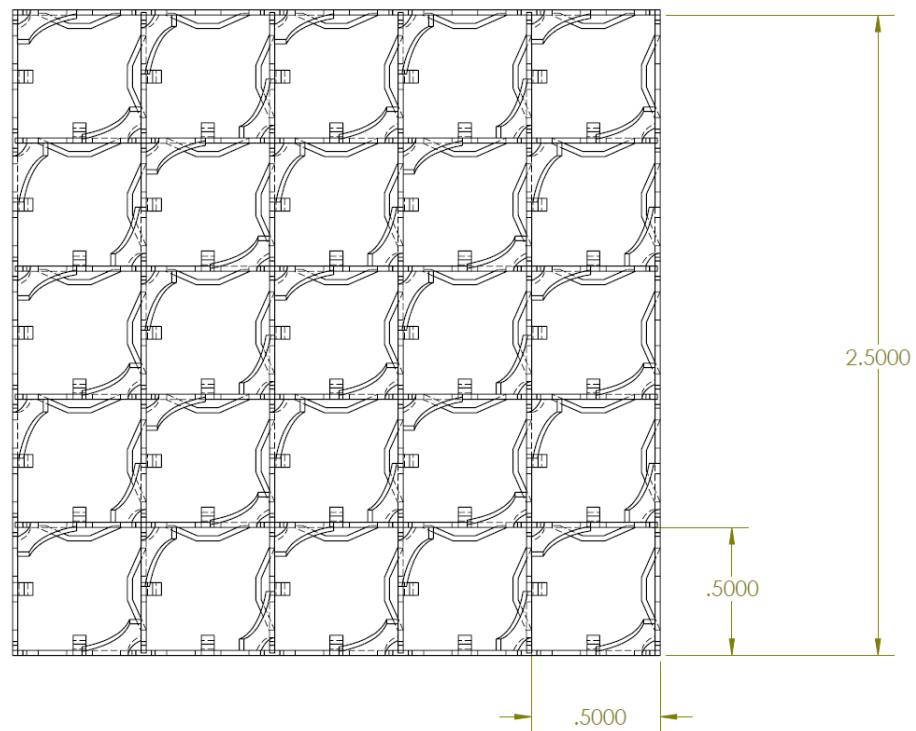


d) Side Views

Figure 3.2.2.2.3 Dimensions of the Mixing Vane Spacer

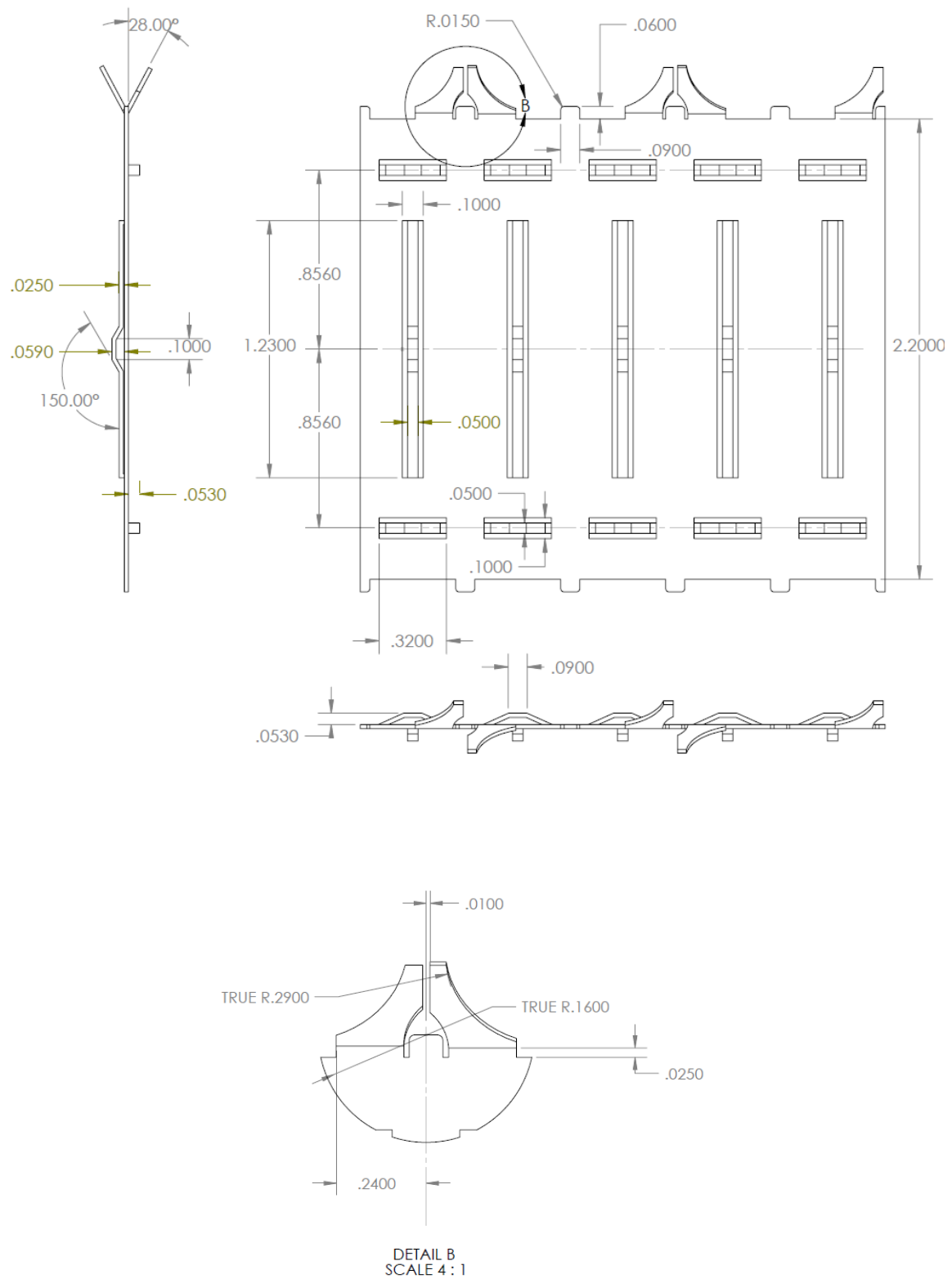


a) Side View



b) Top View

Figure 3.2.2.2.3 Dimensions of the Mixing Vane Spacer – cont.



c) Mixing Vane Dimensions

3.3 DNB Measurement

Table 3.3.1 shows the assembly types used in the DNB measurements. The heater rods used in these bundles are of the same type as those used in the bundles in the void distribution measurement exercises.

Table 3.3.2 gives the geometry and power shape for one of the test assemblies, A0. Similar information for the other assembly types is provided.

Table 3.2.2.1.1 provides information on the heating rods used in the rod bundles for this set of tests.

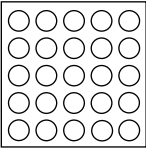
Figure 3.3.5 displays the locations of the thermocouples for the test assemblies.

The DNB measurements were performed with two additional radial power distributions (type C and type D). Radial power distributions A and B are the same as those used in the void distribution measurements.

Table 3.3.1 Test Assemblies for DNB Measurements

Assembly	Reference fuel type	Rods array	Type of cell	Power distribution	
				Radial	Axial
A0	17×17 M	5×5	Typical cell	A	Uniform
A1			Typical cell	C	Uniform
A2			Typical cell	A	Uniform
A3		6×6	Typical cell	D	Uniform
A4		5×5	Typical cell	A	Cosine
A8			Thimble cell	B	Cosine
A11			Typical cell	A	Cosine
A12			Thimble cell	B	Cosine

Table 3.3.2 Geometry and Power Shape for Test Assembly A0

Item	Data
Assembly	 A0
Rods array	5×5
Number of heated rods	25
Number of thimble rods	0
Heated rod outer diameter (mm)	9.50
Thimble rod outer diameter (mm)	-
Heated rods pitch (mm)	12.60
Axial heated length (mm)	3658
Flow channel inner width (mm)	64.9
Radial power shape	A
Axial power shape	Uniform
Number of MV spacers	5
Number of NMV spacers	2
Number of simple spacers	6
MV spacer location (mm)	610, 1219, 1829, 2438, 3048
NMV spacer location (mm)	0, 3658
Simple spacer location (mm)	305, 914, 1524, 2134, 2743, 3353

○ : Heated rod ⊗ : Thimble rod

MV: Mixing vane, NMV: No mixing vane

Spacer location is distance from bottom of heated length to spacer bottom face.

Table 3.3.3 Geometry and Power Shape for Test Assembly A1, A2, and A3

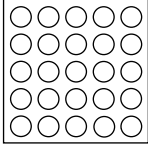
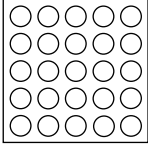
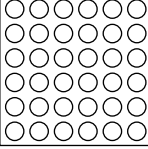
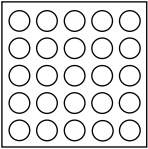
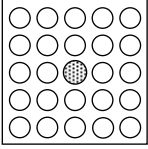
Item	Data		
Assembly			
	A1	A2	A3
Rods array	5×5	5×5	6×6
Number of heated rods	25	25	36
Number of thimble rods	0	0	0
Heated rod outer diameter (mm)	9.50	9.50	9.50
Thimble rod outer diameter (mm)	-	-	-
Heated rods pitch (mm)	12.60	12.60	12.60
Axial heated length (mm)	3658	3658	3658
Flow channel inner width (mm)	64.9	64.9	77.5
Radial power shape	C	A	D
Axial power shape	Uniform	Uniform	Uniform
Number of MV spacers	7	7	7
Number of NMV spacer	2	2	2
Number of simple spacers	8	8	8
MV spacer location (mm)	457, 914, 1372, 1829, 2286, 2743, 3200		
NMV spacer location (mm)	0, 3658		
Simple spacer location (mm)	229, 686, 1143, 1600, 2057, 2515, 2972, 3429		

Table 3.3.4 Geometry and Power Shape for Test Assembly A4, A8, A11, and A12

Item	Data	
Assembly	 A4, A11	 A8, A12
Rods array	5×5	5×5
Number of heated rods	25	24
Number of thimble rods	0	1
Heated rod outer diameter (mm)	9.50	9.50
Thimble rod outer diameter (mm)	-	12.24
Heated rods pitch (mm)	12.60	12.60
Axial heated length (mm)	3658	3658
Flow channel inner width (mm)	64.9	64.9
Radial power shape	A	B
Axial power shape	Cosine	Cosine
Number of MV spacers	7	7
Number of NMV spacer	2	2
Number of simple spacers	8	8
MV spacer location (mm)	471, 925, 1378, 1832, 2285, 2739, 3247	
NMV spacer location (mm)	2.5, 3755	
Simple spacer location (mm)	237, 698, 1151, 1605, 2059, 2512, 2993, 3501	

**Figure 3.3.1 Radial Power Distribution
Type C**

1.00	1.00	0.25	0.25	0.25
1.00	1.00	1.00	0.25	0.25
1.00	1.00	0.25	0.25	0.25
1.00	1.00	1.00	0.25	0.25
1.00	1.00	0.25	0.25	0.25

**Figure 3.3.2 Radial Power Distribution
Type D**

0.85	0.85	0.85	0.85	0.85	0.85
0.85	1.00	1.00	1.00	1.00	0.85
0.85	1.00	1.00	1.00	1.00	0.85
0.85	1.00	1.00	1.00	1.00	0.85
0.85	1.00	1.00	1.00	1.00	0.85
0.85	0.85	0.85	0.85	0.85	0.85

Figure 3.3.5 Location of Thermocouples for Test Assemblies

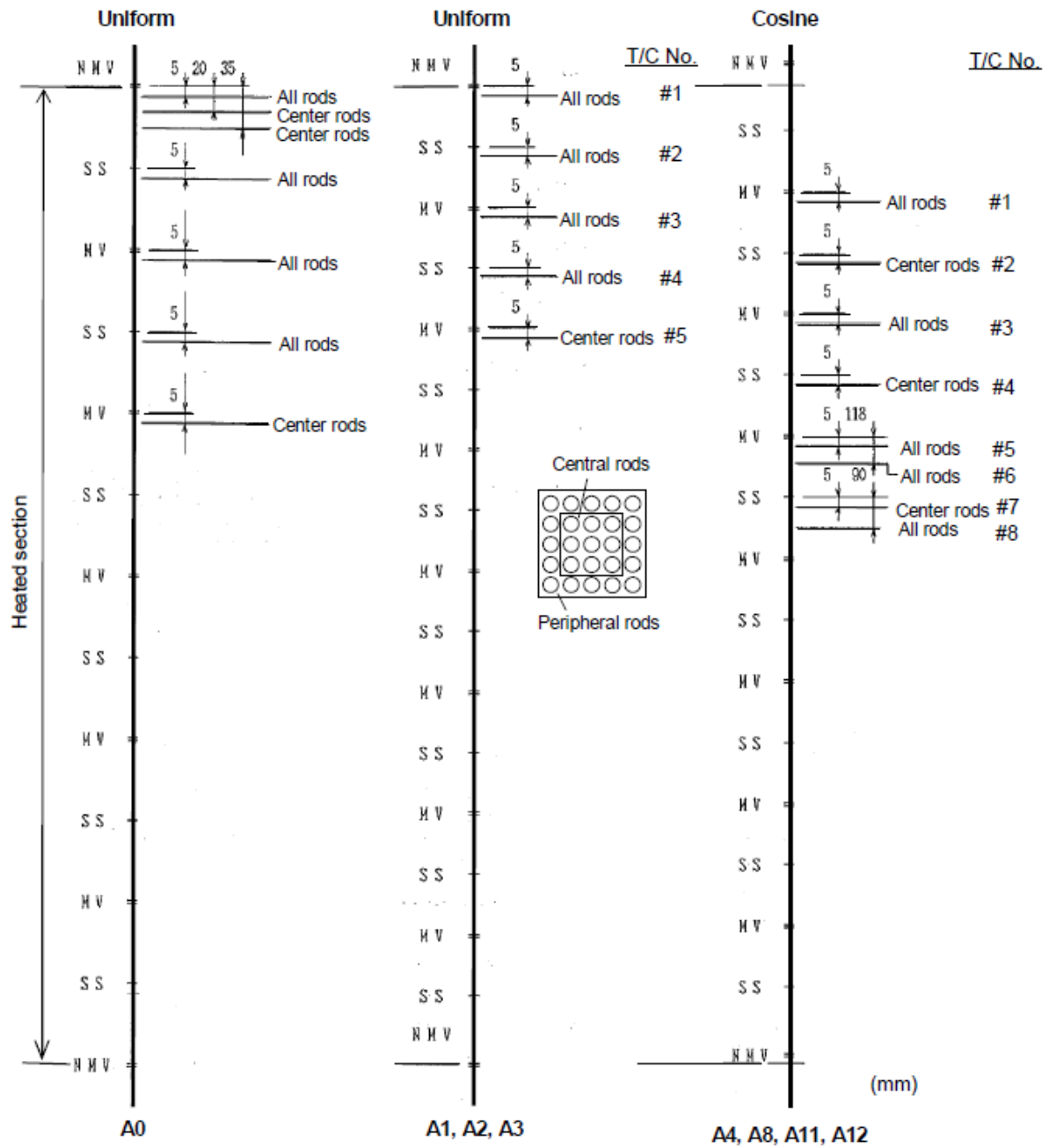
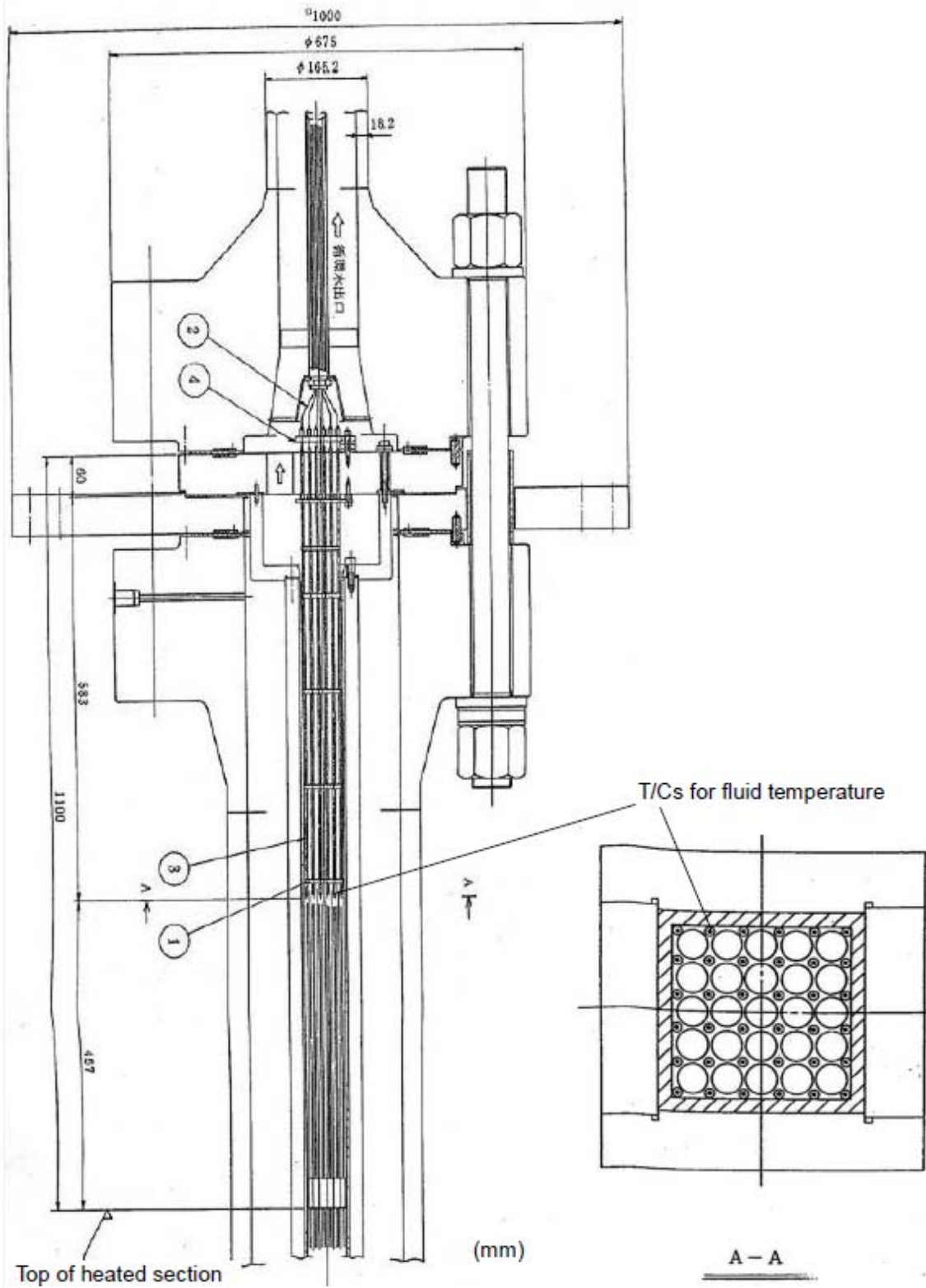


Figure 3.3.6 Fluid Temperature Measurements



3.4 Thermo-Mechanical Properties

The thermo-mechanical properties of Inconel 600 are based on the MATPRO model used in TRAC code. Since no information on the heat loss is available in the NUPEC PSBT database, an adiabatic condition is suggested for the benchmark.

3.4.1 Properties of Inconel 600

(1) Density

The density is:

$$\rho = 16.01846 \times (5.261008 \times 10^2 - 1.345453 \times 10^{-2} T_f - 1.194357 \times 10^{-7} T_f^2),$$

where ρ is the density (kg/m³) and T_f is the temperature (F).

(2) Specific Heat

The specific heat is:

$$c_p = 4186.8 \times (0.1014 + 4.378952 \times 10^{-5} T_f - 2.046138 \times 10^{-8} T_f^2 + 3.418111 \times 10^{-11} T_f^3 - 2.060318 \times 10^{-13} T_f^4 + 3.682836 \times 10^{-16} T_f^5 - 2.458648 \times 10^{-19} T_f^6 + 5.597571 \times 10^{-23} T_f^7)$$

where c_p is the specific heat (J/kg.K) and T_f is the temperature (F).

(3) Thermal Conductivity

The thermal conductivity is:

$$k = 1.729577 \times (8.011332 + 4.643719 \times 10^{-3} T_f + 1.872857 \times 10^{-6} T_f^2 - 3.914512 \times 10^{-9} T_f^3 + 3.475513 \times 10^{-12} T_f^4 - 9.936696 \times 10^{-16} T_f^5)$$

where k is the thermal conductivity (W/m.K) and T_f is the temperature (F).

3.4.2 Properties of Alumina

It is assumed that 99.5% Aluminum Oxide (Al₂O₃) is used.

(1) Density

A constant value of 3.89 g/cm³ is to be used.

(2) Specific heat

A constant value of 880 J/kg-K is to be used.

(3) Thermal Conductivity

A constant value of 35 W/m-K is to be used.

3.4.3 Properties of Titanium

(1) Density

A constant value of 4.54 g/cm³ is to be used.

(2) Specific heat

A constant value of 502.44 J/kg-K is to be used.

(3) Thermal Conductivity

A constant value of 16.44 W/m-K is to be used.

Chapter 4

BENCHMARK PHASES AND EXERCISES

4.1 Introduction

The OECD/NRC PSBT benchmark consists of two phases with each phase consisting of different exercises. The benchmark phases and exercises are described below.

Phase I – Void distribution benchmark

- Exercise I-1 – Steady-state single subchannel benchmark
- Exercise I-2 – Steady-state bundle benchmark
- Exercise I-3 – Transient bundle benchmark
- Exercise I-4 – Pressure drop benchmark

Phase II – Departure from Nucleate Boiling benchmark

- Exercise II-1 – Steady-state fluid temperature benchmark
- Exercise II-2 – Steady-state DNB benchmark
- Exercise II-3 – Transient DNB benchmark

The subchannel approach is categorized as the macroscopic grade approach that can resolve an element as small as a subchannel size mesh. Time and space averaged formulations cause the loss of the heterogeneous and instantaneous void fraction information of the two-phase flow structure within the subchannel. As a consequence, many macroscopic correlations are still indispensable for reproducing the experimental results. In recent years, there have been developed newer two-phase flow numerical approaches that can be categorized into meso-, micro- or molecular dynamics approaches. Taking into account the immaturity of numerical modeling for detailed void distribution, this benchmark specification is designed to accept as many potential numerical approaches as possible.

Table 4.1.1 provides a guide to the participants where to find the conditions for each benchmark case within each benchmark exercise and phase. Table 4.1.2 provides a guide where to find the data for the involved fuel assemblies.

Table 4.1.1 NUPEC PSBT Benchmark Database

Items of Data	Test Series
Void fraction measurements data - Steady-state void fraction in subchannel by CT measurement - Steady-state void distribution image in subchannel by CT measurement - Steady-state void fraction in rod bundle by chordal measurement - Steady-state void distribution image in rod bundle by chordal measurement - Transient void fraction in rod bundle by chordal measurement	1, 2, 3, 4 1, 2 5, 6, 7, 8 5, 6, 7, 8 5T, 6T, 7T
DNB measurements data - Steady-state DNB data in rod bundle - Steady-state DNB detected location in rod bundle - Steady-state fluid temperature distribution in rod bundle - Transient DNB data in rod bundle	0, 2, 3, 4, 8, 13 4, 8, 13 1 11T, 12T

Table 4.1.2 Benchmark Conditions

Phase	Exercises	Assembly ID	Test Series Number	Geometrical Data	Test Matrix / Measurement Conditions	Related Section
Void Distribution	Steady-state subchannel	S1	1	Table 3.2.1.1	Table 4.2.1.1	4.2.1
		S2	2	Table 3.2.1.1	Table 4.2.1.2	4.2.1
		S3	3	Table 3.2.1.1	Table 4.2.1.3	4.2.1
		S4	4	Table 3.2.1.1	Table 4.2.1.4	4.2.1
	Steady-state bundle	B5	5	Table 3.2.2.1	Table 4.2.2.1	4.2.2
		B6	6	Table 3.2.2.1	Table 4.2.2.2	4.2.2
		B7	7	Table 3.2.2.1	Table 4.2.2.3	4.2.2
		B5	8	Table 3.2.2.1	Table 4.2.2.4	4.2.2
	Transient bundle	B5	5T	Table 3.2.2.1	Table 4.2.3.1	4.2.3
		B6	6T	Table 3.2.2.1	Table 4.2.3.1	4.2.3
		B7	7T	Table 3.2.2.1	Table 4.2.3.1	4.2.3
DNB Measurement	Steady-state fluid temperature	A1	1	Table 3.3.3	Table 4.3.1.1	4.3.1
	Steady-state DNB	A0	0	Table 3.3.2	Table 4.3.2.1	4.3.2
		A2	2	Table 3.3.3	Table 4.3.2.2	4.3.2
		A3	3	Table 3.3.3	Table 4.3.2.3	4.3.2
		A4	4	Table 3.3.4	Table 4.3.2.4	4.3.2
		A8	8	Table 3.3.4	Table 4.3.2.5	4.3.2
		A4	13	Table 3.3.4	Table 4.3.2.6	4.3.2
	Transient DNB	A11	11T	Table 3.3.4	Table 4.3.3.1	4.3.3
		A12	12T	Table 3.3.4	Table 4.3.3.1	4.3.3

4.2 Phase I - Void Distribution Benchmark

Table 4.2.1 summarizes the test series used for both the subchannel and rod bundle void fraction measurements.

Table 4.2.1 Test Series for Void Fraction Measurement

Test series	Test section	Assembly	Test mode		Void measurement	
			Steady-State	Transient	CT	Chordal
1	Subchannel	S1	Y		Y	Y
2		S2	Y		Y	Y
3		S3	Y		Y	Y
4		S4	Y		Y	Y
5	5×5 Rod bundle	B5	Y			Y
5T				Y		Y
6		B6	Y			Y
6T				Y		Y
7		B7	Y			Y
7T				Y		Y
8		B5	Y			Y

4.2.1 Exercise I-1 – Steady-State Single Subchannel Benchmark

The available experimental data includes CT scan measurements of the void fraction (subchannel averaged) of four representative subchannel types: typical central, central with a guide tube, side, and corner subchannels and images of the void distribution within two representative subchannel types: typical central and central with a guide tube. The data can be used to assess and improve the current models for void generation (subchannel/system and CFD codes) and void distribution within subchannels (CFD codes).

The test matrixes for the steady-state single subchannel void measurements - Test Series 1, 2, 3, and 4 - are given in Table 4.2.1.1, Table 4.2.1.2, Table 4.2.1.3, and Table 4.2.1.4, respectively.

The highlighted cases are selected to be analyzed as part of benchmark. The cases in *italics* are selected to be modelled for the investigation of the axial variation of the void fraction.

Table 4.2.1.1 Test Conditions for Steady-State Void Measurement Test Series 1

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
1.1221	169.1	10.95	49.9	329.7
1.1222	169.1	10.98	50.0	334.7
1.1223	169.1	11.00	49.9	339.7
1.2121	150.1	14.80	79.9	309.5
1.2122	149.9	14.89	80.1	319.7
1.2211	150.1	10.91	90.0	295.4
1.2212	150.1	10.88	90.0	299.4
1.2221	150.1	10.88	69.8	299.4
1.2223	150.1	10.91	69.8	319.6
1.2231	150.2	10.87	60.0	299.3
1.2233	150.2	10.89	59.9	309.6
1.2234	150.1	10.92	60.1	314.6
1.2235	150.1	10.89	59.9	319.6
1.2236	150.0	10.91	60.0	324.7
1.2237	150.3	10.93	60.0	329.6
1.2421	150.2	5.02	59.8	268.9
1.2422	150.1	5.00	60.0	284.1
1.2423	150.3	4.93	59.9	299.3
1.3221	125.0	11.10	59.9	294.4
1.3222	125.1	10.91	60.0	309.5
1.3223	124.9	11.10	60.1	319.7
1.4121	100.1	10.97	69.9	274.1
1.4122	99.8	10.90	69.8	304.5
1.4311	100.4	5.01	79.9	214.2
1.4312	100.2	5.03	79.8	248.9
1.4321	100.5	5.01	59.9	209.3
1.4323	100.5	5.03	59.9	229.2
1.4324	100.1	5.02	60.1	238.9
1.4325	100.3	5.03	59.8	253.8
1.4326	100.1	5.02	60.1	268.8
1.4327	100.1	4.96	59.9	289.0
1.4411	100.4	1.99	19.9	253.7
1.4412	100.3	1.99	20.0	284.0
1.5221	75.5	5.02	49.9	219.2
1.5222	75.0	5.02	50.0	243.9
1.5223	75.6	5.03	49.9	263.8
1.6211	50.4	5.02	79.7	164.1
1.6212	50.4	5.00	79.8	199.3
1.6221	50.5	5.01	50.0	189.2
1.6222	50.0	5.00	49.9	204.2
1.6223	50.5	5.03	49.9	239.0
1.6311	50.6	1.95	20.1	204.1
1.6312	50.6	1.96	20.1	238.9

Table 4.2.1.2 Test Conditions for Steady-State Void Measurement Test Series 2

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
2.1231	168.9	10.98	37.5	335.0
2.1232	168.8	11.00	37.5	340.0
2.1233	168.8	11.01	37.5	345.0
2.2121	149.5	14.89	60.2	319.9
2.2122	149.5	14.88	60.1	325.1
2.2221	149.9	10.91	60.0	310.4
2.2222	149.9	10.92	60.1	314.9
2.2231	149.9	10.91	52.5	310.3
2.2233	149.9	10.95	52.5	330.6
2.2241	150.0	10.90	45.1	304.3
2.2243	149.9	10.91	45.2	314.5
2.2244	149.9	10.91	45.0	320.0
2.2245	149.9	10.91	45.0	325.5
2.2246	150.0	10.95	45.0	329.8
2.2247	150.0	10.98	45.1	334.8
2.2441	150.4	5.00	45.1	289.8
2.2442	150.4	4.98	45.1	304.8
2.2443	150.4	4.99	45.1	320.1
2.3231	125.2	10.90	45.0	304.6
2.3232	125.1	10.91	45.1	309.8
2.3233	125.0	10.91	45.1	319.9
2.4231	100.3	10.96	52.7	279.3
2.4232	100.2	10.89	52.7	299.6
2.4421	100.4	5.03	60.1	244.0
2.4422	100.5	5.02	60.1	279.2
2.4441	100.5	5.03	45.0	234.1
2.4443	100.5	5.03	45.2	253.9
2.4444	100.6	5.03	45.1	263.9
2.4445	100.5	5.02	45.0	274.0
2.4446	100.5	5.01	45.1	284.2
2.4447	100.5	4.98	45.2	299.4
2.4551	100.7	1.79	15.0	274.0
2.4552	100.7	1.78	15.1	294.3
2.5431	75.6	5.03	37.7	239.0
2.5432	75.7	5.03	37.5	263.9
2.5433	75.6	5.02	37.5	279.2
2.6411	50.5	5.01	60.1	194.2
2.6412	50.5	5.03	60.1	229.2
2.6431	50.6	5.01	37.5	209.2
2.6432	50.6	5.02	37.5	224.2
2.6433	50.5	5.03	37.6	253.9
2.6541	50.7	1.79	15.1	224.1
2.6542	50.7	1.79	15.2	253.7

Table 4.2.1.3 Test Conditions for Steady-State Void Measurement Test Series 3

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
<u>3.2231</u>	149.9	10.90	40.4	309.4
<u>3.2232</u>	149.9	10.90	40.5	314.5
3.2251	150.0	10.89	30.0	304.3
3.2253	149.9	10.89	30.0	314.4
3.2254	149.9	10.90	30.1	319.5
3.2255	150.0	10.90	30.2	324.5
3.2256	150.0	10.93	30.1	329.6
3.2257	150.0	10.96	30.2	334.5
<u>3.2451</u>	150.3	4.96	30.2	283.8
<u>3.2452</u>	150.5	4.93	30.2	299.0
<u>3.2453</u>	150.3	4.94	30.2	314.3
3.4451	100.5	4.98	30.2	228.9
3.4453	100.5	4.98	30.3	248.8
3.4454	100.4	4.98	30.1	258.6
3.4455	100.6	4.98	30.3	268.8
3.4456	100.6	4.96	30.2	283.7
3.4457	100.6	4.93	30.2	299.0
<u>3.6431</u>	50.4	4.96	40.2	189.0
<u>3.6432</u>	50.4	4.97	40.1	223.9
<u>3.6461</u>	50.6	4.96	25.2	203.9

Table 4.2.1.4 Test Conditions for Steady-State Void Measurement Test Series 4

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
4.2231	149.9	11.06	20.3	314.3
4.2232	149.9	11.06	20.3	319.3
4.2251	149.9	11.06	15.2	310.3
4.2253	150.0	11.06	15.1	318.4
4.2254	150.0	11.07	15.2	322.4
4.2255	150.0	11.08	15.2	326.5
4.2256	150.0	11.11	15.1	330.5
4.2257	149.9	11.12	15.1	334.5
4.2451	150.4	5.01	15.2	294.0
4.2452	150.4	5.01	15.2	309.2
4.2453	150.4	5.01	15.2	324.3
4.4451	100.6	5.05	15.2	238.8
4.4453	100.5	5.05	15.2	258.6
4.4454	100.5	5.04	15.1	268.7
4.4455	100.5	5.04	15.2	278.8
4.4456	100.5	5.02	15.2	289.0
4.4457	100.5	5.00	15.2	299.1
4.6431	50.6	5.02	20.2	204.1
4.6432	50.5	5.05	20.3	238.8
4.6461	50.6	5.03	12.5	214.0

4.2.2 Exercise I-2 – Steady-State Bundle Benchmark

The experimental data include X-ray densitometer measurements of void fraction (chordal averaged) at three axial elevations. The averaging is over the four central subchannels. Images of the bundle void distribution are available as well. The data can be applied to thermal-hydraulic subchannel and system codes and CFD codes.

The test matrixes for the steady-state bundle void measurements - Test Series 5, 6, 7, and 8 - are given in Table 4.2.2.1, Table 4.2.2.2, Table 4.2.2.3, and Table 4.2.2.4, respectively.

The highlighted cases are selected to be analyzed as part of benchmark.

Table 4.2.2.1 Test Conditions for Steady-State Void Measurement Test Series 5

Run No	Pressure (kg/cm ² a)	Mass Flux (10 ⁶ kg/m ² hr)	Power (kW)	Inlet Temperature (°C)
5.1121	167.39	14.96	2990	316.9
5.1122	167.25	15.02	2995	322.0
5.1221	168.29	11.00	3000	292.3
5.1222	168.27	10.98	2998	297.3
5.1231	168.29	10.97	2488	311.8
5.1232	168.29	10.98	2486	316.6
5.1331	168.80	7.95	2491	287.2
5.1332	168.71	7.93	2491	292.2
5.1341	168.76	7.90	2005	312.2
5.1342	168.83	7.91	2005	317.2
5.1451	169.19	4.99	1515	302.3
5.1452	169.19	4.98	1516	312.5
5.2111	148.15	15.08	3296	291.9
5.2112	148.04	14.98	3294	296.8
5.2131	148.02	14.99	2360	316.5
5.2132	147.67	15.01	2361	321.8
5.2221	149.06	11.06	2972	287.2
5.2222	149.18	11.04	2965	292.5
5.2231	149.16	10.95	2364	301.7
5.2232	148.83	11.01	2360	311.8
5.2241	148.92	11.00	1901	316.8
5.2242	148.78	11.01	1898	321.6
5.2331	149.57	8.03	2481	277.5
5.2332	149.82	7.94	2523	287.8
5.2341	149.72	7.97	1902	302.0
5.2342	149.51	7.91	1901	312.2
5.2351	149.71	7.89	1431	316.8
5.2352	149.63	7.94	1433	321.9
5.2442	149.97	4.99	2000	263.0
5.2451	150.10	4.94	979	311.4
5.2452	150.16	4.90	964	321.8
5.3111	123.65	15.10	3480	276.7
5.3112	123.52	15.10	3489	281.7
5.3221	124.55	11.06	2997	271.8

Table 4.2.2.1 Test Conditions for Steady-State Void Measurement Test Series 5 – Cont.

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
5.3222	124.31	11.07	2988	277.0
5.3321	124.88	8.00	3000	243.0
5.3322	124.77	8.00	2998	248.0
5.3331	124.89	8.00	2491	267.1
5.3332	124.96	7.99	2494	277.3
<u>5.3441</u>	125.22	5.00	2014	247.9
<u>5.3442</u>	125.13	5.00	2013	257.7
5.4211	98.71	11.12	3478	242.6
5.4212	98.70	11.11	3480	247.7
5.4321	99.28	8.09	2984	237.9
5.4322	99.21	8.07	2985	243.1
5.4431	99.89	4.91	2507	199.2
5.4432	100.09	5.02	2449	204.3
5.4441	100.02	4.97	2003	223.2
5.4442	99.97	5.02	2007	233.4
5.4561	100.33	2.03	1018	204.4
<u>5.4562</u>	100.23	2.02	1016	214.3
5.5201	74.44	11.07	4018	208.2
5.5202	74.36	11.07	4021	213.2
5.5311	74.95	8.07	3506	193.8
5.5312	74.87	8.13	3511	204.0
5.5321	75.00	8.05	3017	203.6
5.5322	74.93	8.03	3016	218.6
5.5431	75.21	5.07	2520	174.2
5.5432	75.18	5.05	2518	184.2
5.5441	75.13	5.01	2020	203.6
5.5442	75.27	5.02	2022	213.6
5.5551	75.41	2.00	1028	183.0
5.5552	75.40	2.00	1026	193.2
5.6311	48.99	7.86	3496	163.8
5.6312	48.85	7.85	3492	173.9
<u>5.6321</u>	49.20	7.87	3000	173.5
<u>5.6322</u>	49.08	7.86	3000	183.6
5.6431	49.82	4.93	2507	149.4
5.6432	49.73	4.96	2506	159.3
5.6441	49.92	4.96	2016	168.6
5.6442	49.84	4.96	2017	178.5
5.6551	50.17	2.00	1028	149.1
<u>5.6552</u>	50.17	2.00	1028	159.1

Table 4.2.2.2 Test Conditions for Steady-State Void Measurement Test Series 6

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
<u>6.1121</u>	167.59	15.16	3372	301.5
<u>6.1122</u>	167.56	15.17	3376	306.7
6.1221	168.24	11.32	3390	274.9
6.1222	168.28	11.32	3391	280.2
6.1231	168.23	11.25	2890	296.7
6.1232	168.26	11.25	2889	301.9
6.1331	168.65	8.23	2891	262.1
6.1332	168.68	8.22	2891	267.2
6.1341	168.75	8.18	2398	286.9
6.1342	168.67	8.18	2398	291.9
<u>6.1451</u>	169.03	5.20	1914	267.4
<u>6.1452</u>	169.06	5.20	1915	272.5
6.2111	148.13	15.32	3882	276.4
6.2112	148.60	15.30	3879	281.5
6.2131	148.50	15.21	2884	303.4
6.2132	148.40	15.23	2886	311.8
6.2221	149.41	11.34	3394	271.7
6.2222	149.21	11.07	3597	277.3
6.2231	149.42	11.31	2894	281.7
6.2232	149.27	11.30	2894	291.9
6.2241	149.29	11.27	2398	301.6
6.2242	149.24	11.25	2396	306.8
6.2331	149.78	8.24	2899	257.2
6.2332	149.64	7.97	3013	267.2
6.2341	149.69	8.21	2387	281.7
6.2342	149.75	8.21	2398	289.0
6.2351	149.72	8.17	1897	306.8
6.2352	149.71	8.17	1897	313.8
<u>6.2441</u>	150.01	5.16	2415	223.5
<u>6.2442</u>	150.02	5.12	2412	228.4
6.2461	150.17	5.08	957	311.6
6.2462	150.20	5.09	957	321.7
6.3121	123.73	15.12	3308	276.2
6.3122	123.62	15.12	3305	286.4
6.3231	124.58	11.09	2846	271.4
6.3232	124.37	11.09	2842	281.8
6.3331	124.90	8.03	2840	242.3
6.3332	124.84	8.02	2839	252.1
6.3341	124.91	8.02	2373	266.9
6.3342	124.91	8.01	2375	282.1
6.3451	125.15	5.01	1924	242.8
<u>6.3452</u>	125.20	5.00	1920	262.5
6.4221	99.66	11.11	3333	227.4
6.4222	99.53	11.11	3317	241.3
6.4331	99.91	8.03	2862	222.9
6.4332	99.94	8.04	2854	242.6

Table 4.2.2.2 Test Conditions for Steady-State Void Measurement Test Series 6 - Cont

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
6.4441	100.19	5.03	2382	183.7
6.4442	100.11	5.03	2833	204.3
6.4451	100.19	5.01	1923	213.1
6.4452	100.22	5.01	1922	232.9
<u>6.4561</u>	100.42	2.05	973	192.6
<u>6.4562</u>	100.41	2.05	972	213.0
6.5211	74.58	11.11	3803	197.9
6.5212	74.53	11.09	3802	213.0
6.5321	75.00	8.07	3335	173.3
6.5322	74.95	8.06	3336	193.5
6.5331	74.96	8.05	2857	187.8
6.5332	75.04	8.03	2862	202.8
6.5441	75.27	5.05	2392	153.7
6.5442	75.25	5.05	2392	173.8
6.5451	75.22	5.02	1921	183.2
6.5452	75.25	5.01	1920	203.8
6.5561	75.41	2.06	977	163.5
6.5562	75.41	2.06	974	178.2
6.6321	49.90	8.09	3351	153.4
6.6322	49.80	8.09	3344	168.5
6.6331	49.94	8.09	2875	162.7
6.6332	49.93	8.08	2876	178.3
6.6441	50.17	5.06	2396	143.7
6.6442	50.19	5.05	2394	158.7
6.6451	50.29	5.04	1925	162.9
6.6452	50.28	5.06	1930	178.2
<u>6.6561</u>	50.38	2.02	979	144.0
<u>6.6562</u>	50.40	2.00	979	158.5

Table 4.2.2.3 Test Conditions for Steady-State Void Measurement Test Series 7

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
7.1121	167.48	15.07	3385	301.8
7.1122	167.41	15.07	3384	306.8
7.1221	168.32	11.05	3391	277.4
7.1222	168.25	11.10	3390	282.3
7.1231	168.27	11.04	2892	297.0
7.1232	168.24	11.01	2891	302.0
7.1331	168.73	7.97	2994	268.0
7.1332	168.75	7.98	2995	272.8
7.1341	168.74	7.92	2391	289.4
7.1342	168.75	7.94	2391	295.3
7.1451	169.17	4.99	1963	273.0
7.1452	169.17	4.96	1963	278.0
7.2111	148.63	15.06	3992	281.8
7.2112	148.68	14.94	3993	286.5
7.2131	148.74	14.77	2993	306.2
7.2132	148.48	14.98	2987	315.5
7.2221	149.29	11.01	3503	272.1
7.2222	149.30	11.06	3506	277.2
7.2231	149.28	11.00	3004	286.8
7.2232	149.28	11.01	3004	294.6
7.2241	149.24	10.96	2407	306.1
7.2242	149.17	10.94	2409	311.3
7.2331	149.73	7.99	2958	257.6
7.2332	149.80	7.96	2957	264.5
7.2341	149.72	7.97	2447	282.0
7.2342	149.75	7.95	2450	289.2
7.2351	149.73	7.91	1958	306.4
7.2352	149.74	7.91	1955	311.4
7.2441	149.99	4.98	2516	223.9
7.2442	150.03	4.96	2516	233.9
7.2461	150.10	4.96	1019	311.2
7.2462	150.13	4.95	1040	321.2
7.3121	123.68	15.20	3502	276.1
7.3122	123.52	15.12	3498	286.0
7.3231	124.36	11.41	3011	270.9
7.3232	124.35	11.09	3006	281.1
7.3331	124.77	8.00	3012	242.7
7.3332	124.75	8.04	3011	252.5
7.3341	124.77	8.01	2480	266.0
7.3342	124.89	8.02	2480	280.2
7.3451	125.07	5.03	2023	242.8
7.3452	125.09	5.03	2021	260.1
7.4221	99.58	11.12	3508	227.7
7.4222	99.46	11.12	3506	242.6
7.4331	100.03	8.02	3017	223.0

Table 4.2.2.3 Test Conditions for Steady-State Void Measurement Test Series 7 – Cont.

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
7.4332	99.87	8.01	3018	242.8
7.4441	100.13	5.05	2520	183.9
7.4442	100.24	5.05	2516	203.9
7.4551	100.19	4.99	2030	213.2
7.4452	100.25	5.02	2026	233.2
7.4561	100.28	2.16	1023	196.8
7.4562	100.29	2.16	1023	214.9
7.5211	74.47	11.15	4032	198.0
7.5212	74.36	11.12	4021	212.9
7.5321	74.81	8.07	3536	173.5
7.5322	74.75	8.09	3537	193.6
7.5331	74.80	8.02	3036	188.5
7.5332	74.83	8.01	3031	203.3
7.5441	75.10	5.14	2534	154.0
7.5442	75.19	5.12	2532	174.4
7.5451	75.13	5.08	2035	183.4
7.5452	75.24	5.03	2039	203.5
7.5561	75.34	2.14	1029	162.7
7.5562	75.36	2.14	1031	178.4
7.6321	49.84	8.10	3541	153.5
7.6322	49.65	8.06	3536	168.6
7.6331	49.88	8.07	3037	163.1
7.6332	49.82	8.05	3036	178.2
7.6441	50.09	5.16	2540	144.0
7.6442	50.08	5.14	2536	158.9
7.6451	50.17	5.04	2046	163.3
7.6452	50.16	5.00	2048	178.4
7.6561	50.68	2.18	1040	143.4
7.6562	50.69	2.16	1037	158.4

Table 4.2.2.4 Test Conditions for Steady-State Void Measurement Test Series 8

Run No	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Power (kW)	Inlet Temperature (°C)
<u>8.1232</u>	168.8	10.98	2511.8	316.8
<u>8.1342</u>	168.9	7.91	2011.7	316.9
<u>8.1452</u>	169.2	4.92	1520.2	312.0
8.2131	149.6	15.14	2354.7	316.3
8.2132	149.6	15.14	2356.4	321.3
8.2222	149.9	11.03	2996.2	292.0
8.2231	149.9	11.04	2356.9	301.5
8.2232	150.1	11.00	2359.3	311.9
8.2332	150.1	7.98	2505.4	287.1
<u>8.2351</u>	149.9	7.90	1426.8	316.6
<u>8.2352</u>	150.0	7.93	1424.9	321.6
8.2441	150.2	4.98	2012.1	257.5
8.2442	150.2	4.98	2012.0	262.5
8.2452	149.6	4.93	965.6	321.1
8.3221	124.9	11.10	3010.7	271.7
8.3222	125.0	11.10	3009.9	277.0
8.3332	125.1	8.00	2522.7	277.0
8.3442	125.3	4.98	2020.7	257.7
<u>8.4211</u>	100.0	11.11	3514.9	242.5
<u>8.4212</u>	100.0	11.11	3510.4	247.4
8.4431	100.2	5.00	2534.9	199.1
8.4432	100.2	4.99	2534.9	203.9
8.4562	99.6	2.10	1028.3	213.7
<u>8.5311</u>	75.3	8.05	3529.9	193.6
<u>8.5312</u>	75.4	8.07	3535.4	203.6
<u>8.5442</u>	75.0	4.99	2034.9	213.4
8.6321	50.5	8.05	3048.7	173.1
8.6322	50.7	8.09	3049.1	183.3
8.6441	50.3	5.05	2045.0	168.4
8.6442	50.5	5.05	2043.5	178.3
<u>8.6551</u>	50.3	2.08	1025.2	148.6

The same assembly type was used in Test Series 5 and Test Series 8. Therefore, test cases performed at similar conditions may be used to assess the repeatability of the measurements. The duplicated cases are listed in Table 4.2.2.5. The highlighted cases are benchmark exercise cases.

Table 4.2.2.5 Duplicated Cases in Test Series B5 and B8

B8 Test Cases	B5 (Duplicated) Test Cases	B8 Test Cases	B5 (Duplicated) Test Cases
8.1232	5.1232	8.3332	5.3332
8.1342	5.1342	8.3442	-
8.1452	5.1452	8.4211	5.4211
8.2131	5.2131	8.4212	5.4212
8.2132	5.2132	8.4431	5.4431
8.2222	5.2222	8.4432	5.4432
8.2231	5.2231	8.4562	5.4562
8.2232	5.2232	8.5311	5.5311
8.2332	5.2332	8.5312	5.5312
8.2351	5.2351	8.5442	5.5442
8.2352	5.2351	8.6321	5.6321
8.2441	5.2442	8.6322	5.6322
8.2442	5.2442	8.6441	5.6441
8.2452	5.2452	8.6442	5.6442
8.3221	5.3221	8.6551	5.6551
8.3222	5.3222		

4.2.3 Exercise I-3 – Transient Bundle Benchmark

The experimental data include X-ray densitometer measurements of void fraction (chordal averaged) at three axial elevations. The averaging is over the four central subchannels. Data is collected for four transient scenarios: power increase; flow reduction; depressurization; temperature increase. The data can be used to assess the subchannel and system codes capabilities of predicting the void generation during transients.

Tables 4.2.3.2 through 4.2.3.13 contain the boundary conditions for each transient. Also included are plots of each parameter during the transient. These plots represent the ratio of the value of each parameter at a certain time to the value at the start of the transient.

All transient cases are to be analyzed as part of benchmark.

Table 4.2.3.1 Test Conditions for Transient Void Measurement Test Series 5T, 6T, 7T

Test Series	Assembly	Initial Conditions				Transients
		Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² h)	Power (kW)	Inlet temperature (Celsius)	
5T	B5	154.2	11.95	2282.0	300.4	Power increase
		153.8	11.93	2244.0	301.2	Flow reduction
		153.0	11.92	2236.0	300.4	Depressurization
		152.5	11.94	2230.0	301.7	Temperature increase
6T	B6	158.2	11.55	2621.0	288.1	Power increase
		158.4	12.03	2574.0	288.8	Flow reduction
		154.6	12.02	2556.0	288.2	Depressurization
		157.2	11.92	2603.0	288.8	Temperature increase
7T	B7	158.2	12.02	2500.0	291.9	Power increase
		158.1	12.04	2405.0	292.0	Flow reduction
		155.0	11.99	2577.0	291.8	Depressurization
		158.8	11.99	2496.0	290.2	Temperature increase

Table 4.2.3.2 Transient Void Fraction in Rod Bundle Test Series 5T (Power Increase)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
30.0	154.2	11.95	2282.0	300.4
30.2	154.4	11.95	2261.0	300.4
30.4	154.3	11.94	2254.0	300.4
30.6	153.9	11.96	2256.0	300.4
30.8	154.1	11.96	2261.0	300.4
31.0	154.0	11.95	2182.0	300.5
31.2	154.1	11.95	2194.0	300.4
31.4	154.2	12.01	2228.0	300.6
31.6	154.3	12.00	2236.0	300.5
31.8	154.4	11.98	2249.0	300.5
32.0	154.4	11.95	2265.0	300.5
32.2	154.5	11.94	2295.0	300.5
32.4	154.3	11.93	2343.0	300.5
32.6	154.3	11.91	2389.0	300.4
32.8	154.3	11.92	2451.0	300.4
33.0	154.3	11.92	2511.0	300.4
33.2	154.4	11.90	2573.0	300.4
33.4	154.4	11.90	2639.0	300.5
33.6	154.5	11.87	2675.0	300.5
33.8	154.4	11.85	2717.0	300.7
34.0	154.5	11.84	2765.0	300.6
34.2	154.6	11.82	2834.0	300.6
34.4	154.5	11.82	2882.0	300.5
34.6	154.6	11.81	2948.0	300.4
34.8	154.4	11.79	2992.0	300.5
35.0	154.7	11.76	3074.0	300.4
35.2	154.8	11.73	3096.0	300.3
35.4	154.9	11.71	3079.0	300.3
35.6	155.0	11.79	3067.0	300.4
35.8	154.9	11.76	3062.0	300.5
36.0	155.2	11.75	3090.0	300.5
36.2	155.0	11.74	3100.0	300.7
36.4	155.0	11.72	3093.0	300.7
36.6	155.3	11.71	3074.0	300.6
36.8	155.1	11.73	3061.0	300.6
37.0	155.3	11.76	3092.0	300.5
37.2	155.0	11.78	3103.0	300.5
37.4	155.0	11.78	3104.0	300.4
37.6	155.0	11.78	3051.0	300.5
37.8	154.9	11.76	3052.0	300.5
38.0	155.1	11.76	3071.0	300.6
38.2	155.0	11.75	3073.0	300.6
38.4	155.1	11.75	3074.0	300.6
38.6	154.9	11.76	3089.0	300.7
38.8	154.9	11.76	3078.0	300.6
39.0	155.0	11.76	3062.0	300.6

Table 4.2.3.3 Transient Void Fraction in Rod Bundle Test Series 5T (Flow Reduction)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
30.0	153.8	11.93	2244.0	301.2
30.2	153.9	11.93	2228.0	301.2
30.4	154.0	11.94	2234.0	301.2
30.6	153.9	11.95	2233.0	301.2
30.8	153.9	11.86	2259.0	301.1
31.0	154.0	11.13	2256.0	301.1
31.2	153.7	9.88	2259.0	301.2
31.4	153.8	8.68	2250.0	301.2
31.6	153.7	7.94	2241.0	301.2
31.8	153.6	7.63	2242.0	301.1
32.0	153.8	7.38	2233.0	301.2
32.2	153.9	7.06	2240.0	301.1
32.4	154.0	6.77	2219.0	301.2
32.6	154.0	6.59	2246.0	301.1
32.8	153.8	6.55	2240.0	301.2
33.0	153.8	6.69	2234.0	301.2
33.2	153.9	6.96	2239.0	301.2
33.4	153.9	7.22	2225.0	301.2
33.6	154.0	7.38	2233.0	301.1
33.8	154.1	7.49	2240.0	301.2
34.0	154.1	7.88	2228.0	301.2
34.2	154.2	8.53	2239.0	301.2
34.4	154.3	9.15	2228.0	301.2
34.6	154.1	9.55	2236.0	301.2
34.8	154.2	9.71	2234.0	301.3
35.0	154.2	9.77	2243.0	301.2

Figure 4.2.3.1 Variation of Properties during Transient for Data Series 5T (Power Increase)

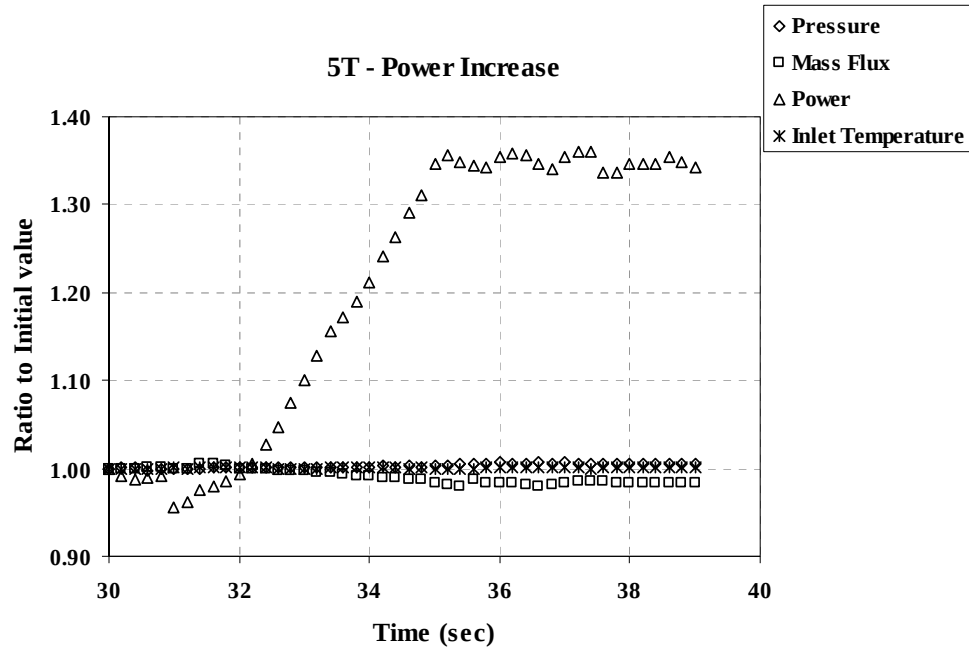


Figure 4.2.3.2 Variation of Properties during Transient for Data Series 5T (Flow Reduction)

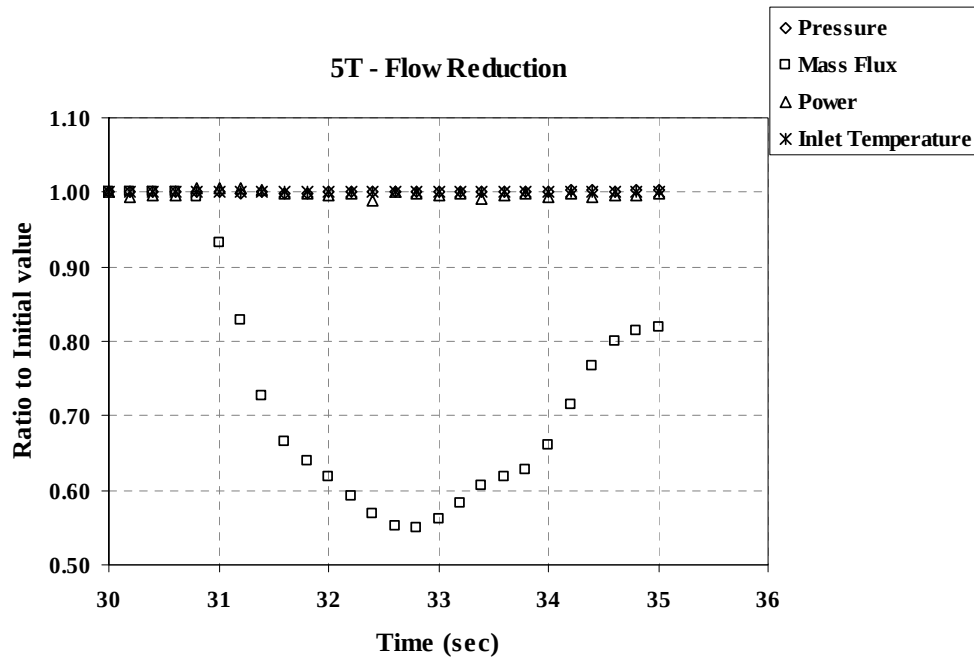


Table 4.2.3.4 Transient Void Fraction in Rod Bundle in Test Series 5T (Depressurization)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
20.0	153.0	11.92	2236.0	300.4
24.0	152.6	11.93	2251.0	300.4
28.0	152.3	11.93	2263.0	300.5
32.0	151.7	11.90	2247.0	300.5
36.0	151.1	11.93	2258.0	300.5
40.0	150.4	11.92	2253.0	300.5
44.0	149.5	11.91	2282.0	300.5
48.0	148.4	11.94	2255.0	300.5
52.0	147.5	11.93	2257.0	300.5
56.0	146.3	11.91	2257.0	300.5
60.0	145.2	11.90	2257.0	300.5
64.0	144.1	11.92	2260.0	300.5
68.0	142.9	11.94	2267.0	300.6
72.0	141.5	11.91	2264.0	300.5
76.0	140.6	11.91	2278.0	300.4
80.0	139.5	11.92	2267.0	300.0
84.0	138.2	11.93	2258.0	299.7
88.0	137.0	11.94	2263.0	299.3
92.0	135.8	11.96	2256.0	298.7
96.0	134.4	11.92	2170.0	298.2
100.0	132.8	11.94	2244.0	297.6
104.0	131.6	11.78	2246.0	297.1
108.0	130.4	11.58	2261.0	296.6
112.0	129.5	11.39	2248.0	296.2
116.0	128.7	11.21	2262.0	295.8
120.0	127.8	10.99	2269.0	295.4
124.0	126.8	10.82	2273.0	295.0
128.0	126.0	10.75	2260.0	294.6
132.0	125.2	10.78	2263.0	294.2
136.0	124.1	10.80	2273.0	293.8
138.0	123.8	10.75	2266.0	293.6
142.0	122.7	10.79	2273.0	293.1
146.0	121.6	10.76	2262.0	292.5
150.0	120.6	10.75	2264.0	291.8
154.0	119.5	10.80	2272.0	290.9
158.0	118.2	10.77	2269.0	290.0
162.0	116.9	10.73	2259.0	289.1
166.0	115.7	10.76	2269.0	288.2
170.0	114.7	10.80	2270.0	287.5
174.0	114.3	10.79	2270.0	286.7
178.0	114.2	10.80	2263.0	286.0

Table 4.2.3.5 Transient Void Fraction in Rod Bundle in Test Series 5T (Temperature Increase)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
60.0	152.5	11.94	2230.0	301.7
62.0	152.5	11.96	2268.0	302.5
64.0	152.7	11.92	2270.0	304.1
66.0	152.3	11.94	2272.0	305.8
68.0	152.3	11.98	2264.0	307.9
70.0	152.5	11.93	2279.0	310.3
72.0	152.8	11.94	2263.0	312.7
74.0	152.7	11.95	2264.0	315.1
76.0	152.5	11.93	2264.0	317.6
78.0	152.7	11.94	2243.0	319.8
80.0	153.0	11.95	2245.0	322.1
82.0	153.5	12.00	2244.0	324.0
84.0	153.7	11.98	2272.0	325.7
86.0	153.7	12.00	2255.0	327.5
88.0	153.8	12.04	2260.0	328.9
90.0	154.8	12.05	2266.0	330.3
92.0	155.2	12.16	2250.0	331.3
94.0	155.3	12.23	2268.0	332.4
96.0	155.4	12.11	2266.0	333.4
98.0	156.0	12.10	2252.0	334.2
100.0	156.3	12.09	2237.0	334.9
102.0	156.4	12.09	2251.0	335.5
104.0	157.5	12.13	2250.0	335.6
106.0	157.6	12.12	2232.0	334.8
108.0	158.0	12.07	2240.0	333.6
110.0	158.4	12.12	2238.0	331.6
112.0	158.8	12.07	2240.0	329.8
114.0	159.0	12.06	2256.0	327.8
116.0	159.6	12.09	2274.0	326.1
118.0	159.7	12.05	2236.0	324.7
120.0	159.8	12.07	2247.0	323.4

Figure 4.2.3.3 Variation of Properties during Transient for Data Series 5T (Depressurization)

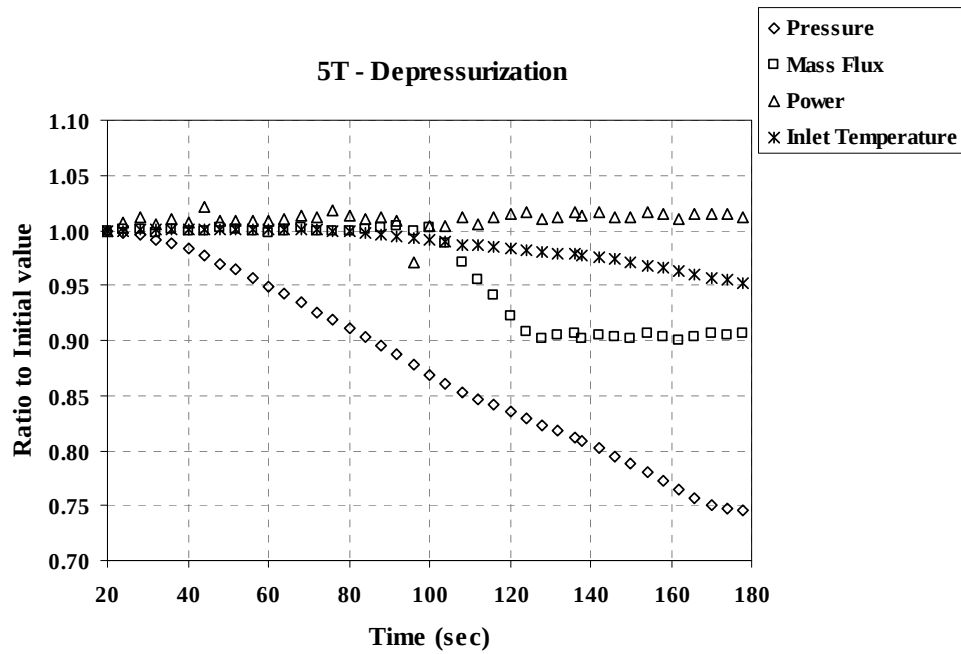


Figure 4.2.3.4 Variation of Properties during Transient for Data Series 5T (Temperature Increase)

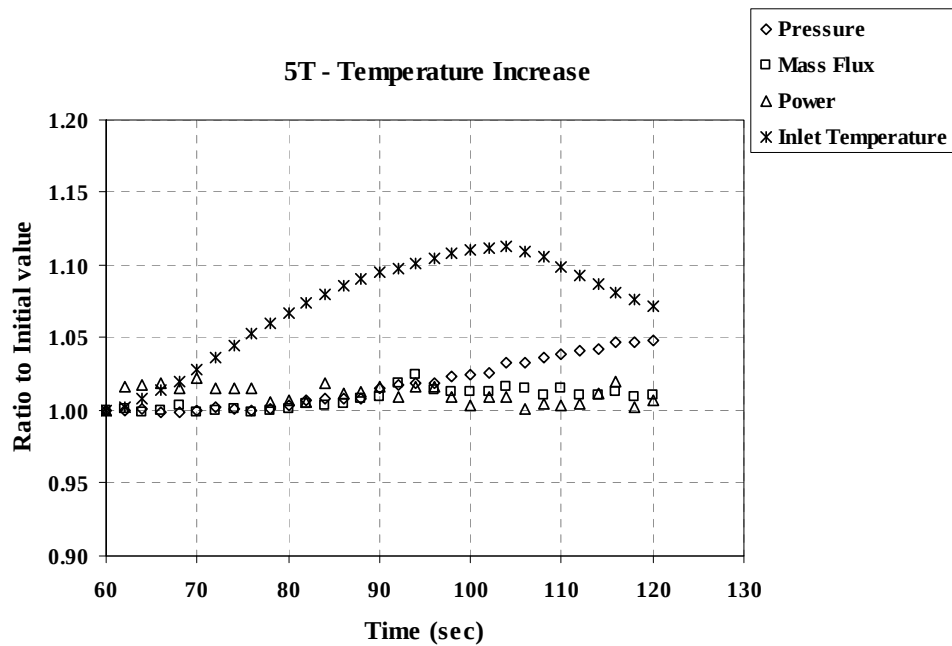


Table 4.2.3.6 Transient Void Fraction in Rod Bundle in Test Series 6T (Power Increase)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
24.0	158.2	11.55	2621.0	288.1
24.2	158.2	11.54	2682.0	288.1
24.4	158.2	11.54	2750.0	288.1
24.6	158.2	11.53	2811.0	288.1
24.8	158.2	11.52	2886.0	288.2
25.0	158.2	11.50	2946.0	288.4
25.2	158.2	11.48	3020.0	288.4
25.4	158.3	11.43	3106.0	288.4
25.6	158.3	11.43	3181.0	288.5
25.8	158.4	11.41	3237.0	288.5
26.0	158.4	11.43	3283.0	288.5
26.2	158.5	11.41	3314.0	288.5
26.4	158.5	11.40	3321.0	288.6
26.6	158.6	11.40	3333.0	288.5
26.8	158.6	11.40	3339.0	288.6
27.0	158.6	11.40	3323.0	288.5
27.2	158.6	11.41	3317.0	288.5
27.4	158.7	11.40	3306.0	288.5
27.6	158.7	11.40	3306.0	288.5
27.8	158.7	11.41	3313.0	288.5
28.0	158.7	11.42	3305.0	288.6
28.2	158.7	11.41	3314.0	288.5
28.4	158.8	11.43	3311.0	288.5
28.6	158.8	11.43	3309.0	288.4
28.8	158.8	11.44	3302.0	288.5
29.0	158.8	11.45	3296.0	288.5
29.2	158.8	11.45	3302.0	288.5
29.4	158.8	11.45	3302.0	288.5
29.6	158.8	11.45	3293.0	288.5
29.8	158.9	11.44	3304.0	288.5
30.0	158.9	11.44	3305.0	288.5
30.2	158.9	11.44	3297.0	288.5
30.4	158.9	11.45	3240.0	288.6
30.6	158.9	11.47	3235.0	288.6
30.8	159.0	11.49	3248.0	288.6
31.0	159.0	11.49	3259.0	288.6
31.2	159.0	11.48	3294.0	288.6
31.4	159.0	11.47	3304.0	288.7
31.6	159.1	11.47	3276.0	288.6
31.8	159.1	11.47	3027.0	288.5
32.0	159.1	11.48	2432.0	288.2
32.2	159.0	11.53	1893.0	288.0
32.4	159.0	11.59	1537.0	287.8
32.6	158.9	11.63	1373.0	287.5
32.8	158.8	11.64	1356.0	287.5
33.0	158.7	11.64	1368.0	287.6

Table 4.2.3.7 Transient Void Fraction in Rod Bundle in Test Series 6T (Flow Reduction)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
19.0	158.4	12.03	2574.0	288.8
19.2	158.5	12.02	2562.0	288.8
19.4	158.4	12.01	2561.0	288.8
19.6	158.3	12.01	2544.0	288.8
19.8	158.3	11.90	2548.0	288.8
20.0	158.1	11.22	2557.0	288.8
20.2	157.8	10.18	2560.0	289.0
20.4	157.7	9.14	2563.0	289.2
20.6	157.6	8.38	2571.0	289.3
20.8	157.6	7.97	2581.0	289.5
21.0	157.7	7.72	2583.0	289.5
21.2	157.8	7.50	2573.0	289.6
21.4	157.8	7.32	2575.0	289.6
21.6	157.8	7.17	2579.0	289.6
21.8	157.8	7.12	2583.0	289.7
22.0	157.8	7.16	2593.0	289.6
22.2	157.9	7.27	2551.0	289.6
22.4	157.9	7.37	2587.0	289.6
22.6	157.9	7.45	2572.0	289.6
22.8	158.0	7.53	2536.0	289.6
23.0	158.0	7.82	2484.0	289.5
23.2	158.0	8.29	2511.0	289.4
23.4	158.1	8.76	2541.0	289.3
23.6	158.2	9.12	2551.0	289.2
23.8	158.2	9.32	2571.0	289.2
24.0	158.3	9.49	2567.0	289.1
24.2	158.3	9.70	2574.0	289.0
24.4	158.4	9.89	2576.0	289.0
24.6	158.4	10.05	2572.0	289.1
24.8	158.4	10.15	2566.0	289.0
25.0	158.4	10.35	2559.0	289.1
25.2	158.4	10.61	2567.0	288.9
25.4	158.5	10.86	2564.0	288.9
25.6	158.5	11.05	2576.0	288.9
25.8	158.5	11.15	2572.0	288.9
26.0	158.5	11.22	2544.0	288.9
26.2	158.5	11.29	2537.0	288.9
26.4	158.5	11.34	2541.0	288.9
26.6	158.5	11.39	2550.0	289.0
26.8	158.5	11.39	2552.0	289.0
27.0	158.6	11.36	2557.0	289.1

Figure 4.2.3.5 Variation of Properties during Transient for Data Series 6T (Power Increase)

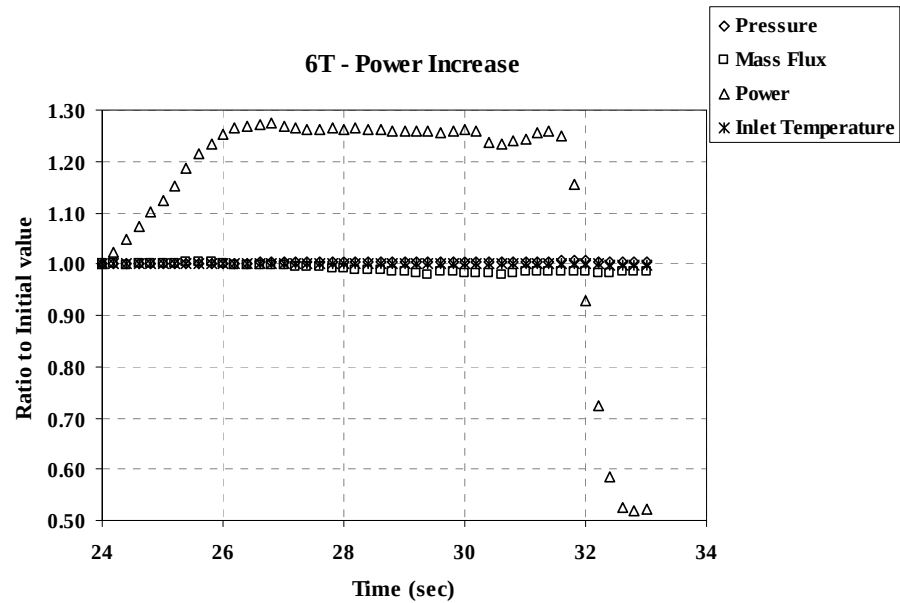


Figure 4.2.3.6 Variation of Properties during Transient for Data Series 6T (Flow Reduction)

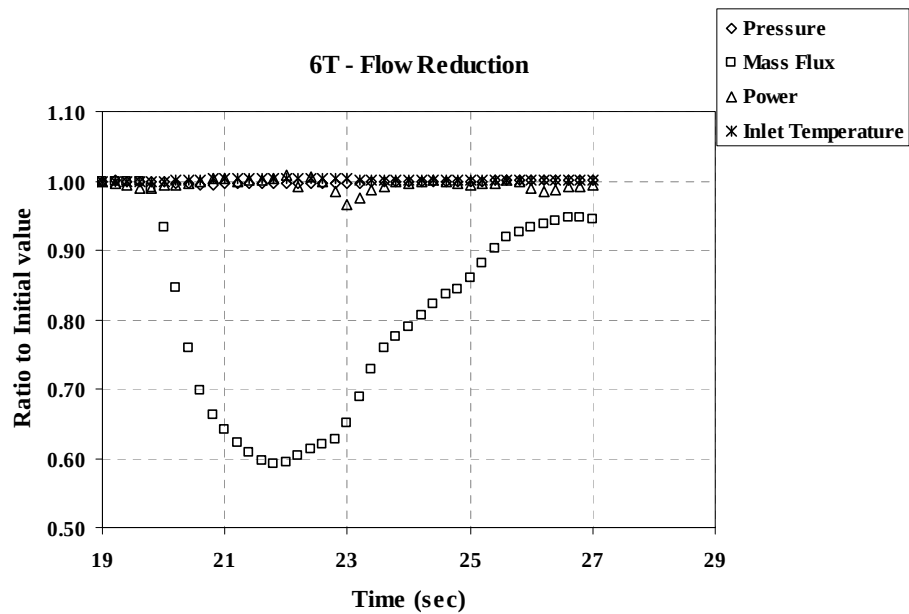


Table 4.2.3.8 Transient Void Fraction in Rod Bundle in Test Series 6T (Depressurization)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
0.0	154.6	12.02	2556.0	288.2
4.0	154.5	11.95	2597.0	288.0
8.0	154.3	11.90	2599.0	288.0
12.0	154.2	11.82	2599.0	287.8
16.0	153.9	11.75	2600.0	287.8
20.0	153.5	11.67	2600.0	287.8
24.0	153.0	11.62	2599.0	287.9
28.0	152.4	11.56	2602.0	287.7
32.0	151.7	11.49	2574.0	287.8
36.0	150.5	11.38	2612.0	287.7
40.0	149.2	11.36	2561.0	287.6
44.0	147.8	11.29	2595.0	287.6
48.0	146.5	11.20	2591.0	287.6
52.0	145.1	11.15	2591.0	287.7
56.0	143.8	11.07	2542.0	287.7
60.0	142.5	11.00	2590.0	287.8
64.0	141.2	10.93	2604.0	288.1
68.0	139.9	10.90	2565.0	288.2
72.0	138.6	10.83	2615.0	288.4
76.0	137.4	10.79	2626.0	288.5
80.0	136.3	10.78	2595.0	288.3
84.0	135.1	10.81	2582.0	287.8
88.0	134.0	10.82	2619.0	287.1
92.0	132.9	10.82	2599.0	286.3
96.0	131.8	10.80	2580.0	285.3
100.0	130.7	10.84	2594.0	284.4
104.0	129.7	10.86	2614.0	283.5
108.0	128.6	10.80	2597.0	282.8
112.0	127.5	10.84	2596.0	282.0
116.0	126.5	10.87	2611.0	281.2
120.0	125.4	10.84	2610.0	280.8
124.0	124.4	10.84	2600.0	280.3
128.0	123.3	10.84	2606.0	279.9
132.0	122.2	10.86	2618.0	279.4
136.0	121.2	10.85	2586.0	279.1
140.0	120.1	10.85	2602.0	278.7
144.0	119.0	10.85	2603.0	278.2
148.0	118.0	10.87	2615.0	277.9
152.0	116.9	10.82	2605.0	277.6
156.0	115.8	10.83	2588.0	277.3
160.0	114.8	10.84	2616.0	277.0
164.0	114.0	10.84	2608.0	276.7
168.0	113.9	10.84	2603.0	276.4
172.0	114.0	10.87	2616.0	276.3
176.0	114.2	10.87	2599.0	275.7
180.0	114.4	10.87	2576.0	275.3
184.0	114.6	10.87	2615.0	274.8
188.0	114.8	10.87	2598.0	274.3
192.0	115.0	10.90	2628.0	273.7
196.0	115.1	10.89	2586.0	272.9

Table 4.2.3.9 Transient Void Fraction in Rod Bundle in Test Series 6T (Temperature Increase)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
56.0	157.2	11.92	2603.0	288.8
58.0	157.2	11.88	2582.0	288.9
60.0	157.1	11.87	2572.0	289.4
62.0	157.0	11.84	2559.0	290.3
64.0	157.0	11.82	2590.0	291.5
66.0	156.9	11.81	2582.0	293.0
68.0	156.8	11.78	2562.0	294.7
70.0	156.7	11.77	2574.0	296.4
72.0	156.7	11.76	2564.0	298.4
74.0	156.7	11.73	2556.0	300.2
76.0	156.6	11.75	2568.0	302.1
78.0	156.7	11.75	2578.0	303.8
80.0	156.7	11.76	2566.0	305.5
82.0	156.8	11.73	2566.0	307.1
84.0	156.9	11.72	2553.0	308.7
86.0	157.0	11.68	2557.0	310.1
88.0	157.1	11.66	2559.0	311.5
90.0	157.3	11.70	2561.0	312.9
92.0	157.4	11.69	2562.0	314.2
94.0	157.6	11.70	2551.0	315.4
96.0	157.8	11.70	2569.0	316.6
98.0	158.0	11.69	2561.0	317.6
100.0	158.2	11.72	2549.0	318.8
102.0	158.6	11.74	2567.0	319.7
104.0	158.9	11.70	2556.0	320.6
106.0	159.2	11.74	2568.0	321.4
108.0	159.6	11.72	2563.0	322.2
110.0	160.0	11.75	2544.0	322.9
112.0	160.4	11.75	2552.0	323.6
114.0	160.8	11.82	2464.0	324.2
116.0	161.3	11.82	2561.0	324.8
118.0	161.8	11.88	2558.0	325.4
120.0	162.2	11.90	2541.0	325.9
122.0	162.8	11.96	2563.0	326.3
124.0	163.3	11.99	2553.0	326.9
126.0	163.9	12.03	2576.0	327.3
128.0	164.5	12.08	2506.0	327.4
130.0	165.1	12.11	2543.0	327.3
132.0	165.7	12.14	2546.0	326.4
134.0	166.3	12.21	2531.0	325.4
136.0	166.8	12.19	2544.0	323.9
138.0	167.3	12.20	2556.0	322.2
140.0	167.7	12.24	2546.0	320.4
142.0	168.0	12.20	2568.0	318.6
144.0	168.3	12.26	2565.0	316.7
146.0	168.5	12.24	2537.0	315.1
148.0	168.6	12.23	2540.0	313.4
150.0	168.7	12.23	2551.0	312.0
152.0	168.7	12.24	2560.0	310.6
154.0	168.7	12.22	2567.0	309.3
156.0	168.6	12.22	2566.0	308.2
158.0	168.4	12.19	2542.0	307.1
160.0	168.3	12.17	2561.0	306.1

Figure 4.2.3.7 Variation of Properties during Transient for Data Series 6T (Depressurization)

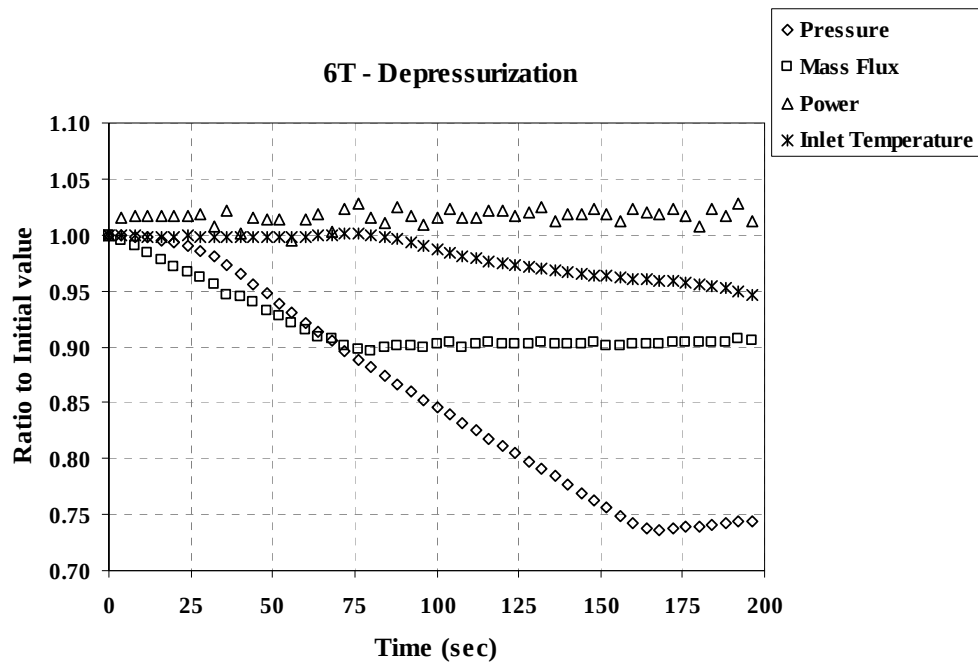


Figure 4.2.3.8 Variation of Properties during Transient for Data Series 6T (Temperature Increase)

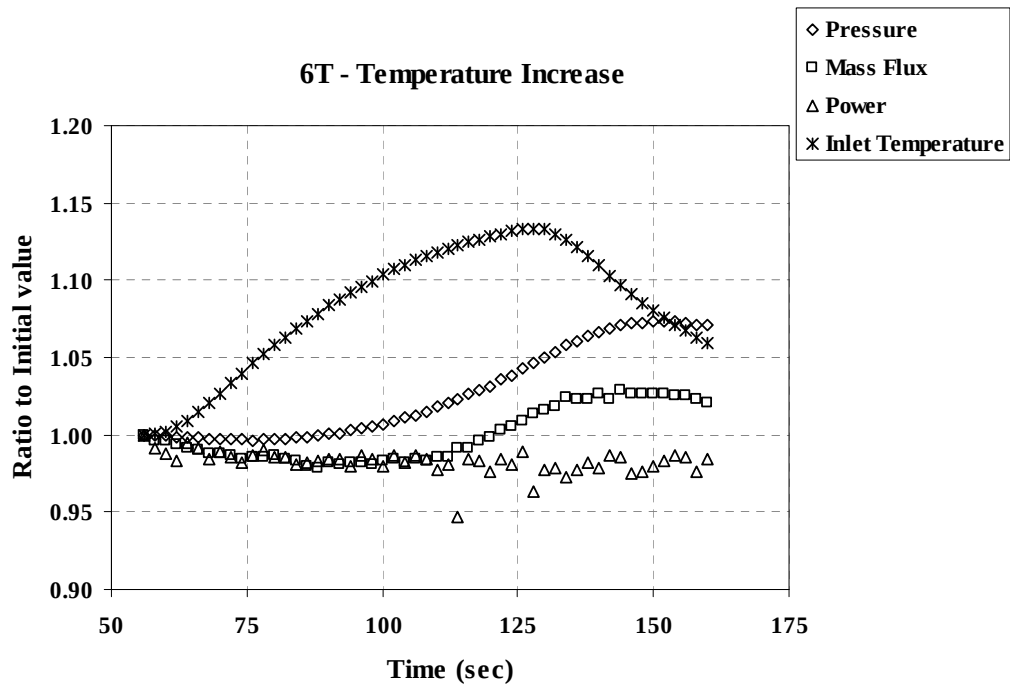


Table 4.2.3.10 Transient Void Fraction in Rod Bundle in Test Series 7T (Power Increase)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
32.0	158.2	12.02	2500.0	291.9
32.2	158.2	12.02	2504.0	291.9
32.4	158.2	12.02	2503.0	291.9
32.6	158.2	12.02	2492.0	291.9
32.8	158.2	12.02	2490.0	291.9
33.0	158.2	12.03	2513.0	291.9
33.2	158.2	12.02	2561.0	291.9
33.4	158.2	12.01	2617.0	292.0
33.6	158.2	12.01	2680.0	292.1
33.8	158.2	11.99	2741.0	292.1
34.0	158.2	11.98	2803.0	292.1
34.2	158.2	11.96	2862.0	292.1
34.4	158.3	11.96	2913.0	292.2
34.6	158.3	11.95	2962.0	292.2
34.8	158.3	11.93	3027.0	292.3
35.0	158.4	11.91	3083.0	292.3
35.2	158.4	11.89	3141.0	292.3
35.4	158.5	11.88	3234.0	292.3
35.6	158.5	11.86	3294.0	292.4
35.8	158.6	11.85	3346.0	292.3
36.0	158.6	11.86	3408.0	292.3
36.2	158.7	11.84	3445.0	292.2
36.4	158.8	11.84	3444.0	292.3
36.6	158.8	11.83	3443.0	292.3
36.8	158.8	11.82	3451.0	292.3
37.0	158.9	11.81	3451.0	292.3
37.2	158.9	11.82	3448.0	292.3
37.4	158.9	11.83	3417.0	292.3
37.6	159.0	11.85	3421.0	292.4
37.8	159.0	11.84	3445.0	292.4
38.0	159.0	11.84	3450.0	292.4
38.2	159.1	11.83	3462.0	292.3
38.4	159.1	11.83	3448.0	292.3
38.6	159.1	11.82	3438.0	292.3
38.8	159.1	11.83	3455.0	292.2
39.0	159.2	11.83	3457.0	292.3
39.2	159.2	11.82	3468.0	292.4
39.4	159.2	11.83	3465.0	292.4
39.6	159.2	11.82	3433.0	292.4
39.8	159.3	11.82	3434.0	292.3
40.0	159.3	11.84	3435.0	292.3
40.2	159.3	11.85	3446.0	292.3
40.4	159.4	11.85	3450.0	292.4
40.6	159.4	11.85	3437.0	292.4
40.8	159.4	11.84	3441.0	292.4
41.0	159.5	11.84	3443.0	292.4
41.2	159.5	11.84	3444.0	292.4
41.4	159.5	11.84	3440.0	292.4
41.6	159.6	11.84	3435.0	292.4
41.8	159.6	11.84	3439.0	292.4
42.0	159.6	11.85	3467.0	292.4

Table 4.2.3.11 Transient Void Fraction in Rod Bundle in Test Series 7T (Flow Reduction)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
157.0	158.1	12.04	2405.0	292.0
157.2	158.1	12.03	2426.0	292.0
157.4	158.1	12.04	2435.0	292.0
157.6	158.1	12.03	2433.0	292.0
157.8	158.1	11.77	2432.0	292.0
158.0	158.0	11.30	2418.0	292.0
158.2	157.9	10.79	2427.0	292.1
158.4	157.8	10.38	2430.0	292.2
158.6	157.7	10.13	2427.0	292.1
158.8	157.7	9.76	2430.0	292.2
159.0	157.5	9.21	2432.0	292.3
159.2	157.5	8.63	2431.0	292.4
159.4	157.5	8.20	2429.0	292.6
159.6	157.5	7.94	2432.0	292.6
159.8	157.6	7.69	2432.0	292.7
160.0	157.6	7.38	2441.0	295.7
160.2	157.6	7.04	2439.0	292.8
160.4	157.6	6.79	2426.0	292.7
160.6	157.7	6.66	2426.0	292.8
160.8	157.7	6.64	2428.0	292.8
161.0	157.8	6.69	2415.0	292.8
161.2	157.8	6.75	2414.0	292.8
161.4	157.9	6.79	2415.0	292.8
161.6	157.9	6.82	2420.0	292.8
161.8	158.0	6.89	2421.0	292.7
162.0	158.0	7.00	2425.0	292.8
162.2	158.1	7.11	2419.0	292.6
162.4	158.2	7.20	2431.0	292.7
162.6	158.2	7.25	2430.0	292.7
162.8	158.2	7.27	2425.0	292.8
163.0	158.3	7.27	2417.0	292.8
163.2	158.3	7.26	2420.0	292.9
163.4	158.4	7.26	2421.0	292.8
163.6	158.4	7.25	2428.0	292.7
163.8	158.4	7.26	2425.0	292.7
164.0	158.5	7.26	2417.0	292.7
164.2	158.5	7.27	2423.0	292.8
164.4	158.6	7.27	2416.0	292.7
164.6	158.6	7.27	2426.0	292.7
164.8	158.6	7.27	2420.0	292.8
165.0	158.7	7.27	2418.0	292.8
165.2	158.8	7.27	2418.0	292.8
165.4	158.8	7.26	2427.0	292.8
165.6	158.9	7.26	2433.0	292.8
165.8	158.9	7.27	2432.0	292.9
166.0	159.0	7.27	2431.0	293.0
166.2	159.1	7.26	2432.0	293.1
166.4	159.1	7.27	2435.0	293.1
166.6	159.2	7.28	2436.0	293.1
166.8	159.2	7.28	2428.0	293.2
167.0	159.3	7.27	2432.0	293.2

Figure 4.2.3.9 Variation of Properties during Transient for Data Series 7T (Power Increase)

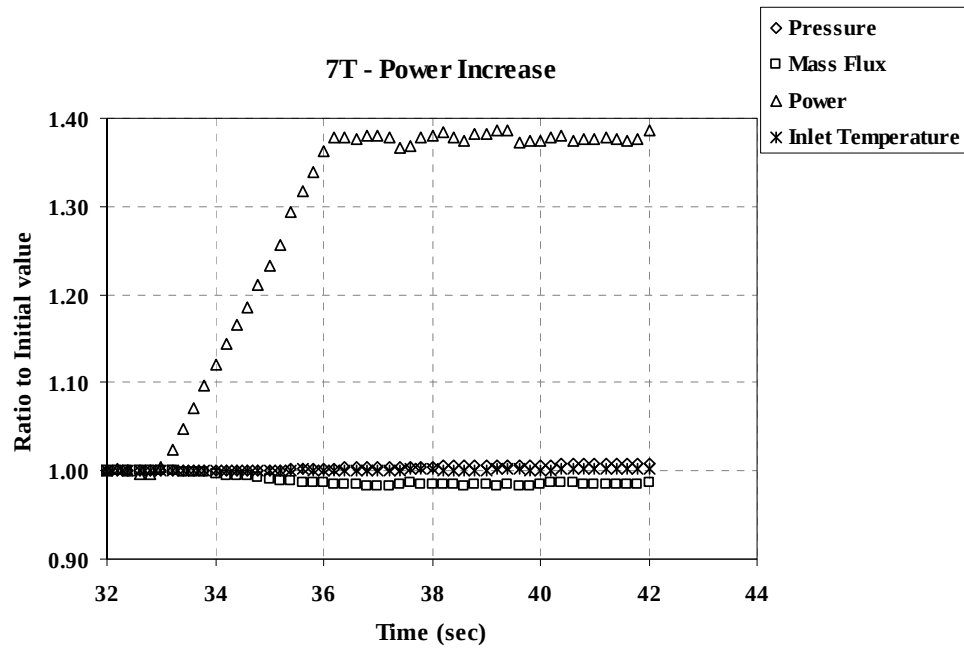


Figure 4.2.3.10 Variation of Properties during Transient for Data Series 7T (Flow Reduction)

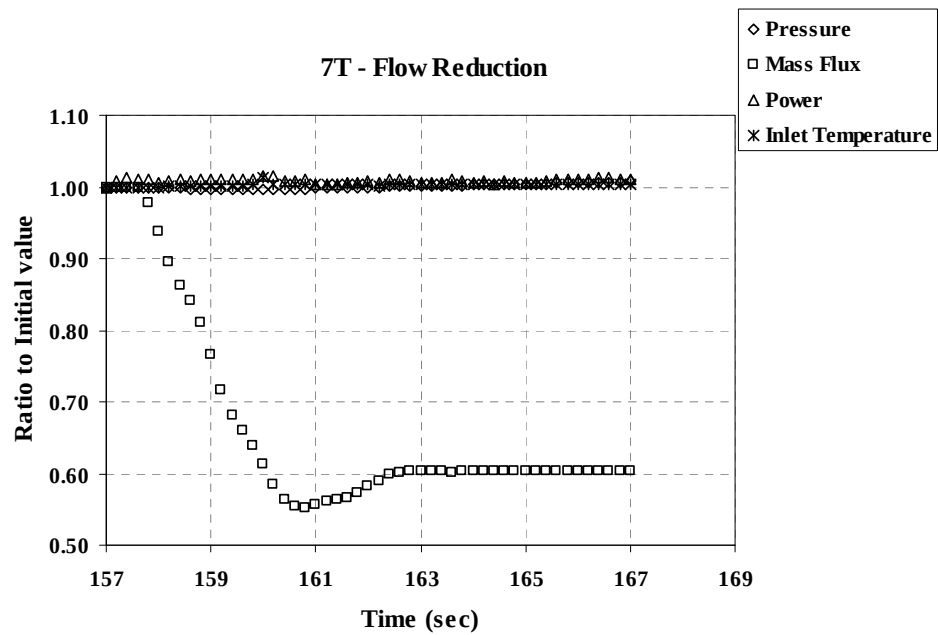


Table 4.2.3.12 Transient Void Fraction in Rod Bundle in Test Series 7T (Depressurization)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
6.0	155.0	11.99	2577.0	291.8
9.0	154.8	11.97	2592.0	291.8
12.0	154.4	11.98	2582.0	291.7
15.0	153.8	11.97	2578.0	291.6
18.0	153.2	11.99	2574.0	291.7
21.0	152.6	11.98	2577.0	291.7
24.0	151.9	11.95	2564.0	291.7
27.0	151.3	11.97	2581.0	291.7
30.0	150.7	11.96	2573.0	291.6
33.0	150.1	11.95	2558.0	291.6
36.0	149.3	11.93	2573.0	291.7
39.0	148.6	11.93	2573.0	291.7
42.0	147.9	11.93	2567.0	291.7
45.0	147.1	11.93	2566.0	291.8
48.0	146.2	11.93	2570.0	291.9
51.0	145.2	11.92	2575.0	292.2
54.0	144.2	11.92	2580.0	292.4
57.0	143.1	11.94	2585.0	292.6
60.0	142.1	11.96	2581.0	292.4
63.0	141.0	11.93	2541.0	292.1
66.0	140.0	11.92	2570.0	291.9
69.0	139.0	11.93	2582.0	291.5
72.0	138.0	11.91	2570.0	291.1
75.0	137.0	11.92	2562.0	290.6
78.0	136.0	11.90	2567.0	290.4
81.0	135.0	11.91	2566.0	290.1
84.0	134.1	11.91	2576.0	290.1
87.0	133.2	11.89	2584.0	290.3
90.0	132.3	11.91	2579.0	290.5
93.0	131.5	11.90	2570.0	290.5
96.0	130.5	11.92	2565.0	290.6
99.0	129.7	11.90	2580.0	290.5
102.0	129.0	11.91	2579.0	290.3
105.0	128.0	11.88	2574.0	290.1
108.0	127.2	11.91	2575.0	289.8
111.0	126.4	11.89	2587.0	289.5
114.0	125.6	11.91	2576.0	289.3
117.0	124.8	11.90	2581.0	289.1
120.0	124.0	11.90	2582.0	288.9
123.0	123.2	11.86	2566.0	288.8
126.0	122.5	11.89	2570.0	288.6
129.0	121.7	11.88	2548.0	288.5
132.0	120.9	11.87	2575.0	288.3
135.0	120.2	11.87	2573.0	288.2
138.0	119.5	11.85	2600.0	288.2
141.0	118.8	11.85	2578.0	288.1
144.0	118.1	11.83	2574.0	288.1
147.0	117.5	11.82	2578.0	288.2
150.0	116.9	11.82	2575.0	288.4
153.0	116.3	11.79	2585.0	288.3
156.0	115.7	11.78	2574.0	288.6
159.0	115.2	11.75	2570.0	288.7
162.0	114.7	11.74	2579.0	288.8

Table 4.2.3.13 Transient Void Fraction in Rod Bundle in Test Series 7T (Temperature Increase)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m² hr)	Power (kW)	Inlet Temperature (°C)
50.0	158.8	11.99	2496.0	290.2
52.0	158.8	11.97	2509.0	291.5
54.0	158.9	11.93	2509.0	292.5
56.0	158.9	11.92	2514.0	293.5
58.0	158.9	11.90	2494.0	294.3
60.0	159.0	11.91	2502.0	295.1
62.0	159.1	11.94	2501.0	296.2
64.0	159.1	11.90	2507.0	297.9
66.0	159.2	11.90	2500.0	300.2
68.0	159.3	11.91	2501.0	303.0
70.0	159.6	11.90	2505.0	305.7
72.0	159.5	11.93	2512.0	309.0
74.0	159.7	11.88	2504.0	312.2
76.0	160.1	11.91	2508.0	315.1
78.0	160.5	11.93	2512.0	318.1
80.0	161.1	11.95	2490.0	320.5
82.0	161.8	12.13	2484.0	322.6
84.0	162.5	12.12	2480.0	323.7
86.0	163.3	12.02	2497.0	324.6
88.0	164.1	12.08	2483.0	325.5
90.0	164.8	12.02	2503.0	326.3
92.0	165.7	12.05	2483.0	327.2
94.0	166.5	12.04	2494.0	328.0
96.0	167.3	12.04	2501.0	328.9
98.0	168.2	12.04	2486.0	329.5
100.0	169.0	12.04	2486.0	329.1
102.0	169.8	12.04	2501.0	327.8
104.0	170.5	12.02	2487.0	325.8
106.0	171.2	12.00	2500.0	323.6
108.0	171.7	12.00	2503.0	321.1
110.0	172.2	12.00	2503.0	319.1
112.0	172.5	11.98	2498.0	317.3
114.0	172.8	11.94	2509.0	315.6
116.0	173.1	11.99	2500.0	314.2
118.0	173.2	11.97	2478.0	313.1
120.0	173.3	11.93	2512.0	311.9
122.0	173.4	11.94	2516.0	311.1
124.0	173.5	11.98	2503.0	310.1
126.0	173.5	11.91	2502.0	309.2
128.0	173.5	11.92	2503.0	308.4
130.0	173.5	11.96	2503.0	307.7
132.0	173.4	11.93	2500.0	307.0
134.0	173.4	11.94	2505.0	306.4
136.0	173.3	11.93	2501.0	305.8
138.0	173.2	11.94	2525.0	305.1
140.0	173.2	11.92	2513.0	304.7
142.0	173.1	11.89	2503.0	304.1
144.0	173.0	11.92	2493.0	303.5
146.0	172.9	11.95	2491.0	303.1
148.0	172.8	11.95	2502.0	302.5

Figure 4.2.3.11 Variation of Properties during Transient for Data Series 7T (Depressurization)

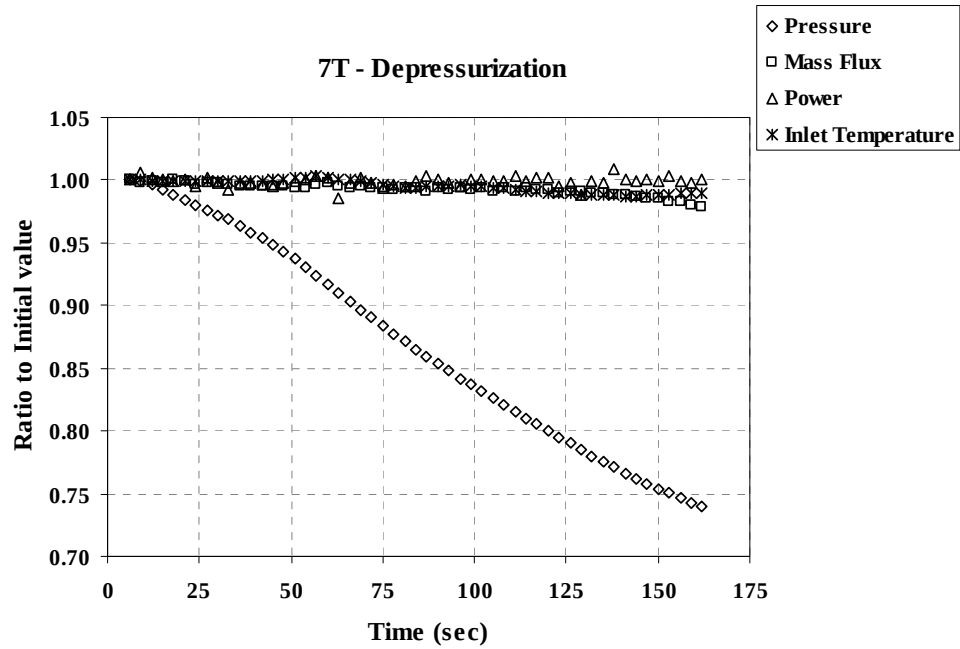
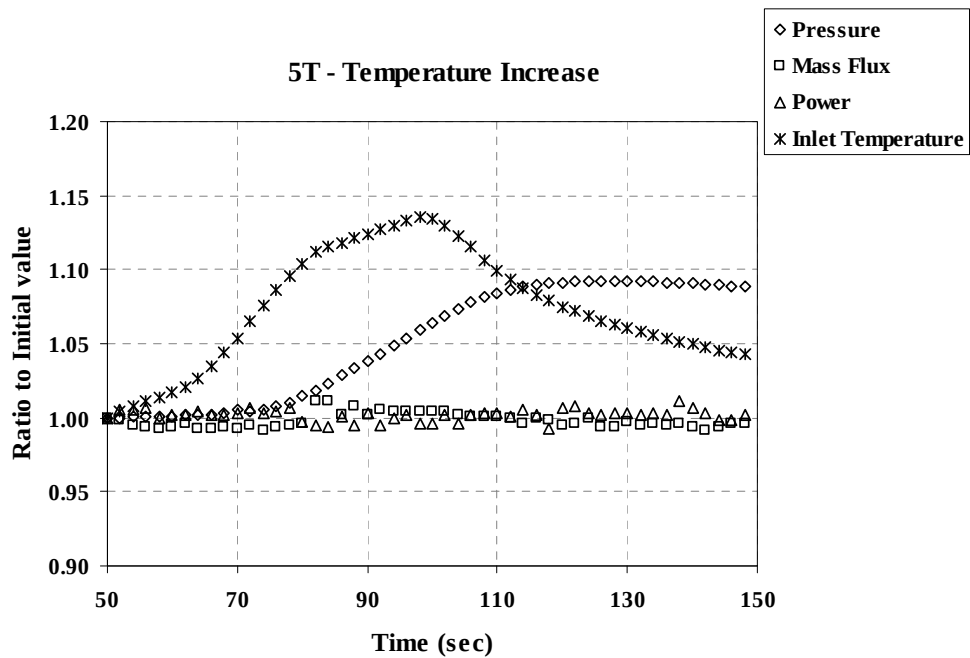


Figure 4.2.3.12 Variation of Properties during Transient for Data Series 7T (Temperature Increase)



4.2.4 Exercise I-4 – Pressure Drop Benchmark

As previously mentioned, no data is available for code-to-data pressure drop comparison. However, code-to-code comparisons will be performed on the pressure drop calculations for the typical central subchannel. The boundary conditions include both single-phase and two-phase conditions. Exercise cases to be calculated correspond to Test Series 1 of the steady-state central subchannel void measurements as given in Table 4.2.1.1. This benchmark will assess the wall friction and the two-phase multiplier models of the codes used.

Code-to-code comparisons will also be performed on the pressure drop calculations for the bundle. Exercise cases to be calculated correspond to Test Series 5 of the steady-state bundle void measurements as given in Table 4.2.1.4. This benchmark will assess the pressure loss and the two-phase multiplier models of the codes used.

4.3 Phase II: DNB Benchmark

Table 4.3.1 summarizes the test series performed for the DNB measurements in Exercises 1, 2, and 3 of Phase 2.

Table 4.3.1 Test Series for DNB Measurement

Test series	Test section	Assembly	Test mode		Measurement	
			Steady-State	Transient	DNB	Fluid temperature
0	5×5	A0	Y		Y	
1		A1	Y			Y
2		A2	Y		Y	
3	6×6	A3	Y		Y	
4	5×5	A4	Y		Y	
8		A8	Y		Y	
11T		A11		Y	Y	
12T		A12		Y	Y	
13		A4	Y		Y	

4.3.1 Exercise II-1 – Steady-State Fluid Temperature Benchmark

In addition to the void distribution and DNB power measurements; data is available for the subchannel exit fluid temperature. The data can be applied to both subchannel/system and CFD codes.

The highlighted cases are selected to be analyzed.

Table 4.3.1.1 Test Conditions for Steady-State Fluid Temperature Benchmark

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet temperature (°C)	Power (MW)
01-3233	100.0	4.62	166.1	0.98
01-3232	100.1	1.94	164.4	0.41
01-3239	100.1	1.24	165.3	0.25
01-3231	100.1	0.44	166.2	0.11
01-5343	150.3	5.03	165.3	1.25
01-5342	150.0	1.92	164.5	0.52
01-5242	149.9	1.92	185.1	0.41
01-5243	150.3	4.89	184.7	0.98
01-5143	150.2	4.66	202.9	0.70
01-5333	150.5	4.60	211.6	1.24
01-5233	150.4	4.91	230.9	0.97
01-5235	150.0	10.97	230.3	2.11
01-5325	150.3	10.99	255.7	2.68
01-5225	150.2	10.97	272.6	2.09
01-5215	150.3	10.95	282.9	2.09
01-5125	150.3	10.94	289.2	1.50
01-5123	150.8	4.91	288.5	0.70
01-5213	150.0	4.98	282.3	0.97
01-5223	150.1	4.99	272.2	0.97
01-5222	150.1	1.99	273.0	0.40
01-5212	150.1	1.97	282.2	0.40
01-5322	150.1	1.96	255.7	0.51
01-5133	150.0	4.98	247.1	0.70
01-5323	150.1	4.97	254.7	1.23
01-5135	150.0	11.03	247.0	1.50
01-5237	150.1	16.95	229.4	3.23
01-5335	150.1	11.01	211.7	2.71
01-5145	150.1	11.03	203.3	1.53
01-5245	150.1	11.06	184.4	2.12
01-5345	150.0	11.08	165.1	2.72
01-3225	100.3	11.07	224.5	2.12
01-3215	100.2	11.10	238.6	2.11
01-5231	150.0	0.69	230.9	0.11
01-6231	169.1	0.54	252.9	0.11
01-6239	169.0	1.18	254.5	0.26
01-5239	151.7	1.23	231.6	0.26
01-5332	150.0	2.00	211.5	0.53
01-5232	150.1	2.04	231.0	0.42
01-6232	169.1	2.10	251.5	0.42
01-6233	169.1	4.90	254.0	1.02
01-6235	169.2	11.00	253.0	2.20
01-6237	169.1	17.08	253.0	3.39
01-3235	101.1	11.05	164.8	2.24
01-3237	101.0	17.06	164.9	3.42
01-1231	50.8	0.73	98.6	0.11
01-1239	50.8	1.15	84.5	0.25

Table 4.3.1.1 Test Conditions for Steady-State Fluid Temperature Benchmark – Cont.

Test number	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m² hr)	Inlet temperature (°C)	Power (MW)
01-1232	50.9	2.00	87.2	0.41
01-1233	50.1	5.03	85.0	1.01
01-1235	50.0	11.05	86.0	2.24
01-1237	50.2	17.00	86.0	3.44
01-3245	100.1	10.99	103.6	2.22
01-5255	150.1	11.02	113.8	2.22
01-5355	150.0	11.03	94.3	2.85
01-5155	150.1	11.02	133.2	1.59
01-5153	150.1	5.05	132.9	0.72
01-5253	150.1	5.02	113.3	1.01
01-5353	150.0	5.03	93.4	1.31
01-5352	150.0	1.99	93.3	0.52
01-5252	150.0	1.95	113.9	0.41

4.3.2 Exercise II-2 – Steady-State DNB Benchmark

The experimental data include the power at which departure from nucleate boiling occurs and the corresponding location in the bundle (for selected test cases only). The data can be applied to thermal-hydraulic subchannel and system codes and CFD codes.

The highlighted cases are selected to be analyzed.

Table 4.3.2.1 Test Conditions for Steady-State DNB Test Series 0

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet Temperature (°C)
00-3230	100.4	5.07	243.2
00-3240	100.7	8.13	242.7
00-3260	101.1	14.28	242.9
00-3270	101.0	17.20	240.9
00-4270	125.8	16.78	262.7
00-4260	126.3	14.27	262.7
00-4240	125.8	8.14	262.4
00-4230	125.2	5.07	263.0
00-5330	150.4	5.08	253.9
00-5340	150.3	8.09	253.1
<u>00-5350</u>	150.1	11.23	252.9
00-5360	150.7	14.28	252.8
00-5370	150.3	17.13	253.3
00-3550	102.6	11.28	147.1
00-4550	125.6	11.27	170.5
00-3460	103.7	14.36	179.8
00-3440	100.1	8.22	179.8
00-3430	100.1	5.12	180.1
00-4430	125.7	5.11	203.6
<u>00-4440</u>	125.6	8.09	202.4
<u>00-4460</u>	125.4	14.20	202.0
00-5560	150.7	14.23	191.8
00-5550	150.5	11.23	191.9
00-5540	152.8	8.19	192.7
00-5530	150.3	5.15	192.0
00-3770	100.8	17.18	193.4
00-3350	99.5	11.14	210.4
00-4770	126.7	17.13	216.0
<u>00-5450</u>	150.7	11.13	223.4
00-3551	103.1	11.31	147.2
00-4551	125.6	11.30	170.2
00-5551	150.1	11.26	192.1
00-6550	169.2	11.23	208.5
00-5451	151.8	11.14	224.0
00-5870	151.5	17.17	227.3
<u>00-6470</u>	169.8	17.16	238.6
00-6460	171.3	14.28	238.7

Table 4.3.2.1 Test Conditions for Steady-State DNB Test Series 0 – Cont.

Test number	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m²hr)	Inlet Temperature (°C)
00-6440	171.0	8.13	238.6
00-6430	169.5	5.12	237.9
00-5331	150.4	5.13	253.4
00-5351	150.5	11.19	252.4
00-5371	150.5	17.21	252.7
00-5352	150.2	11.15	252.2
00-6350	169.3	11.12	278.6
00-6270	170.6	17.09	295.4
00-6260	169.3	14.16	294.4
<u>00-6240</u>	168.9	8.12	295.4
<u>00-6230</u>	169.2	5.12	296.2
00-5250	150.6	11.16	281.8
00-7680	159.7	12.23	287.1
00-5251	150.4	11.80	281.5
00-7681	159.8	12.22	288.2
00-4150	124.9	11.10	305.1
00-5130	150.0	5.08	321.6
00-5140	150.5	8.09	321.2
00-5150	150.7	11.09	321.3
00-5160	150.4	14.15	321.0
<u>00-5170</u>	150.8	17.80	321.4
00-6150	168.9	11.06	331.9
00-5151	150.5	11.09	320.9
00-6151	169.3	11.07	332.4
00-4151	125.5	11.10	304.8
00-3150	101.2	11.20	287.4
00-5252	150.6	11.14	281.0
00-3151	100.3	11.19	286.9
00-4350	125.3	11.20	232.5
00-3351	100.2	11.18	211.7
00-4351	125.3	10.95	232.8
<u>00-3352</u>	100.2	10.93	211.7
00-6551	169.1	10.92	208.8

Table 4.3.2.2 Test Conditions for Steady-State DNB Test Series 2

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet Temperature (°C)
02-2430	75.9	5.03	154.4
02-3430	100.0	5.01	181.0
02-3440	100.5	7.98	178.9
02-2450	76.9	10.90	154.0
02-1450	51.7	10.81	123.2
02-1440	50.9	8.05	123.1
02-1430	50.8	4.97	125.6
02-1420	50.3	2.10	122.6
02-1490	53.5	1.24	128.4
02-4230	125.3	4.89	264.0
02-3230	100.3	4.97	243.4
02-3240	100.7	7.98	243.0
02-3260	100.5	14.13	241.9
02-3270	100.3	16.83	241.9
02-4350	125.7	10.97	233.6
02-4370	125.3	17.04	232.6
02-4270	125.4	16.98	262.9
02-5270	150.0	16.92	281.8
02-7680	158.6	12.03	289.9
02-5250	150.3	10.95	282.3
02-5340	150.0	7.98	253.1
02-5330	149.9	4.93	254.9
02-3420	100.3	2.02	186.4
02-3490	100.5	1.23	175.1
02-1220	50.4	2.09	190.3
02-1230	50.6	4.98	190.8
02-1240	50.9	8.00	189.5
02-1250	52.7	11.03	188.7
02-1260	52.5	14.03	189.0
02-1360	52.7	14.01	155.7
02-1270	53.5	16.89	188.8
02-2270	76.8	16.97	219.5
02-2250	77.4	11.01	219.2
02-1150	52.3	11.06	238.2
02-2150	77.7	11.03	265.4
02-3150	101.6	10.98	287.3
02-3241	100.5	7.98	243.0
02-2230	76.7	4.93	220.1
02-1290	49.4	1.14	200.9
02-3290	100.3	1.18	245.5
02-3220	99.8	1.96	244.7
02-4150	125.5	10.86	306.1
02-5251	150.7	10.93	282.4
02-7681	158.6	12.03	288.9
02-6330	169.8	5.02	268.0

Table 4.3.2.2 Test Conditions for Steady-State DNB Test Series 2 – Cont.

Test number	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m²hr)	Inlet Temperature (°C)
02-6350	170.1	10.88	266.9
02-6370	169.7	16.90	267.4
02-5370	150.6	16.94	254.0
02-5360	150.7	14.04	252.9
02-5350	150.7	11.08	253.0
02-5290	150.0	1.20	274.8
02-5220	150.3	2.03	270.3
02-5130	150.1	4.92	322.4
02-5140	150.6	7.87	321.5
02-5490	151.5	1.27	225.9
<u>02-5420</u>	151.3	2.00	229.2
02-3242	100.9	7.97	242.9
02-5252	149.2	10.96	282.7
02-7682	157.2	11.98	288.5
02-5170	149.2	17.17	321.7
02-5160	150.2	13.94	322.7
02-5150	150.7	10.82	322.5
<u>02-6150</u>	169.5	10.89	333.8
02-6250	170.8	10.91	296.2
02-6450	169.4	10.89	238.8
02-6550	169.0	10.95	206.4
02-5450	149.7	10.92	222.2
02-5460	151.0	13.86	222.7
02-5550	150.1	10.96	189.6
02-5540	150.2	8.02	189.5
<u>02-5530</u>	150.3	5.03	190.3
<u>02-4430</u>	126.0	4.95	201.8
<u>02-3350</u>	101.1	10.76	212.4
02-3460	102.2	14.00	178.5
02-4550	124.9	11.03	168.3
02-3550	100.8	10.97	145.7

Table 4.3.2.3 Test Conditions for Steady-State DNB Test Series 3

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet Temperature (°C)
03-3430	100.3	5.03	180.4
03-3440	100.3	8.05	179.6
03-4550	126.2	11.05	168.8
03-3350	100.6	11.02	210.8
03-3360	99.9	13.77	209.9
03-4350	125.5	10.71	233.3
03-4360	125.4	13.71	232.5
03-4370	126.0	16.76	233.4
03-5460	150.6	13.61	222.5
03-5330	150.3	4.95	254.0
03-5340	150.2	8.02	253.3
03-5250	150.3	11.00	282.5
03-7680	158.1	12.06	289.2
03-5350	150.5	11.02	253.2
03-5360	150.5	13.76	252.7
03-5370	150.7	16.93	252.8
03-5270	150.9	16.72	282.6
03-4270	125.5	16.71	262.8
03-4260	125.1	13.87	262.8
03-4240	125.1	7.99	263.3
03-4230	125.1	4.99	263.9
03-3230	100.1	5.00	243.5
03-3240	100.4	7.94	242.8
03-3260	100.5	13.74	241.5
03-3270	100.2	16.87	242.9
03-3550	100.9	10.83	161.4
03-5530	150.0	5.01	190.9
03-5540	150.3	7.98	190.4
03-4430	125.1	5.02	201.5
03-4440	125.2	7.99	201.4
03-3351	101.0	10.82	212.1
03-5550	150.2	10.92	190.3
03-5450	150.0	10.96	222.4
03-6460	169.6	13.77	238.5
03-6450	169.3	10.73	238.9
03-6440	169.8	7.98	238.5
03-6430	168.8	5.02	238.5
03-6330	169.2	4.96	268.6
03-6350	169.7	10.97	268.4
03-5351	150.5	11.03	252.1
03-5251	150.5	10.97	282.5
03-5352	150.6	11.06	252.9
03-6370	169.8	16.53	268.8
03-6270	169.5	16.62	297.0
03-6260	169.2	13.69	296.7

Table 4.3.2.3 Test Conditions for Steady-State DNB Test Series 3 – Cont.

Test number	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m²hr)	Inlet Temperature (°C)
03-5252	150.3	10.76	282.8
03-6240	169.3	7.91	296.5
03-6230	169.3	4.94	296.9
03-5130	150.3	4.90	323.0
03-5140	150.3	7.79	322.8
03-5150	150.6	10.92	323
<u>03-6150</u>	168.9	10.95	334.5
03-5160	150.4	13.66	322.5
03-5170	150.4	16.71	322.7
03-4150	125.3	10.77	306.2
03-3150	100	10.78	287.8
03-3352	100.3	10.73	211.5

Table 4.3.2.4 Test Conditions for Steady-State DNB Test Series 4

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet Temperature (°C)
04-3330	100.4	4.92	212.0
04-4330	125.1	4.93	233.1
04-3230	100.2	4.91	242.7
04-3240	100.5	7.93	242.0
04-3250	100.2	10.89	241.5
04-3760	100.5	14.09	262.4
04-3750	100.3	10.97	262.3
04-3730	99.9	4.95	263.8
04-4230	125.2	4.96	263.3
04-4250	125.3	10.92	261.8
04-4760	125.3	13.98	282.5
04-4750	125.3	10.95	282.6
04-4730	125.4	4.97	283.4
04-5230	150.1	4.91	281.8
04-5240	150.7	7.87	281.4
04-5250	150.4	10.87	281.5
04-5760	150.3	13.86	300.5
04-5750	150.5	10.84	300.7
04-5730	150.1	4.92	300.9
04-5130	150.1	4.89	321.5
04-5140	150.1	7.86	321.7
04-5150	150.0	10.87	321.3
04-5160	150.2	13.92	321.1
04-5170	150.4	16.78	321.0
04-6170	169.4	16.84	332.9
04-6150	169.3	10.85	332.9
04-6750	169.2	10.83	314.3
04-6250	169.3	10.88	295.0
04-6230	169.4	4.94	295.9
04-7680	158.5	11.94	288.2
04-4150	125.2	10.81	304.8
04-3150	100.3	10.83	286.2
04-3160	100.6	13.90	286.1
04-7681	158.3	12.00	287.2
04-5251	150.8	10.89	281.0
04-6330	169.1	4.94	267.5
04-5330	150.2	4.96	252.6
04-6430	169.4	4.94	237.5
04-5430	150.4	4.95	222.3
04-6530	169.4	4.92	205.9
04-4430	125.5	4.92	201.4
04-3241	100.6	7.93	241.3
04-2150	75.9	10.90	264.6
04-2750	75.3	10.97	236.9
04-2740	75.5	7.95	240.0
04-2730	75.4	4.95	240.8

Table 4.3.2.4 Test Conditions for Steady-State DNB Test Series 4 – Cont.

Test number	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m²hr)	Inlet Temperature (°C)
04-2230	75.6	4.96	219.8
04-1730	50.7	4.94	213.3
04-1230	50.3	4.98	190.3
04-2330	75.5	5.00	188.6
04-1220	53.5	2.16	192.1
04-1320	50.8	2.18	148.2
04-2320	75.7	2.15	188.1
04-3320	100.4	2.09	217.2
04-2220	75.4	2.09	218.8
04-3220	100.2	2.23	244.3
04-4320	124.9	2.08	238.2
04-4220	125.1	2.19	264.3
<u>04-5320</u>	150.3	2.04	257.8
<u>04-5220</u>	150.3	2.15	283.1
04-6770	169.3	16.81	314.6
04-6270	169.5	16.83	294.6
<u>04-5260</u>	150.5	13.94	281.1
<u>04-5270</u>	150.5	16.90	281.6
04-4260	125.5	14.05	261.3
04-4270	126.0	17.01	261.5
04-3770	100.8	17.00	262.1
04-6350	169.0	10.99	267.2
04-6450	168.8	10.96	237.0
<u>04-5350</u>	151.7	10.93	251.9
<u>04-5360</u>	150.5	14.06	251.7
<u>04-5440</u>	150.5	7.93	221.8
04-4350	125.5	10.88	232.0
04-3260	100.8	13.96	240.9
04-3350	100.5	10.86	210.2
04-2250	75.4	10.88	217.6

Table 4.3.2.5 Test Conditions for Steady-State DNB Test Series 8

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet Temperature (°C)
08-3330	100.5	5.05	211.8
08-4330	124.7	5.06	233.1
08-3230	100.2	5.07	242.5
08-3240	100.4	8.09	241.3
08-3250	100.6	11.22	241.3
08-3760	100.4	14.28	262.0
08-3750	100.5	11.09	262.1
08-3740	100.2	8.11	262.4
08-3730	100.2	5.05	263.4
08-4230	125.3	5.03	262.1
08-4240	125.2	8.08	261.9
08-4250	125.4	11.17	262.0
08-4760	125.1	14.21	282.4
08-4750	125.4	11.16	282.2
08-4740	125.2	8.11	282.4
08-4730	125.3	5.02	282.9
<u>08-5230</u>	150.6	4.98	281.6
<u>08-5240</u>	150.4	8.02	281.5
<u>08-5250</u>	149.9	11.05	280.9
08-7680	158.4	12.16	288.0
<u>08-5760</u>	150.1	14.15	300.7
<u>08-5750</u>	150.3	11.04	300.4
<u>08-5740</u>	150.4	8.00	300.5
<u>08-5730</u>	150.2	4.96	300.4
<u>08-5130</u>	150.1	4.95	321.6
<u>08-5140</u>	150.4	8.01	321.3
<u>08-5150</u>	150.2	11.03	321.0
<u>08-5160</u>	150.4	14.16	321.0
<u>08-5170</u>	150.4	17.10	320.9
08-6170	169.6	17.10	332.6
08-6150	168.5	11.04	332.5
08-6750	169.2	11.02	314.0
08-6250	168.9	11.01	295.5
08-6240	168.9	8.03	295.5
08-6230	169.1	4.99	295.6
08-7681	158.3	12.16	288.3
08-4150	125.2	11.04	304.6
08-4140	125.2	8.03	304.8
08-3140	100.6	8.06	286.4
08-3150	100.2	11.08	286.1
08-3160	100.2	14.25	285.9
<u>08-5251</u>	150.1	11.08	281.3
08-6340	169.2	8.06	267.3
08-6330	169.1	5.05	267.7
<u>08-5340</u>	150.4	8.08	252.6
<u>08-5330</u>	150.4	5.04	252.4

Table 4.3.2.5 Test Conditions for Steady-State DNB Test Series 8 – Cont.

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet Temperature (°C)
08-6430	169.1	5.01	237.6
08-5430	150.2	5.04	222.5
08-6530	169.2	5.00	206.4
08-4430	125.2	5.02	201.7
08-3241	100.5	8.09	241.6
08-2150	75.5	11.20	264.0
08-2750	75.3	11.20	239.3
08-2740	75.4	8.15	239.6
08-2730	75.2	5.04	240.8
08-2230	75.2	5.02	219.2
08-1730	50.2	5.02	212.8
08-1230	50.5	5.03	189.8
08-1220	50.2	2.10	189.3
08-1320	50.4	2.09	157.5
08-2320	75.1	2.09	187.8
08-2220	75.3	2.12	222.3
08-3320	100.5	2.12	208.0
08-3220	100.4	2.09	244.8
08-4320	124.8	2.11	230.4
08-4220	125.0	2.07	265.2
08-5320	150.3	2.12	248.7
08-5220	150.3	2.07	279.5
08-5252	150.2	11.13	281.5
08-7682	158.3	12.22	288.3
08-6770	169.5	17.20	314.4
08-6270	168.8	17.18	295.0
08-5260	150.4	14.25	281.3
08-5270	149.9	17.25	281.0
08-3242	100.3	8.06	241.4
08-4260	125.0	14.29	260.7
08-4270	125.8	17.33	262.0
08-6350	169.4	11.12	266.7
08-6450	169.1	11.14	237.0
08-7683	158.1	12.36	288.4
08-5350	150.5	11.35	251.9
08-5360	150.4	14.20	252.8
08-5440	150.0	8.09	222.6
08-4350	125.1	11.15	232.7
08-3770	100.5	17.34	262.2
08-3260	100.4	14.33	241.5
08-3350	100.3	11.14	210.9
08-2250	74.7	11.12	216.4
08-1150	50.2	11.20	237.5
08-1750	50.5	11.15	212.1
08-1250	50.3	11.15	188.4
08-2330	75.2	5.04	188.6
08-1330	50.2	5.08	158.2

Table 4.3.2.6 Test Conditions for Steady-State DNB Test Series 13

Test number	Pressure (kg/cm ² a)	Mass flux (10 ⁶ kg/m ² hr)	Inlet Temperature (°C)
13-4240	125.2	7.79	265.7
13-4250	125.5	10.94	263.6
13-A250	138.1	10.91	273.6
13-A240	137.3	7.93	274.1
13-5340	150.2	8.04	254.3
13-5D40	150.4	8.05	269.1
13-5230	150.4	4.90	283.7
13-52A0	150.1	6.42	285.2
13-5240	150.3	7.91	283.1
13-52B0	150.3	9.45	284.6
13-5250	150.1	10.90	282.8
13-4251	125.0	11.00	262.4
13-4241	125.1	7.80	265.8
13-A241	137.6	8.04	272.1
13-A251	137.9	10.79	275.2
13-5341	150.1	8.04	254.4
13-5D41	150.3	8.05	269.1
13-5231	150.0	4.92	283.9
13-52A1	150.0	6.41	282.2
13-5241	150.1	7.90	283.3
13-52B1	150.3	9.48	284.7
13-5251	150.5	10.95	283.3
13-B240	159.5	7.98	288.6
13-B250	159.7	10.71	291.9
13-6250	169.5	10.84	297.2
13-6240	169.3	7.98	297.2
13-5C40	150.0	7.76	293.1
13-5B40	150.1	7.74	302.6
13-5A40	150.3	7.95	313.3
13-5140	150.2	7.89	323.1
13-51B0	150.2	9.37	322.3
13-5150	150.2	10.84	322.8
13-51C0	150.3	12.32	321.0
13-5160	150.3	13.90	322.7
13-5A50	150.1	10.69	312.8
13-5B50	150.0	10.69	302.0
13-5C50	150.1	10.66	297.5
13-6251	169.0	10.80	297.0
13-6241	169.0	7.99	295.5
13-B251	159.8	10.67	292.0
13-B241	159.7	7.97	290.5
13-5C41	150.0	7.76	293.4
13-5B41	150.4	7.94	302.7
13-5A41	149.7	7.97	313.2
13-5C51	150.3	10.61	293.0
13-52C0	150.6	12.31	282.7

Table 4.3.2.6 Test Conditions for Steady-State DNB Test Series 13 – Cont.

Test number	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m²hr)	Inlet Temperature (°C)
13-5D50	150.1	10.66	269.0
13-5260	150.5	13.78	282.8
13-5230	150.1	10.95	254.0
13-52C1	150.5	12.38	282.3
13-5261	150.8	13.77	283.0
13-5D51	150.1	10.74	269.2
13-5141	150.4	7.86	320.2
13-51B1	150.6	9.35	318.1
13-5151	150.5	10.83	319.7
13-51C1	150.7	12.36	318.1
13-5161	151.4	13.89	319.6
13-5A51	150.0	10.69	310.2
13-5B51	150.2	10.67	299.3
13-5351	150.5	10.94	251.5

4.3.3 Exercise II-3 – Transient DNB Benchmark

The experimental data include the power at which departure from nuclear boiling occurs during four transient scenarios: power increase; flow reduction; depressurization; temperature increase. A summary of the test conditions for transient DNB test series 11T and 12T is given in Table 4.3.3.1. The data can be applied to subchannel and system codes.

Tables 4.3.3.2 through 4.3.3.9 contain the boundary conditions for each transient. Also included are plots of each parameter during the transient (Figures 4.3.3.1 through 4.3.3.8). These plots represent the ratio of the value of each parameter at a certain time to the value at the start of the transient.

All transient cases are to be analyzed.

Table 4.3.3.1 Test Conditions for Transient DNB Test Series 11T and 12T

Test Series	Assembly	Initial Conditions				Transients
		Power (MW)	Mass flux ($10^6 \text{kg/m}^2 \text{h}$)	Pressure ($\text{kg/cm}^2 \text{a}$)	Inlet temperature ($^{\circ}\text{C}$)	
11T	A4	2.50	11.18	156.2	291.0	Power increase
		2.50	11.19	156.1	293.1	Flow reduction
		2.52	11.28	156.3	291.7	Depressurization
		2.48	11.04	154.6	291.6	Temperature increase
12T	A8	2.51	11.40	156.1	291.3	Power increase
		2.51	11.71	156.3	292.5	Flow reduction
		2.50	11.42	156.2	290.6	Depressurization
		2.50	11.38	155.8	291.2	Temperature increase

Table 4.3.3.2 Transient DNB Data in Rod Bundle in Test Series 11T (Power Increase)

Time (sec)	Pressure (kg/cm²a)	Mass Flux (10⁶ kg/m²hr)	Inlet temperature (°C)	Power (MW)
61	156.2	11.18	291.0	2.50
62	156.3	11.16	290.9	2.52
63	156.3	11.18	290.9	2.55
64	156.4	11.16	291.0	2.58
65	156.8	11.18	291.0	2.61
66	156.6	11.16	291.0	2.64
67	156.5	11.16	291.0	2.67
68	156.4	11.16	290.9	2.70
69	156.3	11.16	291.0	2.73
70	156.3	11.14	291.0	2.75
71	156.5	11.14	291.0	2.78
72	156.8	11.13	291.0	2.81
73	156.9	11.11	291.0	2.83
74	156.8	11.14	291.0	2.86
75	157.0	11.13	291.0	2.88
76	157.0	11.10	291.0	2.91
77	156.8	11.13	291.0	2.94
78	156.8	11.12	291.0	2.98
79	156.8	11.13	291.0	3.01
80	157.0	11.10	291.0	3.04
81	157.1	11.12	291.0	3.05
82	157.2	11.11	291.0	3.08
83	157.4	11.10	291.0	3.11
84	157.3	11.11	291.0	3.14
85	157.5	11.14	291.0	3.17
86	157.2	11.09	291.0	3.21
87	157.6	11.07	291.0	3.23
88	157.8	11.10	291.0	3.26
89	157.3	11.06	291.0	3.29
90	157.4	11.06	291.0	3.33
91	157.5	11.06	291.0	3.35
92	157.6	11.07	291.0	3.39
93	157.9	11.08	291.0	3.41
94	157.6	11.06	291.0	3.44
95	157.7	11.05	291.0	3.46
96	158.1	11.05	291.0	3.50
97	157.7	11.03	291.0	3.53
98	157.9	11.06	291.0	3.55
99	157.6	11.05	291.0	3.58
100	157.6	11.04	291.0	3.61
101	157.9	11.03	291.0	3.64
102	157.5	11.02	291.0	3.67
103	157.6	11.00	291.0	3.69
104	157.9	11.00	291.0	3.72
105	157.6	11.01	291.0	3.75
106	157.6	11.00	291.0	3.78
107	157.9	10.99	291.0	3.81
108	157.6	10.95	290.9	3.83

Table 4.3.3.3 Transient DNB Data in Rod Bundle in Test Series 11T (Flow Reduction)

Time (sec)	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m² hr)	Inlet temperature (°C)	Power (MW)
20	156.1	11.19	293.1	2.50
21	156.3	11.19	293.1	2.50
22	156.4	11.19	293.1	2.50
23	156.3	11.16	293.1	2.50
24	156.3	10.88	293.2	2.50
25	156.2	10.58	293.2	2.50
26	156.1	10.34	293.2	2.51
27	156.3	10.20	293.2	2.50
28	156.1	10.13	293.2	2.50
29	156.5	10.05	293.1	2.50
30	156.1	9.89	293.1	2.50
31	156.1	9.66	293.2	2.50
32	156.5	9.46	293.1	2.50
33	156.3	9.26	293.2	2.50
34	156.4	9.13	293.2	2.49
35	156.3	8.99	293.2	2.49
36	156.3	8.82	293.2	2.49
37	156.5	8.57	293.2	2.50
38	156.6	8.31	293.2	2.50
39	156.4	8.10	293.2	2.50
40	156.6	7.98	293.2	2.50
41	156.8	7.90	293.2	2.50
42	156.9	7.76	293.3	2.50
43	157.1	7.49	293.3	2.50
44	157.2	7.21	293.3	2.49
45	156.9	6.98	293.3	2.49
46	156.9	6.89	293.3	2.50
47	157.3	6.77	293.3	2.49
48	157.3	6.53	293.3	2.49
49	157.2	6.36	293.2	2.50
50	157.4	6.31	293.3	2.50
51	157.7	6.25	293.2	2.50
52	157.6	6.05	293.2	2.50
53	157.7	5.75	293.2	2.50
54	158.1	5.51	293.2	2.50
55	157.8	5.34	293.2	2.50

Figure 4.3.3.1 Variation of Properties during Transient for Data Series 11T (Power Increase)

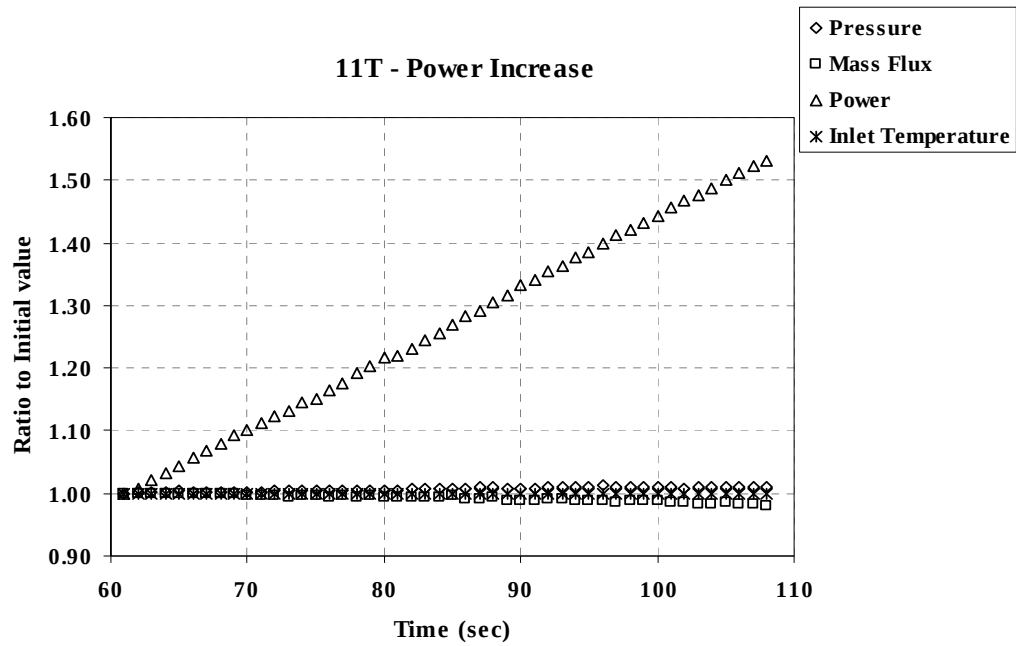


Figure 4.3.3.2 Variation of Properties during Transient for Data Series 11T (Flow Reduction)

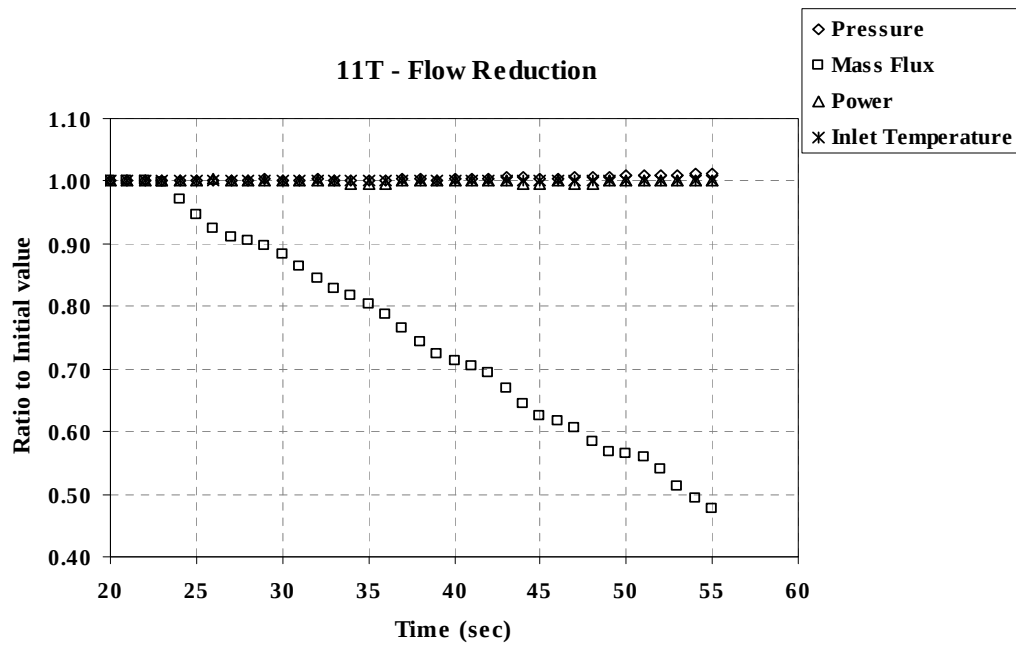


Table 4.3.3.4 Transient DNB Data in Rod Bundle in Test Series 11T (Depressurization)

Time (sec)	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m²hr)	Inlet temperature (°C)	Power (MW)
50	156.3	11.28	291.7	2.52
51	156.4	11.27	291.7	2.53
52	156.4	11.26	291.7	2.53
53	156.0	11.24	291.7	2.54
54	155.3	11.22	291.7	2.57
55	154.9	11.25	291.7	2.60
56	154.6	11.25	291.7	2.63
57	154.2	11.25	291.7	2.66
58	153.9	11.23	291.7	2.69
59	153.5	11.24	291.7	2.72
60	153.2	11.23	291.7	2.75
61	152.9	11.22	291.7	2.77
62	152.5	11.23	291.6	2.80
63	152.3	11.24	291.6	2.83
64	151.8	11.25	291.7	2.85
65	151.2	11.21	291.7	2.88
66	150.8	11.18	291.6	2.92
67	150.4	11.19	291.6	2.95
68	150.0	11.22	291.6	2.98
69	149.7	11.20	291.6	3.01
70	149.2	11.20	291.6	3.04
71	148.7	11.18	291.6	3.07
72	148.3	11.17	291.6	3.10
73	147.9	11.18	291.6	3.13
74	147.5	11.15	291.6	3.17
75	147.3	11.15	291.5	3.20
76	146.7	11.15	291.5	3.23
77	146.3	11.15	291.5	3.25
78	145.9	11.16	291.6	3.28
79	145.4	11.18	291.6	3.31
80	145.2	11.18	291.5	3.34
81	145.0	11.17	291.6	3.37
82	144.4	11.11	291.5	3.39
83	143.8	11.09	291.6	3.42
84	143.5	11.09	291.6	3.45
85	143.2	11.11	292.7	3.47
86	143.0	11.13	291.8	3.50
87	142.3	11.11	291.9	3.53
88	141.9	11.08	292.0	3.56
89	141.4	11.10	292.2	3.59
90	141.3	11.12	292.5	3.62
91	140.9	11.12	292.8	3.65
92	140.3	11.11	293.1	3.68
93	139.9	11.10	293.4	3.71
94	139.6	11.12	293.8	3.74
95	139.3	11.12	294.2	3.76
96	139.1	11.12	294.5	3.79

Table 4.3.3.5 Transient DNB Data in Rod Bundle in Test Series 11T (Temperature Increase)

Time (sec)	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m²hr)	Inlet temperature (°C)	Power (MW)
96	154.6	11.04	291.6	2.48
97	154.4	11.04	291.6	2.48
98	154.3	11.03	291.6	2.49
99	154.5	11.02	291.7	2.50
100	154.2	11.01	291.8	2.51
101	154.2	11.02	291.8	2.53
102	154.6	11.03	292.0	2.55
103	154.3	11.02	292.3	2.56
104	154.3	10.98	292.6	2.57
105	154.2	11.00	293.1	2.59
106	154.4	11.01	293.5	2.60
107	154.1	11.02	294.0	2.62
108	154.3	11.00	294.7	2.64
109	154.2	10.99	295.3	2.65
110	154.2	11.02	296.0	2.66
111	154.2	11.05	296.7	2.68
112	154.2	11.06	297.5	2.70
113	154.4	11.03	298.2	2.72
114	154.9	11.04	299.1	2.74
115	154.9	11.03	299.8	2.75
116	154.6	11.03	300.6	2.77
117	154.5	11.06	301.3	2.79
118	154.6	11.04	302.0	2.80
119	154.7	11.05	302.8	2.82
120	155.0	11.05	303.5	2.84
121	154.9	11.06	304.1	2.85
122	155.0	11.07	304.9	2.87
123	155.1	11.08	305.4	2.88
124	155.3	11.08	306.0	2.89
125	155.5	11.05	306.6	2.91
126	155.6	11.06	307.2	2.93
127	155.6	11.09	307.8	2.94
128	155.8	11.09	308.3	2.96
129	156.0	11.09	308.9	2.97
130	156.3	11.09	309.4	2.99
131	156.4	11.07	309.9	3.00
132	156.8	11.06	310.3	3.02
133	156.4	11.06	310.7	3.04
134	157.0	11.08	311.2	3.05
135	156.6	11.10	311.5	3.07
136	156.6	11.10	311.9	3.09
137	156.6	11.11	312.2	3.10
138	156.7	11.11	312.6	3.11
139	157.1	11.15	312.9	3.12
140	157.1	11.17	313.3	3.14
141	156.8	11.19	313.6	3.16
142	157.0	11.22	313.9	3.18
143	156.9	11.22	314.2	3.19
144	157.1	11.21	314.4	3.21
145	156.9	11.18	314.7	3.22
146	157.1	11.17	314.9	3.24
147	157.0	11.19	315.1	3.25
148	157.0	11.24	315.4	3.27
149	157.3	11.25	315.5	3.28
150	157.1	11.25	315.8	3.30
151	157.3	11.17	315.8	3.30

Figure 4.3.3.3 Variation of Properties during Transient for Data Series 11T (Depressurization)

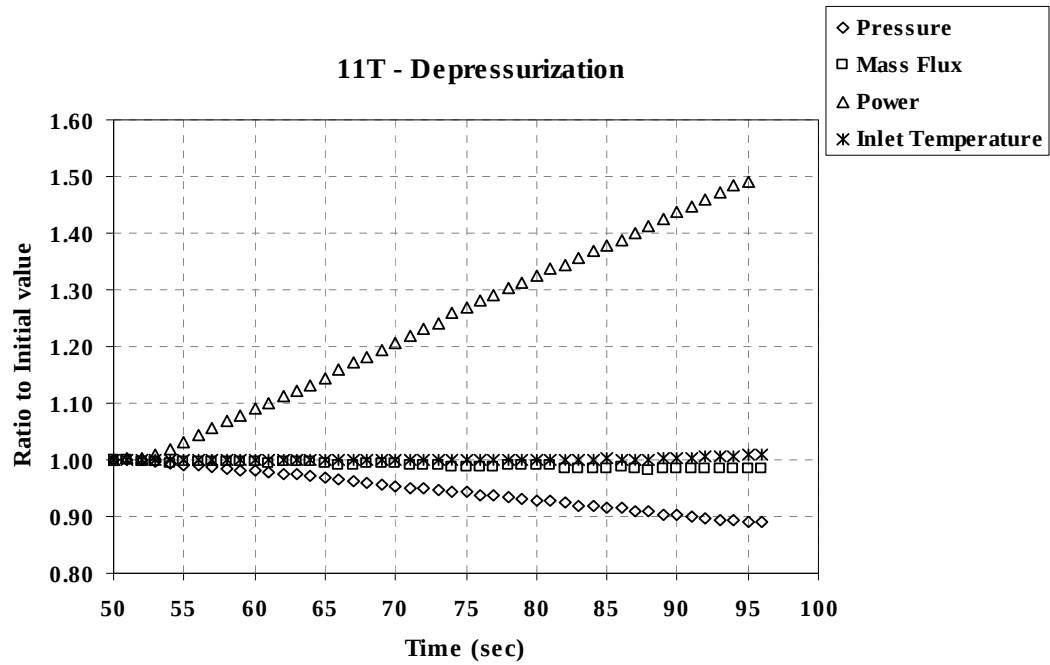


Figure 4.3.3.4 Variation of Properties during Transient for Data Series 11T (Temperature Increase)

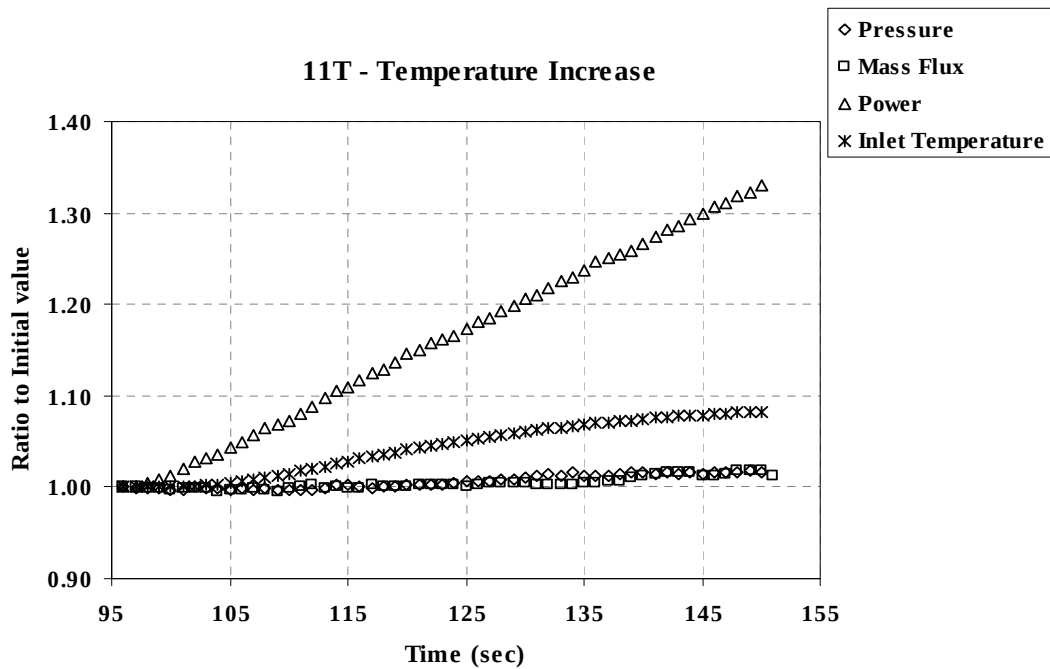


Table 4.3.3.6 Transient DNB Data in Rod Bundle in Test Series 12T (Power Increase)

Time (sec)	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m² hr)	Inlet temperature (°C)	Power (MW)
46	156.1	11.40	291.3	2.51
47	156.1	11.42	291.3	2.52
48	156.1	11.40	291.3	2.56
49	156.1	11.39	291.2	2.59
50	156.2	11.39	291.3	2.62
51	156.2	11.41	291.3	2.65
52	156.2	11.39	291.2	2.68
53	156.6	11.38	291.2	2.71
54	156.3	11.38	291.1	2.75
55	156.3	11.37	291.2	2.78
56	156.4	11.38	291.2	2.81
57	156.4	11.38	291.1	2.85
58	156.5	11.39	291.1	2.87
59	156.5	11.38	291.1	2.90
60	156.6	11.35	291.1	2.93
61	156.8	11.35	291.1	2.96
62	156.7	11.37	291.0	3.00
63	156.8	11.37	291.0	3.04
64	156.9	11.39	291.0	3.07
65	157.2	11.37	291.0	3.10
66	157.3	11.37	291.0	3.13
67	157.1	11.37	290.9	3.17
68	157.2	11.35	291.0	3.20
69	157.3	11.39	290.9	3.24
70	157.3	11.35	290.9	3.27
71	157.6	11.36	290.9	3.31
72	157.6	11.35	290.9	3.34
73	157.6	11.34	290.8	3.37
74	157.7	11.34	290.8	3.40
75	157.8	11.34	290.8	3.43
76	158.0	11.32	290.9	3.47
77	158.0	11.34	290.8	3.50
78	158.1	11.31	290.8	3.52
79	158.3	11.29	290.8	3.57
80	158.4	11.31	290.8	3.59
81	158.6	11.30	290.8	3.62
82	158.5	11.31	290.8	3.65
83	158.8	11.28	290.8	3.68
84	158.7	11.29	290.8	3.71
85	158.8	11.28	290.8	3.75
86	158.9	11.26	290.8	3.78
87	159.1	11.26	290.9	3.81
88	159.2	11.25	290.8	3.84

Table 4.3.3.7 Transient DNB Data in Rod Bundle in Test Series 12T (Flow Reduction)

Time (sec)	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m² hr)	Inlet temperature (°C)	Power (MW)
20	156.3	11.71	292.5	2.51
21	156.3	11.67	292.5	2.51
22	156.4	11.66	292.4	2.51
23	156.2	11.66	292.4	2.51
24	156.3	11.68	292.5	2.51
25	156.3	11.67	292.4	2.51
26	156.2	11.58	292.5	2.51
27	156.1	11.27	292.5	2.51
28	156.6	10.96	292.5	2.51
29	156.1	10.64	292.5	2.51
30	156.2	10.49	292.4	2.51
31	156.2	10.40	292.5	2.50
32	156.2	10.28	292.5	2.51
33	156.1	10.09	292.5	2.51
34	156.1	9.86	292.5	2.51
35	156.1	9.61	292.6	2.50
36	156.2	9.44	292.5	2.50
37	156.3	9.31	292.5	2.50
38	156.3	9.14	292.4	2.50
39	156.4	8.91	292.5	2.50
40	156.4	8.67	292.5	2.50
41	156.6	8.39	292.5	2.50
42	156.7	8.17	292.5	2.50
43	156.8	8.02	292.5	2.50
44	156.7	7.88	292.4	2.50
45	156.8	7.70	292.5	2.50
46	156.8	7.49	292.3	2.49
47	156.9	7.26	292.4	2.49
48	156.9	7.01	292.4	2.49
49	157.1	6.74	292.4	2.49
50	157.2	6.52	292.4	2.49
51	157.3	6.38	292.4	2.49
52	157.3	6.25	292.3	2.49
53	157.5	6.05	292.5	2.49
54	157.5	5.80	292.5	2.49
55	157.5	5.54	292.6	2.48
56	157.6	5.35	292.7	2.48
57	157.8	5.26	292.7	2.48
58	157.9	5.10	292.6	2.48

Figure 4.3.3.5 Variation of Properties during Transient for Data Series 12T (Power Increase)

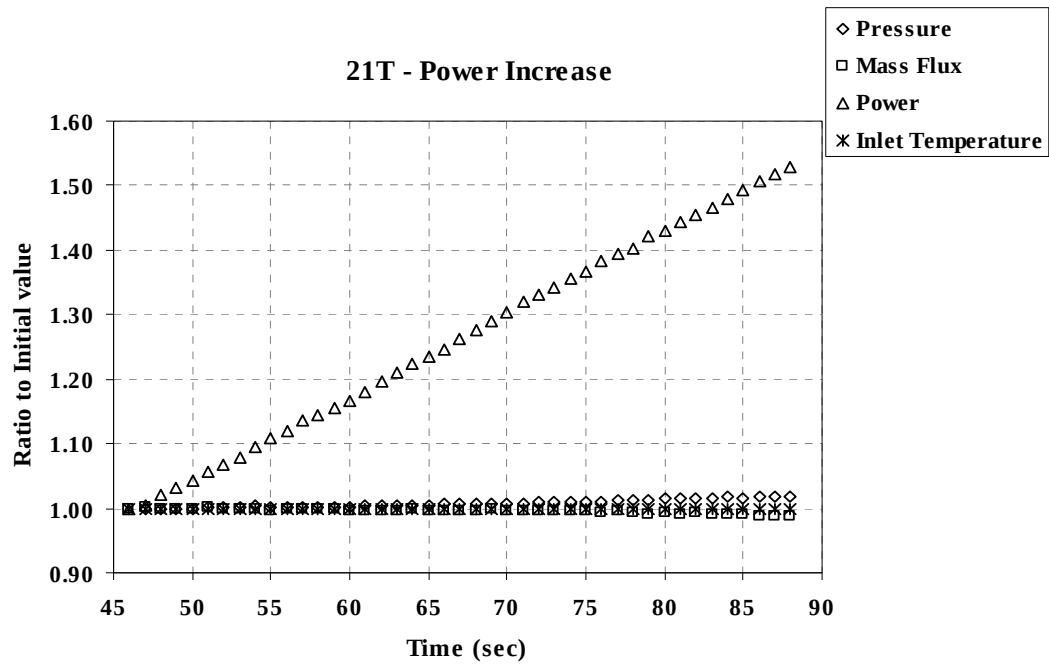


Figure 4.3.3.6 Variation of Properties during Transient for Data Series 12T (Flow Reduction)

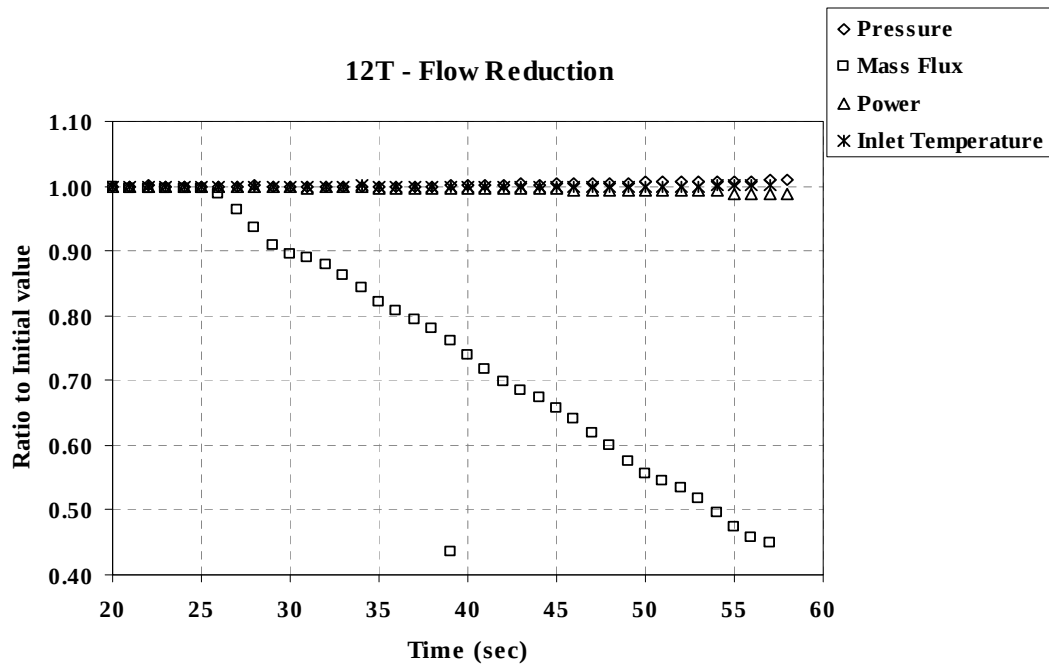


Table 4.3.3.8 Transient DNB Data in Rod Bundle in Test Series 12T (Depressurization)

Time (sec)	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m² hr)	Inlet temperature (°C)	Power (MW)
92	156.2	11.42	290.6	2.50
93	156.3	11.42	290.6	2.50
94	156.1	11.34	290.6	2.50
95	156.1	11.36	290.6	2.50
96	155.9	11.36	290.7	2.51
97	155.7	11.39	290.6	2.53
98	155.6	11.41	290.7	2.55
99	155.4	11.41	290.6	2.56
100	155.2	11.38	290.7	2.58
101	155.0	11.36	290.7	2.60
102	154.8	11.36	290.7	2.61
103	154.8	11.40	290.7	2.63
104	154.4	11.40	290.7	2.65
105	154.2	11.39	290.7	2.67
106	153.9	11.37	290.7	2.68
107	153.6	11.39	290.6	2.70
108	153.3	11.36	290.6	2.72
109	153.0	11.35	290.6	2.74
110	152.7	11.36	290.6	2.76
111	152.4	11.41	290.6	2.78
112	152.0	11.39	290.6	2.80
113	151.7	11.38	290.6	2.81
114	151.4	11.39	290.6	2.83
115	151.0	11.34	290.6	2.85
116	150.7	11.33	290.6	2.87
117	150.4	11.31	290.6	2.89
118	150.2	11.38	290.6	2.91
119	149.8	11.38	290.6	2.92
120	149.4	11.34	290.6	2.93
121	149.1	11.32	290.6	2.95
122	148.8	11.32	290.6	2.97
123	148.5	11.36	290.6	2.99
124	148.0	11.34	290.6	3.01
125	147.7	11.31	290.6	3.03
126	147.3	11.31	290.6	3.05
127	147.0	11.32	290.6	3.08
128	146.6	11.35	290.6	3.09
129	146.4	11.33	290.5	3.11
130	145.9	11.32	290.6	3.13
131	145.8	11.35	290.7	3.15
132	145.4	11.36	290.7	3.17
133	144.9	11.32	290.8	3.19
134	144.7	11.30	290.9	3.20
135	144.3	11.27	291.0	3.22
136	144.1	11.26	291.1	3.23
137	143.6	11.29	291.3	3.25
138	143.3	11.30	291.5	3.27
139	142.9	11.32	291.6	3.28
140	142.6	11.31	291.8	3.30
141	142.3	11.29	292.0	3.32
142	142.0	11.28	292.3	3.33
143	141.8	11.28	292.5	3.35
144	141.4	11.33	292.7	3.37
145	141.1	11.35	293.0	3.39
146	140.9	11.36	293.3	3.41

Table 4.3.3.9 Transient DNB Data in Rod Bundle in Test Series 12T (Temperature Increase)

Time (sec)	Pressure (kg/cm²a)	Mass flux (10⁶ kg/m² hr)	Inlet temperature (°C)	Power (MW)
83	155.8	11.38	291.2	2.5
84	155.8	11.39	291.2	2.5
85	155.6	11.36	291.2	2.5
86	155.6	11.32	291.2	2.51
87	155.5	11.34	291.2	2.51
88	155.5	11.37	291.2	2.51
89	155.4	11.37	291.2	2.53
90	155.5	11.34	291.2	2.54
91	155.4	11.32	291.2	2.56
92	155.5	11.29	291.2	2.58
93	155.3	11.25	291.2	2.6
94	155.2	11.23	291.2	2.61
95	155.2	11.23	291.2	2.63
96	155.2	11.27	291.2	2.65
97	155.2	11.28	291.2	2.66
98	155.1	11.28	291.2	2.67
99	155.4	11.21	291.2	2.69
100	155.1	11.20	291.3	2.71
101	155.2	11.22	291.3	2.73
102	155.2	11.24	291.3	2.75
103	155.2	11.22	291.4	2.77
104	155.0	11.23	291.5	2.79
105	155.0	11.20	291.7	2.81
106	155.2	11.19	292.0	2.83
107	155.0	11.19	292.3	2.84
108	155.0	11.23	292.7	2.86
109	155.0	11.18	293.3	2.88
110	155.0	11.16	293.8	2.9
111	155.1	11.12	294.4	2.91
112	155.1	11.14	295.0	2.92
113	155.3	11.18	295.7	2.94
114	155.3	11.22	296.4	2.95
115	155.4	11.22	297.1	2.97
116	155.3	11.14	297.8	2.99
117	155.4	11.19	298.6	3.01
118	155.4	11.21	299.4	3.02
119	155.4	11.25	300.2	3.04
120	155.5	11.22	301.0	3.06
121	155.7	11.24	301.9	3.08
122	155.7	11.21	302.7	3.1
123	155.8	11.22	303.5	3.12
124	155.9	11.24	304.4	3.14
125	156.1	11.24	305.3	3.15
126	156.1	11.26	306.2	3.17
127	156.4	11.25	307.2	3.18
128	156.4	11.26	308.0	3.2
129	156.7	11.23	308.8	3.22
130	156.9	11.26	309.7	3.24
131	157.0	11.26	310.5	3.25
132	157.2	11.25	311.4	3.28

Figure 4.3.3.7 Variation of Properties during Transient for Data Series 12T (Depressurization)

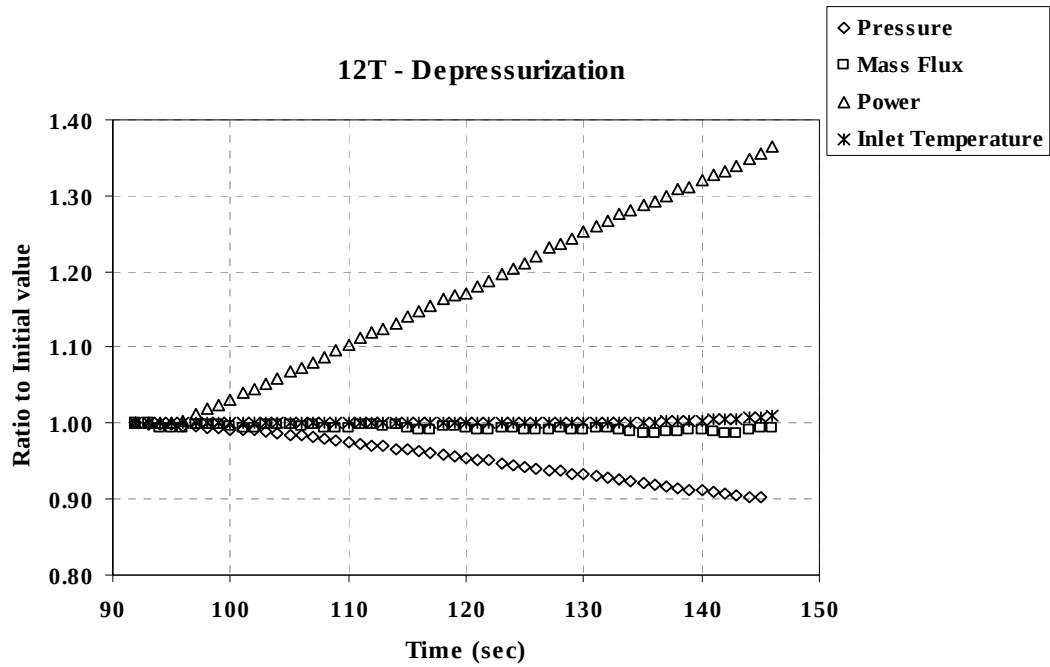
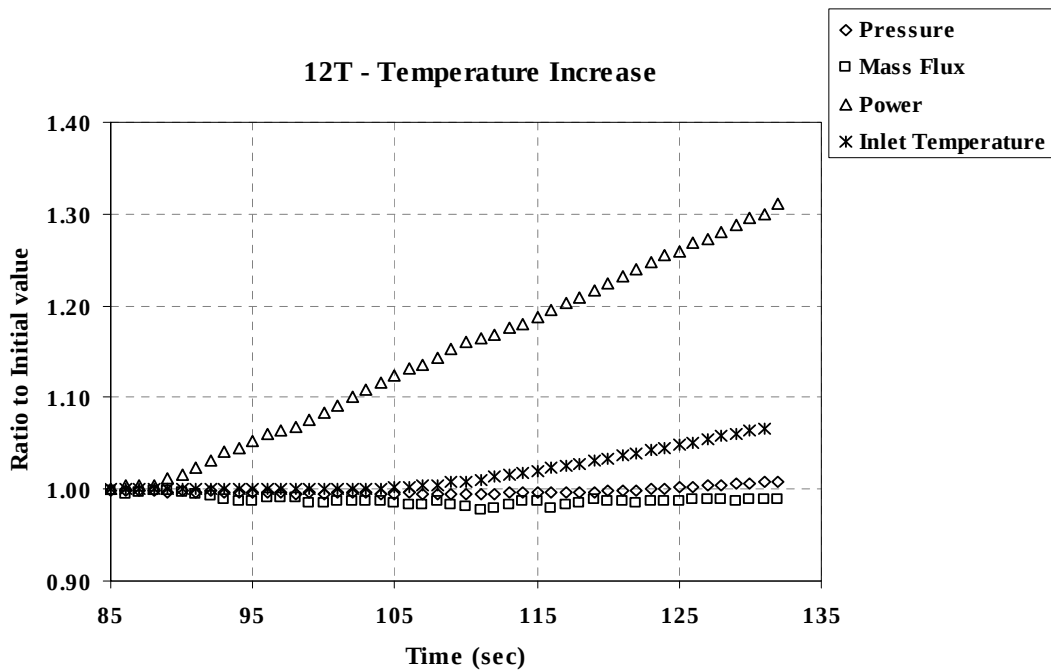


Figure 4.3.3.8 Variation of Properties during Transient for Data Series 12T (Temperature Increase)



Chapter 5

OUTPUT REQUESTED

5.1 Introduction

The participants should submit their results in an electronic format. All output should be in SI units.

Templates of requested output data for each exercise are provided by the benchmark team.

The PSU team has the obligation to:

- Provide code-to-data comparisons for different submittals;
- Provide code-to-code comparisons for different submittals;
- Provide overall conclusions on current code capabilities to represent the key phenomena.

5.2 Void Distribution Benchmark

Exercise I-1: Steady-State Single Subchannel Benchmark

For each of the test series 1, 2, 3 and 4, the calculated subchannel averaged fluid densities and void fractions have to be submitted in the format shown in Table 5.2.1. The data have to be extracted at the axial elevation of 1.4 m from the beginning of the heated length (see Figure 2.3.1.1).

Participants using CFD codes should provide images of the in-channel void distribution for all four (4) test series, in addition to the tabulated output data.

Table 5.2.1 Output Format of Steady-State Single Subchannel Benchmark

Exercise 1, Phase I
Test Series 1 – 4

Run Number	Subchannel Averaged Fluid Density (kg/m ³)	Subchannel Averaged Void Fraction	Thermal Equilibrium Quality
X.XXX	X.XXX	X.XXX	X.XXX
X.XXX	X.XXX	X.XXX	X.XXX
X.XXX	X.XXX	X.XXX	X.XXX
...	...	...	...
X.XXX	X.XXX	X.XXX	X.XXX

Exercise I-2: Steady-State Bundle Benchmark

For each of the test series 5, 6, 7 and 8, the calculated region-averaged void fractions have to be submitted in the format shown in Table 5.2.2. The averaging has to be performed over the four central subchannels of the bundle. The output data have to be extracted at three axial elevations: 2.216 m (lower), 2.669 m (middle), and 3.177 m (upper) from the beginning of the heated length (see Figure 2.3.2.1).

Table 5.2.2 Output Format of Steady-State Bundle Benchmark

Exercise 2, Phase I
Test Series 5 – 8

Run Number	Region-Averaged Void Fraction, (-)			Bundle Average Void Fraction
	Lower Elevation (2.216m)	Middle Elevation (2.669m)	Upper Elevation (3.177m)	
x.xxx	x.xxx	x.xxx	x.xxx	x.xxx
x.xxx	x.xxx	x.xxx	x.xxx	x.xxx
x.xxx	x.xxx	x.xxx	x.xxx	x.xxx
...	...	...	...	...
x.xxx	x.xxx	x.xxx	x.xxx	x.xxx

Exercise I-3: Transient Bundle Benchmark

There are four different transient scenarios simulated for three different 5×5 bundle configurations. These bundle configurations, respectively B5, B6, and B7, are described in Table 3.2.2.1. The transient scenarios include power rise, flow reduction, depressurization, and temperature increase. In total, twelve calculations are to be performed and for each calculation the results have to be submitted in a format shown in Table 5.2.3. Similarly to the Steady-State Bundle Benchmark, the averaging has to be performed over the four central subchannels of the bundle. The output data have to be extracted at three axial elevations: 2.216 m (lower), 2.669 m (middle), and 3.177 m (upper) from the beginning of the heated length (see Figure 2.3.2.1).

Table 5.2.3 Output Format of Transient Bundle Benchmark

Exercise 3, Phase I
Test Series 5T (Power increase)

Time, [sec]	Region-Averaged Void Fraction, (-)		
	Lower Elevation 2.216 m	Middle Elevation 2.669 m	Upper Elevation 3.177 m
xx.x	x.xxx	x.xxx	x.xxx
xx.x	x.xxx	x.xxx	x.xxx
...	...	...	...
xx.x	x.xxx	x.xxx	x.xxx

Exercise I-4: Pressure Drop Benchmark

As discussed previously, although there is no available data, code to code benchmark on pressure drop of heated section for the typical central subchannel is proposed. The reference value for the bundle under the rated condition (single-phase) is about 1.6 kg/cm² which is read from the initial value of the pressure drop over the heated section for the bundle B7 during power increase transient (test series 7T).

Participants are requested to submit the code predicted pressure drop over the heated length of the bundle test assembly B7, test series T7 before the transient is initialized.

In addition, participants are requested to submit the code predicted pressure drops over the heated length of the single subchannel test assembly S1 and bundle test assembly B5 bundle for the exercise cases included in the Test Series 1 and Test Series 5. Results have to be submitted in the format shown in Table 5.2.4 and Table 5.2.5.

Table 5.2.4 Output Format of Single Subchannel Pressure Drop Benchmark

Exercise 4, Phase I
Test Series 1

Test Number	Pressure Drop, (kg/cm ²)
X.XXX	XXX.X
X.XXX	XXX.X
...	...
X.XXX	XXX.X

Table 5.2.5 Output Format of Bundle Pressure Drop Benchmark

Exercise 4, Phase I
Test Series 5

Test Number	Pressure Drop, [kg/cm ²]
X.XXX	XXX.X
X.XXX	XXX.X
...	...
X.XXX	XXX.X

5.3 DNB Benchmark

Exercise II-1: Steady-State Fluid Temperature Benchmark

The calculated results for a given exercise case should be submitted as a 6×6 matrix of space averaged fluid temperature in a subchannel mesh size. An example of the output format of the Steady-State Fluid Temperature Benchmark is given in Table 5.3.1. The data have to be extracted at the axial elevation of 457mm above the end of the heated length, where the actual measurements were conducted (see Figure 3.3.6).

Table 5.3.1 Output Format of Steady-State Fluid Temperature Benchmark

Exercise 1, Phase II
Test No.
Average Fluid Temperature, (°C)

y/x	1	2	3	4	5	6
1	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x
2	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x
3	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x
4	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x
5	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x
6	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x	xxx.x

Cross-sectional averaged: (°C)

Exercise II-2: Steady-State DNB Benchmark

For each of the test series 0, 2, 3, 4, 8 and 13, the predicted DNB powers; the axial elevation and radial position of the first predicted DNBs (test series 4, 8, and 13 only) have to be submitted in the format shown in Table 5.3.2.

Table 5.3.2 Output Format of Steady-State DNB Benchmark

Exercise 2, Phase II
Test Series 0, 2, and 3

Test Number	Predicted DNB Power, (MW)
XX-XXX	X.XX
XX-XXX	X.XX
...	...
XX-XXX	X.XX

Exercise 2, Phase II
Test Series 4, 8 and 13

Test Number	Predicted DNB Power, (MW)	Axial Elevation of First Predicted DNB, (m)	Radial Position of First Predicted DNB, Central or Peripheral Rod
XX-XXX	X.XX	X.XXX	XXXXXX
XX-XXX	X.XX	X.XXX	XXXXXX
...	...	...	...
XX-XXX	X.XX	X.XXX	XXXXXX

Exercise II-3: Transient DNB Benchmark

There are four different transient scenarios simulated for two different 5×5 bundle configurations. The two bundle configurations, respectively A11 and A12, are described in Table 3.3.4 . The transient scenarios include power rise, flow reduction, depressurization, and temperature increase. In total, eight calculations are to be performed and for each calculation the results have to be submitted in the format shown in Table 5.3.3.

Table 5.3.3 Output Format of Transient DNB Benchmark

Exercise 3, Phase II
Test Series 11T and 12T

Test Number	Time of Detected DNB, (sec)
11-0312	X.X
11-0112	X.X
11-0212	X.X
11-0321	X.X
12-0312	X.X
12-0112	X.X
12-0211	X.X
12-0321	X.X

Chapter 6

CONCLUSIONS

The objective of the PSBT benchmark specifications is to provide the participants with information on the benchmark database. Although it contains detailed information on the test cases and measurements for the complete NUPEC PWR database, specific exercises were selected out for the OECD/NRC PSBT benchmark to reduce the amount of work; to make the analysis more precise; and to avoid confusion among the comparison of the participants' results.

6.1 Phase I - Void Distribution Benchmark

Exercise I-1 – Steady-State Single Subchannel Benchmark

The goal of this exercise is to benchmark the subchannel, meso- and microscopic numerical approaches. The supplied measured data includes space averaged void fraction in subchannel mesh size. The test cases are selected at PWR rated conditions. Different types of single subchannel test assemblies are used to investigate the effect of geometry on phenomenon in concern.

Exercise I-2 – Steady-State Bundle Benchmark

This exercise is designed for benchmarking meso-scopic numerical approaches. The test cases for this exercise are chosen at PWR rated conditions and deviations of quality from the rated conditions.

Exercise I-3 – Transient Bundle Benchmark

NUPEC PSBT database includes simulation of four representative transients of PWRs; power increase, flow reduction, depressurization, and temperature increase. All four transients are selected as benchmark cases. Exercise 3 of Phase II is designed for benchmarking subchannel numerical approaches.

Exercise I-4 – Pressure Drop Benchmark

This exercise is designed for performing code-to-code comparisons concerning axial pressure drop. Although no empirical data is available, code results will be compared with relevant graphical data.

6.2 Phase II - DNB Benchmark

Exercise II-1 – Steady-State Fluid Temperature Benchmark

Exercise 1 of Phase II is designed to assess the thermal-hydraulic codes' capabilities of predicting the exit coolant temperature.

Exercise II-2 – Steady-State DNB Benchmark

The goal of Exercise 2 of Phase II is to assess the thermal-hydraulic codes' capabilities of correct prediction of DNB along rod bundles.

Exercise II-3 – Transient DNB Benchmark

Exercise 3 of Phase II is designed to enhance the currently underway development of truly mechanistic models for DNB prediction during the four postulated transients in PWRs.

REFERENCES

- [1] “OECD/NEA Benchmark Based on NUPEC PWR Subchannel and Bundle Tests (PSBT)”, JNES, April 2009
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- [3] Hori, K. et al., “In Bundle Void Fraction Measurement of PWR Fuel Assembly”, ICONE-2, Vol.1, pp.69-76, San Francisco, California, 21-24 March 1993
- [4] Pamphlet of Takasago Engineering Laboratory, Nuclear Power Engineering Center (NUPEC).